

Elliptic squares and Zilber–Pink for curves in $\mathcal{A}_2$

OpenAI

September 24, 2026

Abstract

We prove that every Hodge-generic algebraic curve in $\mathcal{A}_2$ over $\overline{\mathbb{Q}}$ contains only finitely many points whose abelian surface is isogenous to the square of an elliptic curve without complex multiplication (CM). Combining this result with the companion $E \times \text{CM}$ and quaternionic-division finiteness theorems, we prove the curve case of Zilber–Pink in $\mathcal{A}_2$ over $\overline{\mathbb{Q}}$, without a boundary hypothesis.

Contents

1	Introduction	2
1.1	The companion branches and the full theorem	2
1.2	History and the arithmetic step	3
1.3	The proof mechanism	5
1.4	Conventions and organization	7
2	Interpolation with a common small parameter	7
2.1	The quantitative statement	7
2.2	Why the graph gives a height bound	9
3	The geometric systems and a finite list of correspondences	10
3.1	Fixed covers and the center	10
3.2	The five-dimensional orthogonal system	11
3.3	Full and joint monodromy	13
3.4	Finitely many dominating correspondence types	15
4	Two auxiliary markings and exclusion of bad disks	18
4.1	Finite-field positions of two planes	18
4.2	Independent markings on one connected cover	20
4.3	A common toric subspace in a strict local model	21
4.4	Excluding proximity at a bad center place	23
5	Heights and integral frames	24
5.1	Height conventions and small endomorphisms	24
5.2	A bundle whose trait lattices preserve endomorphisms	27
5.3	Global coordinates and the sharper estimate near the center	29
6	Local mass and equations	32
6.1	The actual center and the amount of proximity	32
6.2	Good-reduction comparison on a common tube	34
6.3	The three equations	35
6.4	Uniform grouping and the numerical interpolation data	38

7	Nonidentities and the uniform height bound	41
7.1	Algebraic branches and continuation	42
7.2	The Frobenius obstruction	43
7.3	The three equation types	45
7.4	Uniform height conclusion	47
8	From heights to finiteness	47
8.1	The discriminant and the Galois orbit	47
8.2	All polarizations and all elliptic-square curves	49
8.3	The counting implication	50
8.4	The special-curve classification	50
8.5	The full Zilber–Pink theorem	51
A	Proof of the interpolation theorem	52
A.1	Effective algebra with absolute heights	52
A.2	A uniform boundary multiplicity bound	54
A.3	The graph and its projected boundary	57
A.4	The auxiliary polynomial in the actual ideal	58
A.5	The product formula and dimension descent	60
	References	63

1 Introduction

Let $\mathcal{A}_2$ be the coarse moduli variety of principally polarized abelian surfaces. A curve in $\mathcal{A}_2$ is *Hodge generic* if it is contained in no proper Shimura-special subvariety. The curve case of the Zilber–Pink conjecture predicts that such a curve meets the union of all special subvarieties of dimension at most one in only finitely many points. The word “all” includes every Hecke translate; neither an isogeny degree nor an endomorphism order is fixed.

We first prove the elliptic-square part of this prediction, then combine it with the two companion branch theorems to obtain the full conclusion. For a point $s \in \mathcal{A}_2(\overline{\mathbb{Q}})$, write (A_s, λ_s) for its principally polarized abelian surface, and set

$$\Sigma_{E^2}(C) = \{s \in C(\overline{\mathbb{Q}}) : A_s \sim E^2 \text{ for an elliptic curve } E/\overline{\mathbb{Q}} \text{ without CM}\}.$$

Here $\sim$ denotes isogeny over $\overline{\mathbb{Q}}$, without any requirement that it respect the principal polarization. The condition is equivalent to $\text{End}_{\overline{\mathbb{Q}}}(A_s) \otimes_{\mathbb{Z}} \mathbb{Q} \simeq M_2(\mathbb{Q})$.

Theorem 1.1 (The elliptic-square case). *Let $C \subset \mathcal{A}_2$ be a nonempty reduced irreducible closed algebraic curve over $\overline{\mathbb{Q}}$. If C is Hodge generic, then $\Sigma_{E^2}(C)$ is finite.*

No condition is imposed on the closure of C at the boundary of a compactification. In particular, the theorem includes complete curves. The elliptic curve, its isogeny degree and the principal polarization may all vary.

1.1 The companion branches and the full theorem

The two remaining special-curve loci are treated by the following companion theorems. We state their exact conclusions to specify the inputs to the full finiteness theorem.

Theorem 1.2 (The two companion branches). *For every nonempty reduced irreducible closed Hodge-generic curve $C \subset \mathcal{A}_2$ over $\overline{\mathbb{Q}}$, the following two subsets of $C(\overline{\mathbb{Q}})$ are finite:*

- (A) *the points for which A_s is isogenous over $\overline{\mathbb{Q}}$ to $E_1 \times E_2$, with at least one elliptic factor having CM;*
- (B) *the points for which the full algebra $\text{End}_{\overline{\mathbb{Q}}}(A_s) \otimes_{\mathbb{Z}} \mathbb{Q}$ is an indefinite quaternion division algebra over $\mathbb{Q}$.*

These statements allow all CM fields, endomorphism orders, quaternion algebras, isogeny degrees and principal polarizations. They impose no boundary, reduction, integrality, height or field-degree condition.

In (B), indefiniteness means that the algebra becomes $M_2(\mathbb{R})$ over $\mathbb{R}$; the division condition excludes $M_2(\mathbb{Q})$. An embedding of a quaternion algebra into a larger endomorphism algebra does not satisfy (B). In (A), both-CM products are included. The complete proofs of these conclusions are given respectively in the companion articles [Ope26b, Theorem 1.1] and [Ope26a, Theorem 1.1]. The proof below keeps their use separate from the elliptic-square theorem.

Theorem 1.3 (Zilber–Pink for curves in $\mathcal{A}_2$). *For every nonempty reduced irreducible closed Hodge-generic curve $C \subset \mathcal{A}_2$ over $\overline{\mathbb{Q}}$, the set*

$$\Sigma(C) = C(\overline{\mathbb{Q}}) \cap \bigcup_{\substack{Z \subset \mathcal{A}_2 \text{ special} \\ \dim Z \leq 1}} Z(\overline{\mathbb{Q}})$$

is finite.

Theorem 1.3 resolves positively the Hodge-generic-curve case of the Zilber–Pink conjecture in $\mathcal{A}_2$ over $\overline{\mathbb{Q}}$. Its proof combines Theorems 1.1 and 1.2 with the classification of special curves and André–Oort for special points, as explained in Proposition 8.3 and Section 8.5. Theorem 1.1 and its proof are independent of the two companion results. After removing special points, the three branch statements cover the points of C that are Hodge generic on a special curve; Proposition 8.3 proves this classification with the full endomorphism algebras specified. For a curve that is not Hodge generic in $\mathcal{A}_2$, André–Oort also gives the corresponding special-closure formulation: if S_C is the smallest special subvariety containing C , then C meets the union of special Z with $\dim Z < \dim S_C - 1$ in finitely many points (Corollary 8.4).

1.2 History and the arithmetic step

Zilber’s atypical-intersection principle for semiabelian varieties [Zil02] and Pink’s formulation for mixed Shimura varieties place this problem in a common unlikely-intersection framework. Pink’s formulation predicts finiteness for a Hodge-generic curve meeting special subvarieties of codimension at least two [Pin05, Conjecture 1.3]. In the three-dimensional space $\mathcal{A}_2$, the special curves have three generic types: elliptic products with a CM factor, non-CM elliptic squares, and quaternion-division multiplication. Pila and Tsimerman established André–Oort for $\mathcal{A}_2$, which supplies the special-point case [PT13, Theorem 1.1]. Daw and Orr developed the corresponding unlikely-intersection arguments for these curve loci [DO21, DO22].

The arithmetic-to-geometric strategy compares lower bounds coming from Galois conjugates with upper bounds for rational points in definable sets. Pila and Zannier introduced this strategy in their proof of Manin–Mumford [PZ08], using the counting

theorem of Pila and Wilkie [PW06]. In Siegel moduli, the arithmetic difficulty is to obtain a sufficiently large orbit uniformly as the special subvariety varies. Daw and Orr’s quantitative reduction theory supplies the geometric and counting implication for the elliptic-square and quaternionic loci. Their unconditional quaternionic theorem assumes that the curve meets the zero-dimensional Baily–Borel boundary [DO22, Theorem 1.4]; the orbit-to-finiteness implication we use has no such hypothesis.

Daw and Orr subsequently proved the full special-curve finiteness conclusion, including elliptic squares, under the same total multiplicative-degeneration hypothesis [DO25a, Theorem 1.1 and Corollary 1.2]. Papas studied splitting points using periods centered at an interior $E \times \text{CM}$ or elliptic-square point. For an everywhere potentially-good center, his height bound depends polynomially on the point degree and the number of supersingular places of proximity; his finiteness corollary fixes an upper bound for that number [Pap25, Theorem 1.5 and Corollary 1.6]. His analogous bad-reduction statements are conditional on comparison conjectures [Pap25, Theorem 1.8 and Corollary 1.9]. In the product $Y(1)^3$ of three modular curves, Daw, Orr and Papas obtain finiteness from an interior basepoint, allowing bad reduction but restricting the growth of the number of supersingular places of proximity [DOP26, Theorems 1.6 and 1.8]. The present height estimate permits bad reduction of the center and does not bound that number of places.

For the elliptic-square locus, the implication we use is Daw–Orr’s [DO22, Theorem 1.3]: a suitable polynomial lower bound for Galois orbits in terms of the endomorphism-order discriminant implies finiteness. Their arithmetic hypothesis is stated in [DO22, Conjecture 6.2]. We prove the required bound for the entire elliptic-square locus before applying that theorem. The constants may depend on C ; effectivity is not required.

The main arithmetic result is a polynomial height bound for selected lifts t to one fixed fine-level curve. Write $d = \max\{2, [K(t) : K]\}$, where K contains the fixed data, and let $h(t) \geq 2$ be a shifted ample logarithmic height. If s is the coarse image of t , let Δ_s be the discriminant of the full geometric order $\text{End}_{\overline{\mathbb{Q}}}(A_s)$, using the trace of its regular representation. The two arithmetic estimates are

$$h(t) \leq cd^\kappa, \quad \Delta_s \leq c(dh(t))^{\kappa'}.$$

We prove the first in Theorem 7.5; the second comes from the endomorphism estimates of Masser and Wüstholz [MW94], as explained in Proposition 5.2. Since the cover degree is fixed, $d \leq c[K(s) : K]$. Combining the two estimates therefore gives the required orbit bound $[K(s) : K] \geq c'\Delta_s^\delta$ for some $\delta > 0$. The discriminant belongs to the full order, not just the symmetric plane used in our period equations. This is why we bound endomorphisms spanning the whole algebra as well as a pair spanning that plane. Propositions 8.1 and 8.2 account for every fixed omission and identify Δ_s with the complexity in the counting theorem.

Several classical inputs make this reduction quantitative. André’s normality theorem for connected algebraic monodromy identifies the monodromy group of our Hodge-generic family [And92, §5, Theorem 1]. Strong approximation then permits independent markings at two auxiliary levels [MVW84]. Deligne’s regularity theorem and Schmid’s Hodge-norm estimates control fixed algebraic frames near the punctures [Del70, Sch73], while rational crystalline comparison identifies horizontal transport at nearby good reductions [BO83, Ber82]. The interpolation argument uses effective polynomial ideal membership and arithmetic intersection bounds [Her26, Dub90, Asc04, KPS01]; its graph construction is what turns those bounds into the height estimate needed here.

The use of arithmetic auxiliary functions and period equations belongs to the tradition of Bombieri’s and André’s work on G -functions [Bom81, And89]. For curves in products

of modular curves with multiplicative degeneration, Daw and Orr use relations valid simultaneously at all nonarchimedean places, obtained from Tate uniformization [DO25b, §1.B]. Interior p -adic periods are developed in André’s motivic account [And95]; see also his recent discussion [And25, §§2.2 and 3.3.4] and [Pap25, Theorem 3.4]. Our interpolation theorem is formulated for fixed differential systems at an ordinary point. Section 2 states the theorem and explains its mechanism; Appendix A proves the varying-curve multiplicity estimate and two-argument graph descent.

1.3 The proof mechanism

Suppose the square locus is infinite, and fix one of its points t_0 on a smooth fine-level cover. This interior point will be the center of our local equations. On first homology, put

$$\mathcal{P} = \{D : D^\dagger = D, \operatorname{Tr}(D) = 0\}, \quad \langle D, E \rangle = \operatorname{Tr}(DE),$$

where $\dagger$ is the adjoint for the polarization. This is a rank-five quadratic system. At a non-CM elliptic square, its actual rational endomorphisms form a positive plane. Choose labeled integral pairs $\mathbf{B} = (B_1, B_2)$ and $\mathbf{P} = (P_1, P_2)$ spanning the center and target planes, with the target pair chosen using the endomorphism estimate. Let $a = x(t)$ for a local parameter x vanishing at t_0 , and let $Y(a)$ be horizontal back-transport in a fixed de Rham frame from the target to the center, with $Y(0) = 1$. We study the four cross pairings

$$\langle \mathbf{B}, Y(a)\mathbf{P} \rangle := (\langle B_i, Y(a)P_j \rangle)_{1 \leq i, j \leq 2}.$$

When the two fibers have the same good reduction, rational cohomological comparison identifies these entries with traces of products of actual endomorphisms of that reduction. Frobenius preserves the traces after simultaneously conjugating both fibers and both pairs. Writing primes for those conjugate data gives the equation

$$\langle \mathbf{B}, Y(a)\mathbf{P} \rangle = \langle \mathbf{B}', Y'(aR)\mathbf{P}' \rangle, \quad R = a'/a, \quad |R|_v = 1.$$

The residue characteristic no longer occurs in this equation. Grouping places by the two labeled conjugate tuples gives only polynomially many systems in d , even when the number of proximity primes is unbounded. At the archimedean and finitely many exceptional places we instead use a single equation $\langle \mathbf{B}, Y(a)\mathbf{P} \rangle = \mathbf{r}$ with a rational matrix $\mathbf{r}$.

Two fixed prime levels make these equations usable. At the first, ℓ_1 , the separately saturated center and target plane lattices are marked so that their reductions span a degenerate four-space; at the second, ℓ_2 , their sum is nondegenerate. Both conditions can be imposed on one connected finite cover. The second incidence excludes sufficiently close targets at a place where the center is not potentially good: both planes would preserve the same toric two-space, forcing their sum to be degenerate. Thus every retained finite place admits the good-reduction comparison just described.

The first incidence serves a different purpose. Full monodromy rules out identities among the equations unless the target and its conjugate lie on one of a fixed finite list of polarized Hecke correspondences. On that list, isogenies of fixed multiplier transport the primed pairs back to the unprimed fibers. Monodromy then forces any identity to identify the transported pairs with the original pairs by a scalar on the center pair and its reciprocal on the target pair. The center norms force that scalar to be $+1$ or -1 . The positive sign would make a rational square equal to p/m , where p is the residue characteristic and m is a fixed isogeny multiplier prime to p . The negative sign contradicts the degenerate incidence at ℓ_1 . The Hecke list and its multipliers are fixed before choosing these levels, so ℓ_1 can avoid all those multipliers.

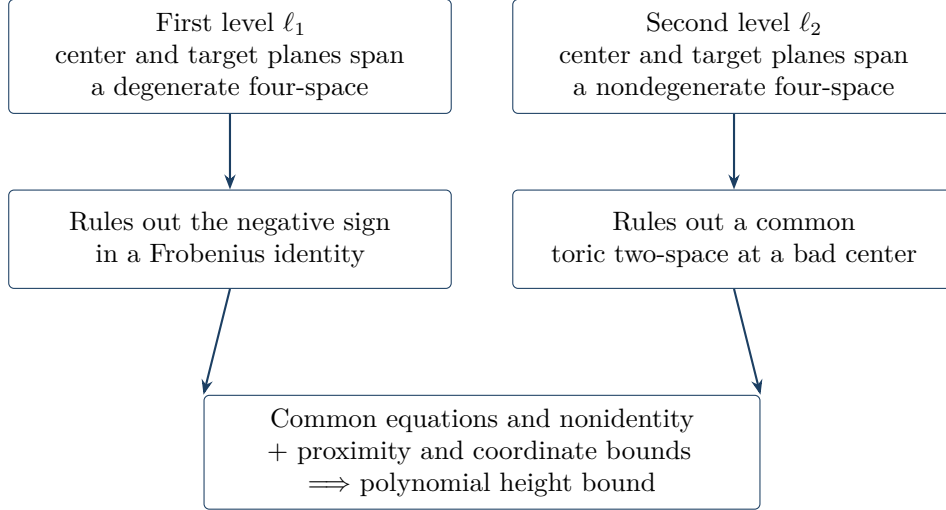


Figure 1: The two level incidences have different roles. The first is used to exclude the negative sign in the nonidentity argument; the positive sign is excluded by the Frobenius multiplier. The second excludes proximity to centers with non-potentially-good reduction. The height conclusion uses common equations at the retained places, nonidentity on algebraic boundary branches, the large proximity sum, and the small positive coordinate logarithms described in the text. The diagram is schematic: all plane positions are in the separately marked finite-field quadratic spaces.

There are two different size estimates in the interpolation step. Let U collect the base coordinates, the target endomorphism coordinates and, when needed, R . Their global height can grow like $h = h(t)$: $h_{\text{aff}}(a, U) \leq Cd^C h$. Unless h is already polynomially bounded in d , we select a group of nearby places at which the weighted positive logarithms of U are at most $Cd^C(1 + \log h)^C$, while the weighted sum of $\log |a|_v^{-1}$ is at least $h/(Cd^C)$. Here the weights are the normalized absolute-height weights. The center divisor supplies the large proximity sum; integral lattices preserved by endomorphisms and bounded center frames supply the much smaller coordinate sum. These estimates hold before any bound for h in terms of d is known.

[Theorem 2.1](#) turns this imbalance, together with the nonidentity just explained, into $h \leq Cd^C$. Its proof constructs a polynomial whose degree is small compared with its order of local smallness. The product formula either bounds h or forces the target into a proper algebraic subvariety. The process repeats in that subvariety, which explains why nonidentity must hold on boundary branches of arbitrary algebraic varieties through the target, even when their operator coordinates are not Hodge classes. [Section 2](#) explains how a graph construction makes this dimension descent quantitative.

The quaternionic companion develops Frobenius cross-pairing interpolation and the degenerate auxiliary incidence [[Ope26a](#), Sections 2–5]. The arguments needed here, including the second incidence, the integral coordinate bounds and the nonidentity analysis for split endomorphism algebras, are proved below. The interpolation theorem and the separation of bad-center disks from good-reduction comparisons are the parts of the construction that may be useful in other height problems.

Figure 1 distinguishes the two uses of the auxiliary levels and their roles alongside the arithmetic estimates.

1.4 Conventions and organization

All heights are absolute logarithmic heights. Place sums use normalized absolute-height weights, so extending a field preserves the sum for pulled-back quantities. Constants may depend on the curve, the center, fixed levels, frames and other fixed data. They never depend on the varying target, its height or degree, or a varying residue characteristic. In an estimate described as polynomial, the exponent is fixed.

We use covariant first homology and its de Rham and prime-to-residue- characteristic étale realizations. The term *ordinary point* for a differential equation means a point where the connection has no singularity; it places no ordinarity condition on the reduction of an abelian variety. Every reduction of an endomorphism plane uses its saturation in the specified integral lattice.

Section 2 states the interpolation theorem and explains its proof mechanism. Section 3 constructs the geometric systems and controls the exceptional Hecke correspondences. Section 4 gives the two incidences and excludes bad-center disks. Section 5 proves the coordinate estimates, and Section 6 constructs the local equations with their quantitative bounds. Section 7 rules out identities and obtains the height bound. Finally, Section 8 proves Theorems 1.1 and 1.3 and the special-closure corollary. The complete proof of the interpolation theorem is given in Appendix A.

2 Interpolation with a common small parameter

The height theorem below applies to equations assembled from a fixed collection of ordinary differential systems. Its hypotheses distinguish the global height of the target from the much smaller positive coordinate logarithms at the places used for interpolation. We state the theorem and explain the graph construction that exploits this distinction here. The full proof is given in Appendix A.

2.1 The quantitative statement

For a tuple $x = (x_1, \dots, x_r)$ over a number field F , write

$$h_{\text{aff}}(x) = \sum_v w_v \log \max(1, |x_1|_v, \dots, |x_r|_v), \quad w_v = \frac{[F_v : \mathbb{Q}_v]}{[F : \mathbb{Q}]}.$$

At an archimedean place we use the ordinary absolute value, so that a complex place has weight $2/[F : \mathbb{Q}]$. At a finite place above p , the absolute value extends $|p|_p = p^{-1}$. The same notation for a polynomial means the height of its coefficient tuple. A weighted subcollection of places may split a place into its embeddings, with the corresponding weights. When F is enlarged, these weights are split among the extensions of each embedding. In particular all sums for quantities defined over the old field are unchanged. We write $\sum_V w_v$ with this convention even when the subcollection is described in terms of embeddings.

An *ordinary germ* below is a horizontal analytic solution in a rational frame in which the connection is regular at the specified center. This is a condition on a differential equation. The rank, frame, center, and curve carrying each such equation are fixed.

Theorem 2.1 (Conditional height bound). *Fix an ambient dimension N , a bound s_0 for the number of equations, an integer b bounding the degrees of the polynomial combinations below, and a real constant $C_0 \geq 2$. Fix also a finite list of algebraic regular germs and ordinary horizontal matrix germs on fixed algebraic curves over number fields. Dual and inverse matrices are allowed in this list. Then there is a constant C , depending only on these fixed data, with the following property.*

Let $d, h \geq 2$ and let $z = (a, U) \in \mathbb{A}^N(\mathbb{Q})$ with $a \neq 0$. Suppose a weighted subcollection V and $1 \leq s \leq s_0$ equations

$$E_\alpha(a, U) = \sum_{j \geq 0} a^j e_{\alpha, j}(U) = 0 \quad (1 \leq \alpha \leq s) \quad (2.1)$$

satisfy the following conditions.

- (i) *The point and the selected places satisfy*

$$h_{\text{aff}}(z) \leq C_0 d^{C_0} h, \quad |a|_v < 1 \quad (v \in V), \quad (2.2)$$

$$\sum_{v \in V} w_v \log |a|_v^{-1} \geq \frac{h}{C_0 d^{C_0}}, \quad \sum_{v \in V} w_v \log^+ \|U\|_v \leq C_0 d^{C_0} (1 + \log h)^{C_0}. \quad (2.3)$$

Here $\|U\|_v$ is the maximum of the coordinate absolute values.

- (ii) *The coefficients are polynomials over a number field containing the target and equation data, with*

$$\deg e_{\alpha, j} \leq C_0(1 + j), \quad h_{\text{aff}}(e_{\alpha, j}) \leq C_0(1 + j + d)^{C_0} (1 + \log h)^{C_0}. \quad (2.4)$$

The same equations are used at every selected embedding. They converge there at the target, and there are numbers $b_v \geq 0$ such that, for all $\ell \geq 0$ and all α ,

$$\left| \sum_{j \geq \ell} a^j e_{\alpha, j}(U) \right|_v \leq |a|_v^\ell \exp((\ell + 1)b_v), \quad (2.5)$$

$$\sum_{v \in V} w_v b_v \leq C_0 d^{C_0} (1 + \log h)^{C_0}. \quad (2.6)$$

- (iii) *As analytic functions in the variables (a, U) , the E_α are polynomial combinations of degree at most b of the coordinates and the fixed germs evaluated at a , and possibly at aR , where R is one of the coordinates of U . Their constant coefficients may depend on the target. The algebraic germs are regular functions on the specified local curve charts. Thus every expression is also a convergent complex analytic germ near any finite point of $a = 0$.*
- (iv) *Let $W \subseteq \mathbb{A}_{\mathbb{Q}}^N$ be any irreducible closed subvariety containing z on which a is nonconstant. For each prime divisor D in its finite normalization lying above a component of $W \cap \{a = 0\}$, at least one E_α is not identically zero as an analytic germ on the local branch at a general point of D .*

Then $h \leq Cd^C$.

The quantifier in condition (iv) includes varieties defined using the coefficients of the equations and the auxiliary polynomials constructed below. It concerns the germ on a whole local branch, rather than its value on the boundary divisor. No bound on the degree of the field of equation coefficients is part of this theorem.

For the elliptic-square application, the inputs to conditions (i)–(iii) are assembled in Proposition 6.5, and condition (iv) is verified in Proposition 7.4; the resulting height bound

is stated in [Theorem 7.5](#). We next explain why the theorem requires nonidentity on every algebraic branch, and why small positive local logarithms are needed in addition to a global height bound.

2.2 Why the graph gives a height bound

The proof constructs an auxiliary polynomial of degree L whose value at the target gains a factor a^n at the selected places. To make the product formula useful, the construction must allow a sufficiently large ratio n/L . The graph construction makes this possible by lowering the dimension of the boundary on which the interpolation conditions are imposed.

Start with $W = \mathbb{A}^N$. During the proof, W will be an irreducible algebraic subvariety containing z , with controlled degree and defining heights. Write $k = \dim W$ and first suppose that a varies on W and that $W \cap \{a = 0\}$ is nonempty. Condition (iv) and an ordinary differential multiplicity estimate give an integral linear combination $E = \sum c_\alpha E_\alpha = \sum_{j \geq 0} a^j e_j(U)$ and an integer m such that its truncation

$$F = E_{<m} := \sum_{j=0}^{m-1} a^j e_j(U) \quad \text{satisfies} \quad \text{ord}_D F < m \text{ord}_D a$$

on every normalized boundary divisor D . The coefficients c_α and m are polynomially bounded in $\deg W$. This strict inequality means that F/a^m has a pole at the generic point of every such divisor; a nonzero restriction of E to the boundary itself is not required.

Let W' be the affine closure of the graph of $q = F/a^m$ over $a \neq 0$. It has dimension k . Its boundary over $a = 0$ has no point above a general point of any boundary divisor of W , because q has a pole there. Thus the projection of $W' \cap \{a = 0\}$ to W has dimension at most $k - 2$. Imposing $q = -e_m(U)$ fixes the graph coordinate, so

$$\dim(W' \cap \{a = 0, q + e_m(U) = 0\}) \leq k - 2.$$

For $k = 1$ this intersection is empty. This is a bound on the indicated incidence, not on the whole graph boundary, whose fibers may have positive dimension.

The lifted target $z' = (a, U, F(z)/a^m)$ has a different useful property. At each selected place $E(z) = 0$, and therefore

$$q(z') = -a^{-m} \sum_{j \geq m} a^j e_j(U).$$

The tail estimate bounds the positive local logarithm of this new coordinate by $(m + 1)b_v$, with the errors enlarged to accommodate the chosen integral combination. Hence z' still has small positive local logarithms, although its global height can be proportional to h .

Choose equations generating an ideal I whose zero set is W' and put

$$F_n = q + \sum_{j=0}^{n-1} a^j e_{m+j}(U).$$

The dimension bound above permits a polynomial P , nonzero on W' , with

$$\deg P \leq L, \quad P \in (I, a^n, F_n), \quad n = cL,$$

for any prescribed positive integer c , with L polynomially bounded in c and the degree bounds of the construction. Here membership is in the displayed ideal itself, even if I is nonradical. The reason for the freedom in c is a count of linear conditions. For $k \geq 2$,

reducing successively through n powers of a imposes $O(n(L+n)^{k-2})$ conditions, with a constant polynomial in the defining degree bounds. Restrictions of degree $\leq L$ polynomials to W' span a space of dimension at least $\binom{L+k}{k}$. For $n = cL$ these grow respectively as L^{k-1} and L^k . When $k = 1$ there are no conditions. This saving of one power of L is the purpose of the graph.

At the selected places, evaluating the ideal-membership certificate and using the tail bound gives the factor $|a|_v^n$. Even if the certificate has degree larger than L , its degree multiplies only the small positive local logarithms of z' . At the remaining places one evaluates P itself, so the global height cost is proportional to Lh . Consequently, if $P(z') \neq 0$, the product formula gives an inequality of the form

$$0 \leq \left(-\frac{n}{A(d)} + LC(d) \right) h + D(d)(1 + \log h)^B,$$

where A, C, D are fixed polynomials and B is fixed. The polynomials A, C are determined before c is chosen. Taking c sufficiently large, still polynomially bounded in d , makes the coefficient of h negative and yields a polynomial bound for h in d .

If that bound does not hold, $P(z') = 0$. Substituting $q = F/a^m$ and clearing denominators gives a polynomial nonzero on W and zero at z . An irreducible component through z of its zero set in W has smaller dimension and controlled degree and height. This explains the universal quantifier in condition (iv): the next variety is chosen using the equations and the auxiliary polynomial. After at most N such steps, either the height is bounded or one reaches a variety disjoint from $a = 0$. If $G_1, \dots, G_r$ define that variety, a polynomial identity $1 = aA + \sum_{\rho} G_{\rho} A_{\rho}$ and the small positive local logarithms give the bound directly. Appendix A proves these degree, height, and multiplicity assertions and the descent in detail.

3 The geometric systems and a finite list of correspondences

We first fix the curves, the family, and an elliptic-square center. The trace-free self-adjoint endomorphisms form a five-dimensional orthogonal system. Its monodromy gives an alternative for two varying points: the two systems have product monodromy, or their joint image lies in a polarized isogeny correspondence. The final result of this section reduces the latter alternative to a fixed finite list. This list is chosen before the auxiliary levels of Proposition 4.2.

3.1 Fixed covers and the center

Proposition 3.1. *In proving Theorem 1.1, we may work with a smooth geometrically irreducible curve S over a fixed number field, a principally polarized abelian scheme $\mathcal{A} \rightarrow S$, and a nonconstant finite map from S onto a dense open of the original curve C . We may equip the family with full symplectic levels at two distinct fixed primes at least three. Under the supposition that the required elliptic-square locus is infinite, we may fix a point $t_0 \in S(\overline{\mathbb{Q}})$ in that locus.*

Further fixed finite covers, normalization, and removal of finite sets preserve this reduction. After enlarging a fixed field of definition K , there is a constant b such that every selected lift t of a retained coarse point s satisfies

$$[K(s) : K] \leq [K(t) : K] \leq b[K(s) : K]. \quad (3.1)$$

The smooth projective completion of S and the divisor $[t_0]$ are available even when C is complete.

Proof. Choose distinct primes $r_1, r_2 \geq 3$ and put $N = r_1 r_2$. These preliminary levels give a fine moduli cover. The auxiliary incidence primes will be chosen separately in Proposition 4.2, after the correspondence list has been fixed. After adjoining the requisite roots of unity, the fine Siegel moduli space $\mathcal{A}_2(N)$ carries a universal principally polarized abelian scheme and has a finite surjective forgetful map to the coarse space $\mathcal{A}_2$; its analytic uniformization and the algebraicity of the universal family are described in [Mil13, Theorem 4.6 and Corollary 4.7]. Take an irreducible component of its pullback to C that dominates C , and normalize. Normalization is finite for a curve over a field of characteristic zero. The resulting curve is smooth, and pullback supplies the abelian scheme. Removing the preimages of any finite exceptional set of coarse points leaves a finite surjective map onto the complementary open of C .

All these varieties, components, morphisms, and level structures are defined over a number field after a fixed enlargement. The fibers of a fixed finite map have bounded length; on a dense open this length is its degree. The residue field degree of any point in such a fiber is at most that length. This proves (3.1), with the degrees of any additional fixed covers included in b . It also shows that the image of every omitted finite set is finite. An infinite coarse locus therefore has infinitely many lifts after every such fixed omission, and we choose one of them as t_0 . Later covers may be supplied with a fixed lift of this point; we then reuse S, t_0 for the refined data. Open charts and frames used at the center are chosen to contain it.

A smooth curve over a number field has a smooth projective completion. The point t_0 is an interior point of the moduli family, and its divisor has positive degree on that completion. This construction uses no geometric boundary point and imposes no condition on the arithmetic reduction of $\mathcal{A}_{t_0}$. $\square$

Fix a number field K_C defining the original coarse curve. There are only finitely many coarse conjugate curves

$$C_1, \dots, C_e \subset \mathcal{A}_2, \quad (3.2)$$

obtained by applying embeddings of K_C into $\overline{\mathbb{Q}}$. All conjugate families used below have coarse images in this list. Enlarging the field to define centers, levels, or individual endomorphisms does not add coarse images: the image of C depends only on the restriction of the embedding to K_C . We always retain the closed coarse curves in (3.2), even when their working parameter curves have been restricted to open subsets.

3.2 The five-dimensional orthogonal system

Let $H = R_1 \mathcal{A}_* \mathbb{Q}$ denote rational covariant first homology. The polarization gives an alternating form ψ , with its usual Tate twist in the different realizations. For an operator D , write $D^\dagger$ for the adjoint defined by $\psi(Dv, w) = \psi(v, D^\dagger w)$. Define

$$\mathcal{P} = \{D \in \text{End}(H) : D^\dagger = D, \text{Tr}_H(D) = 0\}, \quad \langle D, E \rangle = \text{Tr}_H(DE). \quad (3.3)$$

These definitions commute with all the realization functors used later. On actual endomorphisms, $\dagger$ is the Rosati involution of the given principal polarization.

Proposition 3.2. *The system $\mathcal{P}$ has rank five and a split nondegenerate trace pairing. Conjugation induces a surjective morphism*

$$\mathrm{GSp}_4 \longrightarrow \mathrm{SO}(\mathcal{P})$$

with scalar kernel. Its restriction to Sp_4 is the spin covering, with kernel $\{1, -1\}$, and $\mathcal{P}$ is absolutely irreducible. For every odd prime ℓ , the lattice

$$\Lambda_\ell = \{D \in \mathrm{End}_{\mathbb{Z}_\ell}(T_\ell \mathcal{A}_t) : D^\dagger = D, \mathrm{Tr}(D) = 0\} \quad (3.4)$$

is free of rank five with unimodular trace pairing.

If $\mathcal{A}_t$ is isogenous to the square of a non-CM elliptic curve, then

$$F_t = \mathrm{End}^0(\mathcal{A}_t) \cap \mathcal{P}_t \quad (3.5)$$

is a rational two-dimensional plane on which the trace pairing is positive definite. Here the intersection denotes Hodge endomorphisms in the Betti realization, equivalently actual rational endomorphisms in every realization.

Proof. Use a symplectic basis, with matrix $J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$. The equations $D^\dagger J = JD$ and $\mathrm{Tr} D = 0$ give exactly

$$D = \begin{pmatrix} A & uK_2 \\ vK_2 & A^\dagger \end{pmatrix}, \quad A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \quad K_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (3.6)$$

Multiplication gives

$$D^2 = (a^2 + bc - uv)I_4, \quad \mathrm{Tr}(D^2) = 4(a^2 + bc - uv). \quad (3.7)$$

Thus the form is the orthogonal sum of a nonzero one-dimensional form and two hyperbolic planes. Its Gram determinant in the five coordinates displayed above is 64, a unit over every $\mathbb{Z}_\ell$ with ℓ odd. A principal polarization makes the Tate module symplectically unimodular, so the same calculation proves (3.4).

Conjugation by a symplectic similitude preserves both conditions in (3.3) and the trace pairing. Since GSp_4 is connected, its orthogonal image has determinant one. The induced action of Sp_4 is nontrivial, as is already seen by conjugating the matrices in (3.6). The simple Lie algebra $\mathfrak{sp}_4$ consequently has zero kernel in this action. Both Sp_4 and SO_5 have dimension ten, so the image is all of SO_5 . The finite kernel is central in the connected group Sp_4 and is exactly $\{1, -1\}$. Over an algebraic closure a similitude is a scalar times a symplectic element, which proves the scalar-kernel assertion for GSp_4 and identifies the restriction as the spin covering. The natural representation of SO_5 is irreducible over an algebraically closed field of characteristic zero: the orbit of any nonzero vector, whether isotropic or not, spans the entire quadratic space. This proves absolute irreducibility. Equivalently, $D \mapsto \psi(D, \cdot)$ identifies the self-adjoint space with alternating two-forms, and the trace-free summand with the primitive exterior-square representation.

For the last assertion, put $A = \mathcal{A}_t$. The equivalence between complex abelian varieties up to isogeny and polarizable rational Hodge structures of type $(-1, 0), (0, -1)$ identifies $\mathrm{End}^0(A)$ with the Hodge endomorphisms of $H_1(A, \mathbb{Q})$ [Mil95, Theorem 4.1]. Write $H_1(A, \mathbb{Q}) = H_1(E, \mathbb{Q}) \otimes \mathbb{Q}^2$ using the given unpolarized isogeny. Since E has no CM, its endomorphism algebra is $\mathbb{Q}$; the Hodge endomorphism algebra of A is therefore $\mathrm{Mat}_2(\mathbb{Q})$. An alternating polarization on this tensor product has the form $\psi_E \otimes B$, where B is a symmetric rational form on $\mathbb{Q}^2$. Indeed, each of its four matrix entries is a Hodge

alternating pairing on $H_1(E, \mathbb{Q})$, hence a scalar multiple of ψ_E ; alternation of the total form makes these four scalars symmetric. Positivity of the polarization, after choosing the sign of ψ_E , makes B positive definite. The Rosati involution on $\text{Mat}_2(\mathbb{Q})$ is consequently the adjoint for B .

Its self-adjoint part has dimension three and contains the identity. The homology trace on this algebra is twice the ordinary matrix trace, so imposing trace zero leaves dimension two. Over $\mathbb{R}$, conjugation by the positive square root of B turns a self-adjoint matrix D into a symmetric matrix D_0 . Hence, for $D \neq 0$,

$$\text{Tr}_{H_1(A)}(D^2) = 2 \text{tr}(D_0^2) > 0.$$

This proves the asserted positivity for every principal polarization, without requiring the initial isogeny from E^2 to preserve it. $\square$

Fix a labeled pair $\mathbf{B} = (B_1, B_2)$ of linearly independent integral endomorphisms spanning $F_0 := F_{t_0}$; clear denominators in a rational basis to obtain it. At any target and prime, the plane lattice used for reduction is

$$\Pi_{t,\ell} = (F_t \otimes \mathbb{Q}_\ell) \cap \Lambda_\ell. \quad (3.8)$$

It is saturated and has rank two, so its image in $\Lambda_\ell / \ell \Lambda_\ell$ has dimension two. We do not identify this image with the span of a chosen pair reduced modulo ℓ : that pair can fail to be a lattice basis. At the fixed center, the restriction of the pairing to $\Pi_{t_0,\ell}$ is nondegenerate modulo ℓ outside finitely many primes. To see finiteness, intersect F_0 with the integral Betti operator lattice, choose a $\mathbb{Z}$ -basis of this intersection, and exclude the prime divisors of its nonzero Gram determinant. Betti–Tate comparison gives precisely the lattices above.

3.3 Full and joint monodromy

Proposition 3.3. *The connected algebraic monodromy of H is Sp_4 , and that of $\mathcal{P}$ is SO_5 . This remains true on every fixed finite cover, on a dense open, and for the conjugate families under consideration. The same connected monodromy is obtained by pullback to a smooth connected algebraic variety mapping dominantly to the base curve.*

Suppose a smooth connected algebraic variety W maps dominantly to two such curves. The connected joint monodromy on the two five-dimensional systems is either $\text{SO}_5 \times \text{SO}_5$, or the coarse joint image of W is contained in a polarized isogeny correspondence. If only the first projection varies, its connected monodromy is still SO_5 .

Proof. Hodge genericity in Siegel moduli means that the generic Mumford–Tate group of H is GSp_4 . The variation is pure and polarizable and comes from an abelian scheme over a smooth algebraic curve. Thus it is among the good variations to which the connected-monodromy normality theorem applies: connected algebraic monodromy is a normal subgroup of the derived generic Mumford–Tate group [And92, §5, Theorem 1]. Since Sp_4 is almost simple, the connected monodromy is trivial or all of Sp_4 .

Here is a direct exclusion of the trivial possibility. It would make the monodromy group finite, and a finite cover would trivialize the integral local system. The period map would then be a single-valued holomorphic map from a smooth algebraic curve to Siegel space. Realize Siegel space as a bounded domain in $\mathbb{C}^3$. Each coordinate of this map is bounded near each of the finitely many missing points of the smooth projective completion, so has a removable singularity there. Every extended coordinate is constant on the projective curve. The family would have constant moduli image, contrary to its domination of C . This also explains concretely the applicability of [And92, §6, Remark 1]. The monodromy conclusion

is independent of the Mumford–Tate group at the chosen center. Proposition 3.2 now gives the assertion on $\mathcal{P}$.

A nonconstant morphism between smooth algebraic curves restricts, after removal of finitely many points, to a finite étale morphism onto a dense open. Its topological fundamental group therefore has image of finite index in that of this open; the latter maps surjectively to the fundamental group of the original curve. Finite-index subgroups have the same connected Zariski closure. This proves invariance under finite covers and open restrictions. For a dominant map from a higher-dimensional smooth variety, choose an algebraic curve in that variety whose projection is nonconstant, and normalize it. Such a curve is obtained by successive general affine hyperplane sections through a point where the differential of the projection is nonzero. Its monodromy is contained in the pullback monodromy and is already full by the curve case. This proves the pullback assertion.

For completeness, full monodromy persists under field conjugation without a choice of a conjugation-compatible complex period map. Choose a prime ℓ . The Betti monodromy is Zariski dense in Sp_4 , so its image on $H \otimes \mathbb{Q}_\ell$ is Zariski dense in $\mathrm{Sp}_{4, \mathbb{Q}_\ell}$. The comparison of topological and geometric étale fundamental groups identifies the latter image with a dense subgroup of the geometric étale monodromy image. A field conjugation identifies the corresponding geometric étale families and their symplectic Tate systems. Applying the same comparison on the conjugate complex curve proves that its Betti algebraic monodromy is again full. In particular its coarse image is Hodge generic: otherwise its monodromy would lie in the derived group of a proper generic Mumford–Tate subgroup. One can also use the preservation of Siegel special subvarieties by conjugation.

It remains to prove the assertion concerning W . Work at a generic Mumford–Tate point of the direct sum of the two pulled back homology variations, in rational symplectic bases. Both projections of its connected algebraic monodromy are full, by the pullback assertion just proved. Pass to the adjoint groups G_1, G_2 , each isomorphic to PGSp_4 and to the corresponding SO_5 . Let $M \subset G_1 \times G_2$ be the connected joint image. The kernel of either projection is a normal subgroup of the other adjoint simple group. Thus either one kernel is the whole factor, giving $M = G_1 \times G_2$, or both are trivial and

$$M = \{(x, \phi(x)) : x \in G_1\} \tag{3.9}$$

for a $\mathbb{Q}$ -isomorphism $\phi : G_1 \rightarrow G_2$. This is the group form of Goursat’s argument; no passage to a merely real graph has been made.

The remaining task is to lift this adjoint graph to a positive rational similitude identifying the homology Hodge structures. This requires both the graph relation and the polarization characters. Let T be the generic Mumford–Tate group of the direct sum. The normality theorem applies on the smooth algebraic variety W and says that T normalizes connected monodromy. Its adjoint image therefore normalizes (3.9). The normalizer of this graph in $G_1 \times G_2$ is the graph itself. Indeed, normalization by (u, v) says that $v\phi(x)v^{-1} = \phi(u)\phi(x)\phi(u)^{-1}$ for all x ; the trivial center of G_2 gives $v = \phi(u)$.

The root diagram of type C_2 has no nontrivial automorphism, so every automorphism of its adjoint split group is inner. The conjugating element is unique, hence is rational when the automorphism is rational. Finally the sequence

$$1 \longrightarrow \mathbb{G}_m \longrightarrow \mathrm{GSp}_4 \longrightarrow \mathrm{PGSp}_4 \longrightarrow 1$$

and $H^1(\mathbb{Q}, \mathbb{G}_m) = 1$ lift it to a rational similitude $g \in \mathrm{GSp}_4(\mathbb{Q})$. Consequently the two Hodge homomorphisms, after conjugation by g , agree in the adjoint group. Their remaining difference is a scalar character c of the Deligne torus. Each polarization is a morphism of Hodge structures $\psi_i : H_i \otimes H_i \rightarrow \mathbb{Q}(1)$. Thus the full similitude character $\nu \circ h_i$ is the

character of the same Tate structure $\mathbb{Q}(1)$ for both i , on the whole Deligne torus. Since a scalar c has multiplier c^2 , the equality $h_2 = cgh_1g^{-1}$ therefore gives $c^2 = 1$. The Deligne torus is connected as an algebraic group, whence $c = 1$. We have obtained a rational Hodge isomorphism on H_1 , not just an identification of its adjoint representation.

Its multiplier is positive. For if J_i are the complex structures at this point, then $gJ_1 = J_2g$, and positivity of the two polarization metrics in

$$\psi_2(gv, J_2gv) = \nu(g)\psi_1(v, J_1v)$$

forces $\nu(g) > 0$. By the rational Hodge equivalence [Mil95, Theorem 4.1], an integral multiple of g is a polarized isogeny.

This same g describes an analytic open of the joint image. Indeed, trivialize the rational local system in a simply connected open. For each of its countably many rational tensors, the locus where it is Hodge is closed analytic. A point outside the union of those loci that are proper has the generic Mumford–Tate group. Every tensor Hodge at that point is therefore Hodge throughout the open. By the tensor-stabilizer description of the Mumford–Tate group, all Hodge homomorphisms there factor through the transported generic group T . Applying the graph relation for that group makes g a Hodge isomorphism throughout the open. The joint period map is therefore locally in the graph of g , and its coarse image lies locally in the Hecke correspondence T_g . This correspondence is closed and algebraic: it is the image of the finite polarized-isogeny moduli correspondence, or equivalently of the quotient for $\Gamma \cap g^{-1}\Gamma g$, where $\Gamma = \mathrm{Sp}_4(\mathbb{Z})/\{1, -1\}$. An algebraic equation for T_g vanishing on a nonempty analytic open of the irreducible image vanishes on that image. Thus the entire joint image lies in T_g . The claim when the second projection is constant follows already from full monodromy of the first projection. $\square$

3.4 Finitely many dominating correspondence types

For $g \in \mathrm{GSp}_4(\mathbb{Q})$ of positive multiplier, write T_g for its coarse Hecke correspondence in $\mathcal{A}_2 \times \mathcal{A}_2$. Scalar multiples and integral changes of symplectic markings on either side give the same correspondence. A *dominating relation* between closed curves C_i, C_j means an irreducible algebraic curve contained in $(C_i \times C_j) \cap T_g$ with both projections dominant.

Proposition 3.4. *For the fixed finite collection (3.2), only finitely many coarse Hecke correspondences admit a dominating relation between a pair of these curves. In particular there is a fixed finite list $\mathcal{T}$ containing every exceptional joint image in Proposition 3.3. We may take this list closed under exchanging the two factors and choose, for every member, positive integral multipliers for isogenies realizing all its pairs. The list and these multipliers are fixed before any additional auxiliary levels are chosen.*

Proof. We prove finiteness, including the analytic stabilizer assertion on which it depends. Use one fixed torsion-free principal level N as in Proposition 3.1. There are finitely many irreducible closed curves $D_\alpha \subset \mathcal{A}_2(N)$ lying above the curves C_i . All of them have full connected monodromy by Proposition 3.3. Write $\mathfrak{H}_2$ for Siegel space and

$$G = \mathrm{GSp}_4(\mathbb{R})^+/\mathbb{R}^\times \simeq \mathrm{Sp}_4(\mathbb{R})/\{1, -1\}, \quad \Gamma_N \subset G$$

for the effective level group. The map $\pi_N : \mathfrak{H}_2 \rightarrow \mathcal{A}_2(N)(\mathbb{C})$ is a covering: the choice of torsion-free level eliminates stabilizers.

Lift components and their integral stabilizers. Remove the finitely many singular points of D_α , and choose a connected component U_α of its inverse image. Let Y_α be its closure in $\mathfrak{H}_2$. It is a closed irreducible analytic curve. To check this last assertion, the inverse image of D_α is locally analytically identical to D_α under the covering. It has a locally finite collection of curve branches, and closure of a chosen connected smooth part joins precisely its continued branches. The normalization of that closure is connected, so the closure is irreducible. Conversely every irreducible component arises this way. The deck group acts transitively on these components, by path lifting along the connected nonsingular part of D_α .

Put $\Delta_\alpha = \Gamma_N \cap \text{Stab}_G(Y_\alpha)$. This is also the subgroup preserving U_α , and covering-space monodromy identifies it with the image of $\pi_1(D_\alpha^{\text{reg}})$ in Γ_N . In particular it is Zariski dense in G . Moreover

$$\Delta_\alpha \backslash U_\alpha \simeq D_\alpha^{\text{reg}}. \quad (3.10)$$

The identification follows because any two lifts of a point in the chosen component differ by a deck transformation preserving that component. Removing additional finite sets, if needed, has no effect on these statements.

Closedness and discreteness of the real stabilizer. Let $L_\alpha = \text{Stab}_G(Y_\alpha)$. If $g_n \in L_\alpha$ and $g_n \rightarrow g$ in G , closedness of Y_α gives $gY_\alpha \subset Y_\alpha$. Applying the same argument to g_n^{-1} proves the reverse inclusion. Thus L_α is a closed subgroup of the real Lie group G .

Its Lie algebra is invariant under $\text{Ad}(\Delta_\alpha)$. The stabilizer of a linear subspace of $\mathfrak{sp}_4(\mathbb{R})$ under the adjoint action is algebraic. Zariski density therefore makes $\text{Lie}(L_\alpha)$ invariant under all of G , hence an ideal in the simple real Lie algebra $\mathfrak{sp}_4(\mathbb{R})$. It is zero or the entire Lie algebra. The second possibility would imply $L_\alpha = G$, since G is connected, and would make the nonempty curve Y_α invariant under the transitive action of G on the three-dimensional domain $\mathfrak{H}_2$. This is impossible. Thus $\text{Lie}(L_\alpha) = 0$ and L_α is discrete.

Finite area of the integral quotient. Equip $\mathfrak{H}_2$ with its invariant Hermitian metric, and use the induced area on the regular part of Y_α . We claim

$$\text{area}(\Delta_\alpha \backslash Y_\alpha^{\text{reg}}) < \infty. \quad (3.11)$$

By (3.10), it suffices to estimate the induced metric on the normalization of D_α . Its smooth projective completion adds finitely many punctures.

Here is the required uniform disk estimate. Move the value at zero of a holomorphic disk map into $\mathfrak{H}_2$ to iI_2 by an element of G , and use the Cayley transform to the bounded domain of symmetric two-by-two matrices of operator norm less than one. The resulting map Z has $Z(0) = 0$. Scalar Schwarz applied to u^*Zv for unit vectors u, v gives $\|Z'(0)\|_{\text{op}} \leq 1$. At the origin the invariant metric is a fixed positive multiple of $\text{tr}(dZ dZ^*)$. Hence the derivative has a uniform bound. Precomposing with disk automorphisms gives, with an absolute constant c , the metric inequality

$$f^* ds_{\mathfrak{H}_2}^2 \leq c \frac{|dz|^2}{(1 - |z|^2)^2}$$

for every holomorphic disk map f .

On a punctured disk $0 < |q| < R$ in the normalization, lift its universal covering disk to $\mathfrak{H}_2$ and apply this inequality. It descends to

$$ds^2 \leq c' \frac{|dq|^2}{|q|^2 \log^2(R/|q|)}. \quad (3.12)$$

Its area integral for $0 < |q| < r < R$ is bounded by a constant times

$$\int_0^r \frac{d\rho}{\rho \log^2(R/\rho)} = \frac{1}{\log(R/r)} < \infty.$$

The compact remainder has finite area by smoothness. Zeros of the derivative only decrease the area, and singular points and their discrete lifts have area zero. This proves (3.11).

Finite index in the full stabilizer. The action of G on $\mathfrak{H}_2$ is proper: this symmetric space is G/K with K compact. Its restriction to the closed discrete subgroup L_α is consequently proper, with finite point stabilizers. Choose a smooth point $y \in Y_\alpha$ and write $F = (L_\alpha)_y$, a finite group. Properness provides a sufficiently small neighborhood U of y in the regular curve of positive area a , such that $hU \cap U = \emptyset$ unless $h \in F$.

For representatives h_i of distinct double cosets in $\Delta_\alpha \backslash L_\alpha / F$, the images of $h_i U$ in $\Delta_\alpha \backslash Y_\alpha$ are disjoint. Indeed, an intersection would imply $h_i^{-1} \delta h_j U \cap U \neq \emptyset$ for some $\delta \in \Delta_\alpha$, hence $h_i^{-1} \delta h_j \in F$. Each image has area at least $a/|F|$, since only members of a conjugate of F can identify points within it. Infinitely many such double cosets would contradict (3.11). There are therefore finitely many of them; because F is finite, this proves

$$[L_\alpha : \Delta_\alpha] < \infty. \tag{3.13}$$

Transporters and Hecke double classes. For two fixed lift components, consider the real transporter

$$\mathcal{R}_{\alpha\beta} = \{g \in G : gY_\alpha = Y_\beta\}.$$

If it is nonempty and g_0 belongs to it, then $\mathcal{R}_{\alpha\beta} = g_0 L_\alpha$. By (3.13) this set has finitely many right Δ_α -cosets. Among the cosets that meet the image of $\mathrm{GSp}_4(\mathbb{Q})^+$ choose rational representatives. These give finitely many integral double classes for all rational elements of the transporter, because $\Delta_\alpha \subset \Gamma_N$ and passage to the full integral group only identifies more classes.

Now suppose T_g admits a dominating relation between two coarse curves. Over a smooth point of that relation, both projections have nonconstant local parametrizations. Lift them to the fixed torsion-free level and then locally to Siegel space. A local sheet of the Hecke correspondence identifies the two lifted arcs by a rational positive similitude in the integral double class of g . The arcs lie on some translates of Y_α, Y_β ; applying deck transformations makes these the chosen components. The translated similitude takes a nonempty open arc of Y_α into Y_β . Its image of Y_α and Y_β are closed irreducible analytic curves with a one-dimensional intersection, so the analytic identity principle makes them equal. The similitude thus belongs to the transporter just considered. There are finitely many pairs (α, β) and finitely many double classes for each. This proves the claimed finiteness of coarse correspondences.

Finally choose a rational similitude representative g of each retained class. Multiplying it by a positive integer clears its matrix denominators. Its multiplier m is then a positive integer: applying the equality of alternating forms to a pair of integral vectors of pairing one shows that the multiplier is integral. It yields an isogeny f with $f^* \lambda' = m \lambda$ at every pair of its correspondence. This also follows from the finite moduli description by kernels of such isogenies in the m -torsion. Repeat the choice for the inverse representatives to include both directions. There are only finitely many resulting positive integers. The construction has involved only the fixed coarse curves and an initial torsion-free level; further levels change their parametrizations, not these curves or the list. Thus all multipliers are fixed before the auxiliary incidence primes are selected. $\square$

The finiteness just proved concerns dominating relations. It places no bound on the isogeny degrees of isolated pairs of points on the two curves. A pair on a retained correspondence, whether or not it lies on a dominating component, does have an isogeny of that correspondence's fixed multiplier. Also, if an irreducible algebraic joint image is contained in a retained correspondence on a dense open, its whole image, including any specified target pair where the maps are defined, is contained there by closedness. These are precisely the two uses of the finite list in [Propositions 6.4 and 7.4](#).

4 Two auxiliary markings and exclusion of bad disks

We impose two conditions on the relative positions of the center and target endomorphism planes. The first will be used with Frobenius in [Lemma 7.3](#). The second rules out proximity to the center at every place where it is not potentially good. Both conditions concern reductions of saturated planes; they impose no primitivity condition on the particular endomorphisms later chosen to span those planes.

4.1 Finite-field positions of two planes

We first give the linear algebra uniformly in the target plane. Write V for the split quadratic five-space obtained from a symplectic four-space by the representation in [Proposition 3.2](#). We use the associated symmetric bilinear form, and call the rank of its restriction to a subspace the rank of that subspace. The target is reduced from a saturated rank-two lattice, so it remains a plane, but its restricted form may have rank zero, one, or two. The lemma therefore allows an arbitrary target plane H .

Lemma 4.1. *There is an absolute constant q_0 with the following property. Let k be a finite field of odd cardinality $q > q_0$, let $H_0 \subset V_k$ be a nondegenerate plane, and let $H \subset V_k$ be any plane. There are $g_-, g_+ \in \mathrm{Sp}_4(k)$ such that*

$$\dim(H_0 + g_- H) = \dim(H_0 + g_+ H) = 4,$$

the first sum is degenerate, and the second is nondegenerate. Here the action is the quadratic representation, so the assertion does not require surjectivity of $\mathrm{Sp}_4(k) \rightarrow \mathrm{SO}(V_k)$.

Proof. We will use the following elementary observation about rational charts. The big Bruhat cell of the split group Sp_4 has coordinates $\mathbb{A}^4 \times \mathbb{G}_m^2 \times \mathbb{A}^4$: multiply the four negative root subgroups, a diagonal torus element, and the four positive root subgroups, in a fixed order. Their matrices, their inverses, and their actions on V have Laurent-polynomial entries of bounded degrees in these ten coordinates, independently of k . The same assertion holds for a big cell of a split parabolic, using its Levi root subgroups and unipotent radical. Conjugating a parabolic or inserting a fixed group element changes coefficients and does not change these degree bounds.

Consequently, an open condition expressed by minors of a bounded-size matrix in this representation can be tested on either chart by a polynomial of degree at most an absolute constant D . If the condition holds at some geometric point, it holds on the corresponding dense chart: both the group and the parabolic are geometrically irreducible. Choose a minor that is not identically zero, clear its torus denominators, and multiply by the torus coordinates. The resulting nonzero polynomial has a degree bounded by another absolute constant. A nonzero polynomial of total degree less than q cannot vanish on all of k^n . This follows by induction on n , applying the univariate root bound to its nonzero coefficients. Thus every such geometrically nonempty open condition has a k -point once $q > q_0$, for one

constant q_0 that works for all planes under consideration. No coefficient height or isometry type enters this bound.

For the nondegenerate sum, work first over $\bar{k}$. Choose a nondegenerate four-space M containing H_0 , and write $M = H_0 \perp K$ with $\dim K = 2$. Choose bases in which both forms are I_2 . The plane that is the graph of $A = \text{diag}(a, b) : K \rightarrow H_0$ is disjoint from H_0 and has Gram matrix

$$I_2 + A^t A = \text{diag}(1 + a^2, 1 + b^2).$$

Choices of $a, b \in \bar{k}$ give each of the ranks 0, 1, 2. Over $\bar{k}$, bilinear forms on a two-space are classified by rank, so there is a graph isometric to H , including when H is degenerate. Witt's extension theorem for subspaces of a nondegenerate space extends this isometry to an orthogonal transformation of $V_{\bar{k}}$. Its determinant can be changed by a reflection in a nonisotropic vector perpendicular to the graph. Such a vector exists: that perpendicular has dimension three, whereas the largest totally isotropic subspace of $V_{\bar{k}}$ has dimension two. We obtain an element of $\text{SO}(V_{\bar{k}})$, which lifts to $\text{Sp}_4(\bar{k})$ by the spin covering. Nonvanishing of the Gram determinant of the four concatenated basis vectors is an open condition and implies their independence. The chart argument just proved supplies $g_+ \in \text{Sp}_4(k)$ uniformly.

For the degenerate sum, choose an isotropic line $L \subset H_0^\perp$ over k . The perpendicular is a nondegenerate ternary space. Every such space over a finite field of odd cardinality is isotropic: after diagonalization, the two sets $\{ax^2 : x \in k\}$ and $\{-c - by^2 : y \in k\}$ each have $(q+1)/2$ elements and therefore meet, giving an isotropic vector with third coordinate 1. The three-space $H^\perp$ also has an isotropic line over k . If it is nondegenerate the same argument applies; if it is degenerate its nonzero radical supplies one.

The group $\text{Sp}_4(k)$ is transitive on the isotropic lines of V_k . Indeed, under the primitive exterior-square realization these lines are the Plücker lines of the Lagrangian two-spaces in the symplectic four-space; symplectic bases show transitivity over k itself. Move the chosen line in $H^\perp$ to L . After this preliminary move, $H \subset L^\perp$. Let P be the stabilizer of L in Sp_4 and put

$$Q = L^\perp / L.$$

This is a nondegenerate ternary space. The projection of H_0 is a plane $\bar{H}_0 \subset Q$ because $H_0 \cap L = 0$. Over $\bar{k}$, the Levi action of P on Q includes $\text{SO}(Q)$. If H contains L , its projection is a line. This line can be moved outside $\bar{H}_0$, since the standard representation of $\text{SO}(Q)$ is irreducible. Its inverse image then meets H_0 trivially.

If H does not contain L , its projection is a plane $\bar{H} \subset Q$. Move it so that $\bar{H} \neq \bar{H}_0$. Such a move exists since a plane cannot be invariant under the irreducible group $\text{SO}(Q)$. The two projected planes now meet in a line J . The unipotent radical permits an arbitrary change of the lift $Q \rightarrow L^\perp$ by a linear map $Q \rightarrow L$. Explicitly, choose a hyperbolic pair e, f , with $L = \bar{k}e$ and $\langle e, f \rangle = 1$, and identify Q with $(\bar{k}e + \bar{k}f)^\perp$. For $z \in Q$ the orthogonal unipotent transformation is

$$e \mapsto e, \quad n \mapsto n - \langle n, z \rangle e, \quad f \mapsto f + z - \frac{1}{2} \langle z, z \rangle e.$$

These transformations lift to the unipotent radical in Sp_4 ; equivalently, in its symplectic realization that radical consists of the matrices $\begin{pmatrix} I & Z \\ 0 & I \end{pmatrix}$ with Z symmetric. Since the form on Q is nondegenerate, $n \mapsto -\langle n, z \rangle$ runs through all linear forms on Q . Choose it so that the lifts of the nonzero vectors of J in the moved H and in the fixed H_0 are distinct. Then $H \cap H_0 = 0$.

We have exhibited a geometric point of P where a four-by-four minor of the concatenated bases of H_0 and the moved H is nonzero. The parabolic chart argument gives such a

point over k for the same uniform lower bound on q . Their sum, contained in $L^\perp$ and of dimension four, is exactly $L^\perp$. Its radical is L , so it is degenerate. This proves the first assertion and completes all three possible target-rank cases. $\square$

4.2 Independent markings on one connected cover

For a non-CM elliptic-square point u , denote its rational Hodge endomorphism plane by F_u . At an odd prime ℓ , let $\Lambda_{u,\ell}$ be the lattice of trace-free self-adjoint operators on $T_\ell A_u$ from Proposition 3.2, and set

$$\Pi_{u,\ell} = (F_u \otimes \mathbb{Q}_\ell) \cap \Lambda_{u,\ell}, \quad \overline{\Pi}_{u,\ell} = \Pi_{u,\ell} / \ell \Pi_{u,\ell} \subset \Lambda_{u,\ell} / \ell \Lambda_{u,\ell}. \quad (4.1)$$

The quotient $\Lambda_{u,\ell} / \Pi_{u,\ell}$ is torsion free, so the displayed reduction is an embedded two-space. A full symplectic ℓ -marking identifies its ambient five-space with the fixed $V_{\mathbb{F}_\ell}$. This definition is independent of the integral pair used to span F_u rationally.

Proposition 4.2. *After one fixed finite cover of the curve in Proposition 3.1, there are distinct fixed odd primes ℓ_1, ℓ_2 and a fixed lift of the center t_0 with the following properties. Every non-CM elliptic-square target on the previous curve has a lift t for which, in the respective marked five-spaces,*

$$\overline{\Pi}_{t_0, \ell_1} + \overline{\Pi}_{t, \ell_1} \text{ is a degenerate four-space,} \quad (4.2)$$

$$\overline{\Pi}_{t_0, \ell_2} + \overline{\Pi}_{t, \ell_2} \text{ is a nondegenerate four-space.} \quad (4.3)$$

The prime ℓ_1 divides none of the fixed Hecke multipliers in Proposition 3.4. The prime ℓ_2 is different from every residue characteristic at which the fixed center is not potentially good. Both center-plane reductions are nondegenerate. The conditions remain valid upon simultaneous conjugation of all the data, and the degree of any chosen lift increases by at most a fixed factor.

Proof. Choose a symplectic integral Betti lattice. Its geometric monodromy group $\Gamma \subset \mathrm{Sp}_4(\mathbb{Z})$ is finitely generated, because the fundamental group of a smooth algebraic curve is finitely generated, and is Zariski dense by Proposition 3.3. Strong approximation for a finitely generated Zariski-dense subgroup of a connected, simply connected, absolutely almost simple rational group gives

$$\Gamma \longrightarrow \mathrm{Sp}_4(\mathbb{F}_\ell) \quad \text{surjective for every sufficiently large prime } \ell;$$

these are precisely the hypotheses of [MVW84, Introduction, Theorem, p. 515] for Sp_4 .

Fix $\ell_1 > q_0$ outside this theorem's finite exceptional set, the finitely many primes of degeneracy of the fixed center plane, and the primes dividing the already fixed multipliers in Proposition 3.4. Its reduction kernel Γ_1 has finite index in Γ . Schreier's theorem gives finite generation of Γ_1 ; its Zariski closure is still Sp_4 , since a connected algebraic group cannot be a finite union of translates of a proper closed subgroup. Apply strong approximation to Γ_1 , and choose $\ell_2 > q_0$ distinct from ℓ_1 , outside its exceptional set, outside the center-plane degeneracy set, and outside the residue characteristics of all non-potentially-good places of A_{t_0} . The last set is finite because the fixed abelian variety has good reduction away from finitely many places. All these exclusions use fixed data.

It follows that

$$\Gamma \longrightarrow \mathrm{Sp}_4(\mathbb{F}_{\ell_1}) \times \mathrm{Sp}_4(\mathbb{F}_{\ell_2}) \quad \text{is surjective.} \quad (4.4)$$

To check this explicitly, first realize a prescribed element in the first factor by an element of Γ ; correct its second factor using an element of Γ_1 . The cover parametrizing the two full

markings with fixed Weil pairings is therefore connected: its monodromy acts transitively on the product torsor of markings. After a fixed extension of the field of definition, it is a fixed geometrically connected finite étale cover of our curve, carrying both markings. Fix any lift of t_0 to it.

For each target, [Lemma 4.1](#) prescribes a marking at ℓ_1 and, independently, a marking at ℓ_2 satisfying (4.2) and (4.3). The full product torsor lies in the connected cover, so these choices give a point of that cover. We continue to write S, t_0, t for the resulting curve, center, and selected target. Selection need not be an algebraic rule: every point in a fiber of a finite morphism of fixed degree has degree over the original point bounded by that degree. Subsequent finite omissions on the curve omit only finitely many points downstairs, as in [Proposition 3.1](#).

Finally, the planes come from actual rational endomorphisms. Their étale realizations, intersections with integral Tate lattices, and induced marked reductions all commute with simultaneous conjugation. A conjugation changes the pairing trivialization by the same similitude on the center and target; on the operator five-space this is an isometry. Dimensions and degeneracy are unchanged. Potential good reduction is also invariant under finite extension, so adjoining levels and a field of definition creates no new residue characteristics to be excluded from the choice of ℓ_2 . $\square$

4.3 A common toric subspace in a strict local model

We specify the integral models that will also be used in [Proposition 5.3](#). For each $i = 1, 2$, forget all levels except the full ℓ_i -level and its Weil pairing. The Siegel datum is principally polarized, its integral symplectic lattice is self-dual, and $\ell_i \geq 3$ is invertible on $\text{Spec } \mathbb{Z}[1/\ell_i, \zeta_{\ell_i}]$. Thus it is the unramified Siegel case of the integral toroidal construction. A smooth admissible projective cone decomposition exists by [[Lan13](#), Proposition 7.3.1.4]; the principal level is neat by [[Lan13](#), Remark 1.4.1.9]. At this neat level the construction yields a smooth proper compactification and a semiabelian extension of the universal abelian scheme; the projective choice makes the compactification a projective scheme. We use [[Lan13](#), Theorems 6.4.1.1 and 7.3.3.4], with the smooth admissible cone condition of Definition 6.3.3.4, or the Siegel construction of [[FC90](#), Chapter IV]. In particular we are not identifying an arbitrary toroidal compactification with a smooth scheme.

After base change to the appropriate ring of integers of our fixed number field, each base is regular: it is smooth over the regular Dedekind base with ℓ_i inverted. The semiabelian family is smooth and separated, of relative dimension two, with geometrically connected integral fibers. One of these two models is available at every residue characteristic, since their levels are coprime. No additional common set of primes is inverted here.

We also use the homomorphism extension theorem in its normal-base form: a homomorphism between two semiabelian schemes on a dense open of a noetherian normal base extends uniquely to the base [[FC90](#), Chapter I, Proposition 2.7], [[Lan13](#), Proposition 3.3.1.5]. It applies in particular to the discrete valuation rings below, after any finite extension needed to define an endomorphism. Equivalently, over a trait the semiabelian extension with abelian generic fiber is the connected part of the Néron model in the semistable case [[BLR90](#), §7.4, Proposition 3 and Corollary 4]. We use the extension theorem directly, so no Néron mapping property over a higher-dimensional ramified base is being assumed.

Lemma 4.3. *Let X be one of the regular integral toroidal models just specified, let $\mathcal{G} \rightarrow X$ be its semiabelian family, and suppose its interior carries a full ℓ -marking with ℓ invertible on X . Let $\bar{z}$ be a geometric point such that $\mathcal{G}_{\bar{z}}$ is a two-dimensional torus. Any two traits through $\bar{z}$ whose generic points lie in the interior have the same marked two-space*

$$W_{\bar{z}} \subset (\mathbb{Z}/\ell\mathbb{Z})^4$$

coming from the special torus torsion. Along each trait it lifts to a saturated rank-two lattice $\mathcal{T} \subset T_\ell A$. Every actual integral endomorphism of the generic abelian variety, after extension of its field of definition, preserves $\mathcal{T}$. If $F \subset \text{End}^0(A)$ is a rational space of endomorphisms, the reduction of $(F \otimes \mathbb{Q}_\ell) \cap \Lambda_\ell$ therefore preserves $W_{\bar{z}}$ as well.

Proof. Take $R = \mathcal{O}_{X,\bar{z}}^{\text{sh}}$. This ring is noetherian and regular, hence is an integral domain. The inverse image U of the moduli interior in $\text{Spec } R$ is a nonempty open in an integral scheme, hence is connected. We make this construction on the regular toroidal base before pulling back to either trait.

For every $n \geq 1$, multiplication by ℓ^n on the smooth semiabelian scheme is étale: its differential is multiplication by the invertible integer ℓ^n . Its kernel $\mathcal{G}[\ell^n]$ is consequently separated, quasi-finite and étale over R . It need not be finite over R . The finite-part decomposition over a henselian local base writes it as its finite open-and-closed part and a part with empty closed fiber. The finite part is a subgroup; its formation commutes with local morphisms of henselian local schemes. More generally this is the construction of the finite part and the torus part of quasi-finite flat subgroups in [Lan13, §3.4.1, equations (3.4.1.1)–(3.4.1.4)].

Here the whole special fiber is a torus, so the finite part and the torus part coincide. Denote this subgroup by E_n . It is finite étale of rank ℓ^{2n} , with special fiber $\mathcal{G}_{\bar{z}}[\ell^n]$. Since R is strictly henselian, E_n is constant and each of its special points has a unique lift to an R -section. The multiplication and transition maps among the E_n are the unique lifts of those on the special torus. Thus the system is isomorphic to the standard system $(\mathbb{Z}/\ell^n\mathbb{Z})^2$, with its usual reduction transition maps, after choosing compatible generators in the special torus.

Over U , the full ℓ -marking identifies $\mathcal{G}[\ell]_U$ with the constant group $(\mathbb{Z}/\ell\mathbb{Z})^4$. Every section of E_1 has a constant label under this identification, because U is connected. The labels form a subgroup of dimension two; call it $W_{\bar{z}}$. Lift either trait to the strict henselizations using the chosen geometric specialization. Compatibility of the finite part under these local base changes shows that its generic toric torsion has exactly these labels. This proves that the two traits have the same marked subspace. In particular, uniqueness is being used over a common strict local base, rather than separately to compare unrelated trait liftings.

On a trait, passage to the inverse limit of the generic fibers of E_n gives a rank-two lattice $\mathcal{T} \subset T_\ell A$. Its reduction modulo ℓ is the embedded two-space $W_{\bar{z}}$. It is saturated. Indeed, lift a basis of $W_{\bar{z}}$ to compatible generators of the system E_n ; the resulting two Tate vectors have independent reductions in the free rank-four Tate module. They extend to a $\mathbb{Z}_\ell$ -basis, so their span is a direct summand. The inverse-limit construction shows that this span is precisely $\mathcal{T}$.

An integral endomorphism of the generic fiber extends to the semiabelian scheme on the trait by the normal-base extension theorem above. On the special fiber it preserves the torus and its torsion. Uniqueness of the lifted torsion sections implies preservation of each E_n and therefore of $\mathcal{T}$. This argument is compatible with extending the trait, even ramifiedly. Rational linear combinations preserve $\mathcal{T}_{\mathbb{Q}_\ell}$. Since $\mathcal{T}$ is saturated,

$$\mathcal{T} = \mathcal{T}_{\mathbb{Q}_\ell} \cap T_\ell A.$$

Every integral Tate operator lying in their rational span preserves this intersection. Reducing that assertion modulo ℓ proves the last statement, in particular for the full saturation of an endomorphism plane even if its chosen spanning pair becomes dependent modulo ℓ . $\square$

4.4 Excluding proximity at a bad center place

Lemma 4.4. *Let V_4 be a symplectic four-space over a field of characteristic different from two. No nondegenerate four-space of trace-free symplectic-self-adjoint operators can preserve a two-space $W \subset V_4$.*

Proof. Write ψ for the symplectic form. Its restriction to W has rank two or zero. In the first case $V_4 = W \perp W^\perp$. A self-adjoint operator preserving W also preserves $W^\perp$, since $\psi(Xv, w) = \psi(v, Xw)$. A self-adjoint operator on a symplectic two-space is scalar, as is seen by solving $D^t J = J D$ for a two-by-two matrix. Our operator is consequently $aI_W \oplus bI_{W^\perp}$, and its trace-zero condition is $2a + 2b = 0$. The space of such operators has dimension one.

In the rank-zero case W is Lagrangian. Choose it as the first Lagrangian in a symplectic basis, with form matrix $\begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$. The self-adjoint trace-free operators preserving W are exactly

$$\left\{ X(D, J) = \begin{pmatrix} D & J \\ 0 & D^t \end{pmatrix} : \operatorname{tr} D = 0, \quad J^t = -J \right\}. \quad (4.5)$$

This is a four-space, since $\dim \mathfrak{sl}_2 = 3$ and the skew-symmetric two-by-two matrices have dimension one. Its trace pairing is

$$\operatorname{Tr}(X(D, J)X(D', J')) = 2 \operatorname{tr}(DD').$$

The trace pairing on $\mathfrak{sl}_2$ is nondegenerate in odd characteristic: in coordinates $D = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ it pairs (a, b, c) with (a', b', c') as $2aa' + bc' + cb'$. Hence the radical of (4.5) is exactly the one-dimensional J -space. Any four-space of operators contained in this stabilizer equals it and is degenerate. Both cases give the required contradiction. $\square$

Proposition 4.5. *Fix a finite place of the field of the center at which A_{t_0} is not potentially good. There is a sufficiently small disk about the actual point t_0 in the completed local curve such that no selected target from Proposition 4.2 lies in this disk. The radius is independent of the finite, possibly ramified extension over which a target is defined. The same conclusion holds for every conjugate of the fixed data, with a fixed finite collection of radii.*

Proof. We first identify the special group at the center. After a finite local extension the elliptic curve E in an isogeny $A_{t_0} \sim E^2$ has semistable reduction. One may obtain this extension by choosing a full elliptic level invertible at the place and applying the same proper toroidal construction in dimension one. An elliptic curve with semistable reduction is either good or multiplicative. Toric rank is invariant under isogeny and is additive on products: extend an isogeny and a quasi-inverse, with composites multiplication by an integer, to the semiabelian models. On special fibers their restrictions map tori into tori, since a homomorphism from a torus to an abelian variety is zero. The two composites on the tori are multiplication by that integer, so these restrictions are isogenies and the torus dimensions agree. Thus the potential toric rank of A_{t_0} is zero or two. Rank zero in the semistable model is good reduction; in that case A_{t_0} would be potentially good. Our hypothesis therefore gives rank two.

Use the full ℓ_2 -level toroidal model, which is available at this residue characteristic by [Proposition 4.2](#). Properness extends the center point to a section over the valuation ring, after a fixed local extension if needed. Its pullback semiabelian scheme is the semistable model of the center, by uniqueness of semiabelian extension over a normal trait. Its geometric special fiber is therefore a two-dimensional torus. Denote the specialization in the toroidal base by $\bar{z}$.

We show that a disk of fixed radius forces this very same specialization for every target. Choose an affine open of the toroidal scheme containing $\bar{z}$, and enlarge the fixed local field so that the center section and the required residue point are defined. The generic image of t_0 lies in this affine open. Finitely many affine coordinate generators pull back to rational functions on the curve that are regular at t_0 ; shrink its parameter neighborhood also to exclude their denominators and the complement of this affine open. Their values at t_0 are integral. Choose a local parameter x at t_0 with $x(t_0) = 0$. Continuity gives a real $r > 0$ such that

$$|x(t)| < r \implies |f_j(t) - f_j(t_0)| < 1 \quad \text{for every coordinate generator } f_j.$$

The assertion uses the branch of x through t_0 . Each f_j has a convergent expansion on a fixed smaller disk, so the same inequalities hold in the fixed algebraic closure of the local field, for points in every finite extension. They do not involve a valuation normalization by a varying ramification index. The values $f_j(t)$ are integral and satisfy the equations of the chosen affine chart, and have the center's residues. They consequently define an integral section with specialization $\bar{z}$. Separatedness identifies its generic point with the given moduli point of t .

If a selected target lay in this disk, the two traits would therefore satisfy [Lemma 4.3](#) with $\ell = \ell_2$. Both $\bar{\Pi}_{t_0, \ell_2}$ and $\bar{\Pi}_{t, \ell_2}$ would preserve the same two-space $W_{\bar{z}}$ in the marked symplectic four-space. Their sum is a nondegenerate four-space by (4.3), in direct contradiction to [Lemma 4.4](#). This proves exclusion of the target point from the disk itself.

The center has only finitely many non-potentially-good places, and the fixed data have only finitely many conjugates. Apply the argument at each of them. Simultaneous conjugation preserves (4.3), so it gives finitely many fixed radii with the asserted properties. All of this argument uses integral models and prime-to-characteristic torsion. No comparison with de Rham horizontal transport at a bad center is needed. $\square$

5 Heights and integral frames

The endomorphisms used in the local equations must satisfy two different size estimates. Their coordinates may have height proportional to the height of the target, whereas their positive local logarithms near the fixed center must have only logarithmic growth in that height. We first choose small endomorphisms, then construct integral lattices to control their finite-place coordinates, and finally compare these lattices and the archimedean Hodge metrics with fixed rational frames.

5.1 Height conventions and small endomorphisms

Convention 5.1. Fix the smooth curve S , its smooth projective completion $\bar{S}$, the abelian scheme $\mathcal{A} \rightarrow S$, and the center t_0 supplied by [Propositions 3.1](#) and [4.2](#). Enlarge a fixed number field K to contain all these data and the finitely many fixed charts, levels and center endomorphisms. Write

$$d = \max\{2, [K(t) : K]\}, \quad h = h_{[t_0]}(t) + c_0 \geq 2.$$

Here $h_{[t_0]}$ is an absolute logarithmic Weil height. The divisor $[t_0]$ is ample on $\bar{S}$, and c_0 is a fixed constant. The letters C and κ below denote positive constants depending only on the fixed data; they may be increased in successive estimates.

Choose rational frames of $H_{\text{dR}} = H_1^{\text{dR}}(\mathcal{A}/S)$ and of $\mathcal{P}_{\text{dR}}$ that are regular and invertible on an affine neighborhood of t_0 . Choose a rational parameter x with a simple zero at t_0 , and put $a = x(t)$. Throughout, a sufficiently small disk means the actual branch about t_0 , not all points with small $|x|$. Remove the finitely many poles, frame singularities, and unwanted zeros of the rational functions being used. Perform the same choices for the finitely many conjugate data. Such finite omissions are covered by [Proposition 3.1](#).

For a number field L containing the varying data, put $n_v = [L_v : \mathbb{Q}_v]/[L : \mathbb{Q}]$ and use the absolute values extending the usual real or p -adic absolute values on $\mathbb{Q}$. In particular,

$$h_{\text{aff}}(z_1, \dots, z_r) = \sum_v n_v \log \max\{1, |z_1|_v, \dots, |z_r|_v\}.$$

At a real or complex place the local degree is understood in the usual sense. The sum of the archimedean weights is one, and the weights above any rational prime sum to one. When passing to a finite extension, the weights of places above a given place sum to its old weight. This also allows computations after finite local extensions. Weighted sums below use this convention, including when only a subcollection of embeddings and places is retained.

Every fixed rational map from $\bar{S}$ to a projective space has height at most Ch wherever it is defined. Indeed its pulled-back hyperplane divisor is dominated, for height inequalities, by a sufficiently large multiple of the ample divisor $[t_0]$: their difference can be chosen ample and its height is bounded below. This proves in particular $h(a) \leq Ch$ and the same estimate for all fixed base coordinates.

Proposition 5.2. *Let t be a target with $\text{End}_{\overline{\mathbb{Q}}}(\mathcal{A}_t) \otimes \mathbb{Q} \simeq \text{Mat}_2(\mathbb{Q})$, and let $\mathcal{O}_t = \text{End}_{\overline{\mathbb{Q}}}(\mathcal{A}_t)$. For the principal Rosati involution $\dagger$, set*

$$Q_t(f) = \text{Tr}_{H_1}(ff^\dagger).$$

Then

$$h_F(\mathcal{A}_t) \leq Ch, \quad |\text{disc } \mathcal{O}_t| \leq C(dh)^\kappa. \quad (5.1)$$

The discriminant here is that of the full geometric order for the trace of the regular representation of $\text{Mat}_2(\mathbb{Q})$. There are integral endomorphisms $f_1, \dots, f_4$ spanning $\mathcal{O}_t \otimes \mathbb{Q}$ and an independent labeled pair $\mathbf{P} = (P_1, P_2)$ spanning its Rosati-symmetric trace-free plane, all with

$$Q_t(f_i), Q_t(P_j) \leq C(dh)^\kappa. \quad (5.2)$$

All geometric endomorphisms, and hence this pair and its de Rham coordinates, are defined over an extension $L/K(t)$ of bounded degree. The bound depends only on the dimension and the fixed data.

Proof. After a fixed finite theta-level cover, a symmetric ample line bundle and its theta data give an algebraic theta-null map. On that cover use the pullback of $[t_0]$ as the ample divisor; its height at a lift is $h_{[t_0]}(t) + O(1)$. The theta-null projective height is therefore at most Ch by the preceding height comparison. Pazuki's comparison of theta height and stable Faltings height [[Paz12](#), Corollary 1.3(1)] gives $h_F(\mathcal{A}_t) \leq Ch$. The logarithmic error in that comparison is absorbed into this upper bound. The cover has fixed degree, so using it does not impose a bound on the varying degree d .

We spell out the field-degree dependence in the endomorphism estimate. The Proposition of [MW94, §4], with its induction parameter $m = n$ and with principal polarization, gives integral endomorphisms spanning the geometric endomorphism algebra with Rosati quadratic sizes at most

$$c \max\{2, [L : \mathbb{Q}], h_F(\mathcal{A}_t)\}^{\kappa_2},$$

where c and κ_2 depend only on the dimension $n = 2$. The degree dependence is exposed in that proposition and in [MW94, §5, equation (5.3) and Lemma 5.1]; it is not an unspecified constant allowed to depend arbitrarily on $[L : \mathbb{Q}]$. The estimates apply to arbitrary abelian varieties and thus include the split algebra $\text{Mat}_2(\mathbb{Q})$.

Here one can take $[L : K(t)]$ bounded. The continuous Galois action on the rank-four lattice $\mathcal{O}_t$ has finite image. To see finiteness, choose an integral basis; each basis endomorphism is defined over a finite extension, so an open Galois subgroup fixes the entire basis. Finite subgroups of $\text{GL}_4(\mathbb{Z})$ have bounded order. One elementary bound follows by reduction modulo 3: its kernel has no nontrivial torsion. In fact a torsion element in that kernel has a prime-order power; writing it as $1 + 3^s A$, with some entry of A a 3-adic unit, the binomial expansion rules out order prime to 3, and also rules out order 3 by comparing the valuations of the linear and higher terms. Thus reduction embeds the finite image in $\text{GL}_4(\mathbb{F}_3)$. Taking the fixed field of its kernel proves the assertion about L . Since K is fixed, $[L : \mathbb{Q}] \leq Cd$, which gives the generator bound in (5.2).

For completeness, the generator bound controls the full-order discriminant without requiring the chosen generators to be an integral basis. Put

$$b(f, g) = \text{Tr}_{H_1}(fg), \quad q(f, g) = b(f, g^\dagger).$$

The form q is positive definite and integral. Principal polarization makes $\dagger$ an automorphism of the integral order. Consequently, in an integral basis, the matrices of b and q differ by right multiplication by an integral matrix of determinant ± 1 . Their determinants have the same absolute value. Select four independent small generators and let Λ be their span over $\mathbb{Z}$. Hadamard's inequality gives

$$\det(q|_\Lambda) \leq \prod_{i=1}^4 Q_t(f_i) \leq C(dh)^\kappa, \quad \det(q|_{\mathcal{O}_t}) = \frac{\det(q|_\Lambda)}{[\mathcal{O}_t : \Lambda]^2}.$$

On $\text{Mat}_2(\mathbb{Q})$ the rank-four homology representation is the sum of two copies of its standard module. Thus $\text{Tr}_{H_1}(f) = 2\text{tr}_2(f)$, which also equals the trace of the regular representation. This proves the discriminant convention and its bound.

The q -orthogonal projection onto the symmetric trace-free subspace is

$$\pi(f) = \frac{f + f^\dagger}{2} - \frac{\text{Tr}_{H_1}(f)}{4} \text{id}.$$

The two projections commute, and neither increases the q -norm. Moreover

$$4\pi(f) = 2(f + f^\dagger) - \text{Tr}_{H_1}(f) \text{id} \in \mathcal{O}_t. \tag{5.3}$$

By Proposition 3.2, the image has dimension two. The images of the chosen rationally spanning generators span it. Select two independent nonzero images and use (5.3) to obtain P_1, P_2 . Their quadratic sizes increase by at most 16. All these operations take place over L . $\square$

Fix once and for all an analogous integral pair $\mathbf{B} = (B_1, B_2)$ at t_0 . Neither $\mathbf{B}$ nor $\mathbf{P}$ is required to be an integral basis of its plane. The reductions used in Proposition 4.2 concern the saturated plane lattices.

5.2 A bundle whose trait lattices preserve endomorphisms

To bound finite-place coordinates, we need a lattice preserved by every integral endomorphism, including at fibers with bad reduction. We construct one from the polarization biextension. Its Lie bundle agrees with de Rham homology on the abelian characteristic-zero open and remains locally free on the semiabelian model.

Proposition 5.3. *Let T be a regular noetherian integral scheme, $U \subset T$ a dense open, and $\mathcal{G} \rightarrow T$ a smooth separated semiabelian scheme of relative dimension two with geometrically integral fibers. Suppose that $\mathcal{G}_U$ is an abelian scheme with principal polarization. The Poincaré bundle, viewed on $\mathcal{G}_U \times_U \mathcal{G}_U$ using that polarization, has a unique extension as a normalized biextension $\mathcal{B}$ on $\mathcal{G} \times_T \mathcal{G}$.*

This biextension defines a smooth vector-group extension

$$0 \longrightarrow \omega_{\mathcal{G}/T} \longrightarrow \mathcal{E} \longrightarrow \mathcal{G} \longrightarrow 0, \quad \omega_{\mathcal{G}/T} = e^* \Omega_{\mathcal{G}/T}^1, \quad (5.4)$$

where a locally free module denotes the additive group of its sections. The vector bundle $\mathcal{H} = \text{Lie}(\mathcal{E})$ has rank four and, on the characteristic-zero abelian open, identifies functorially with $H_1^{\text{dR}}(\mathcal{G}_U/U)$.

For any trait $\text{Spec } R \rightarrow T$ whose generic point belongs to that open, and any integral endomorphism f of its generic abelian variety, the action of f preserves the lattice $\mathcal{H}_R$. This remains true after every finite extension of traits on which f is defined.

Proof. We first extend the normalized biextension, then construct its vector extension, and finally show that endomorphisms act integrally on the resulting Lie bundle. Every power $\mathcal{G}^r$ is regular. A line bundle on $(\mathcal{G}^r)_U$ has a rational section; the closure of its divisor is a Weil divisor on $\mathcal{G}^r$ and is Cartier by regularity. This extends the line bundle.

Every prime divisor of $\mathcal{G}^r$ supported over $T \setminus U$ is the reduced pullback of a unique prime divisor of T . Indeed flatness implies that the image of a codimension-one point has codimension at most one. It cannot be the generic point of T . The fiber above a codimension-one point is geometrically integral, so its generic point is unique; smoothness makes the pullback divisor reduced. It follows that two extensions, with their generic identification, differ by a divisor pulled back from T .

Start with any extension $\mathcal{B}_0$ of the Poincaré bundle and write $p : \mathcal{G}^2 \rightarrow T$. Replacing it by

$$\mathcal{B} = \mathcal{B}_0 \otimes p^*((e, e)^* \mathcal{B}_0)^{-1}$$

gives the prescribed trivialization at the simultaneous origin. The generic axis rigidifications extend uniquely: the divisor of each rational rigidification is pulled back from T , and restriction to the origin, where it is already the identity, makes that divisor zero. For the same reason, the two generic biextension isomorphisms extend on $\mathcal{G}^3$. For example, the ratio line bundle in

$$(m, 1)^* \mathcal{B} \simeq p_{13}^* \mathcal{B} \otimes p_{23}^* \mathcal{B}$$

has a rational trivialization whose vertical divisor is detected at (e, e, e) . It is therefore zero. A rational section of a line bundle on a regular scheme with zero divisor is an everywhere invertible section. Associativity, compatibility of the two laws, and the unit identities hold on the dense open, hence everywhere. The same argument proves uniqueness of a normalized extension. This is the divisorial argument used over a trait in [EL23, Proposition 6.7.1]; the argument just given establishes the stated version over T .

Let N be the first infinitesimal neighborhood of the identity in the second copy of $\mathcal{G}$. Its ideal is square-zero, and $\mathcal{O}_N = \mathcal{O}_T \oplus \omega_{\mathcal{G}/T}$ as an augmented $\mathcal{O}_T$ -module. Over a test

scheme $V \rightarrow T$, an element of $\mathcal{E}(V)$ is a point $g \in \mathcal{G}(V)$ together with a trivialization of $\mathcal{B}|_{\{g\} \times N_V}$ extending its given value at the origin. Such trivializations exist locally on V . Two of them differ by an element of $1 + \omega_{\mathcal{G}/T} \otimes \mathcal{O}_V$, whose multiplication is addition because the ideal is square-zero. The resulting torsor is represented explicitly as follows. If $q : \mathcal{G} \times_T N \rightarrow \mathcal{G}$ is the finite flat projection and $\pi : \mathcal{G} \rightarrow T$ is the structure map, axis normalization gives an exact sequence of vector bundles

$$0 \longrightarrow \pi^* \omega_{\mathcal{G}/T} \longrightarrow q_*(\mathcal{B}|_{\mathcal{G} \times_T N}) \longrightarrow \mathcal{O}_{\mathcal{G}} \longrightarrow 0.$$

The inverse image of the section 1 is the required affine torsor. Its points are actual sections of this fixed bundle; no quotient by automorphisms of semiabelian extensions is taken. The first biextension law supplies its group law. Its fiber over e is the vector group $\omega_{\mathcal{G}/T}$, with its canonical origin. This proves (5.4), including smoothness and compatibility with arbitrary base change. Taking Lie algebras gives an exact sequence of vector bundles of ranks 2, 4, 2.

Over the abelian open, this is the universal vector extension of $\mathcal{G}_U$, expressed as infinitesimally rigidified line bundles on its dual; see [MM74, I, §2.6 and §4] and [Cai10, Proposition 2.2]. In cohomological notation the classical Lie realization is

$$\mathrm{Lie} E(A) = H_{\mathrm{dR}}^1(A^\vee).$$

The Poincaré de Rham pairing identifies the right side with $H_1^{\mathrm{dR}}(A)$, with the usual Tate line understood over the base field. This is the covariant realization: a map $f : A \rightarrow A$ acts via $(f^\vee)^*$ on $H_{\mathrm{dR}}^1(A^\vee)$ and via f_* on homology. The Hodge filtration and functoriality in this statement are made explicit in [Cai10, Proposition 5.1 and Corollary 5.2]. Only this characteristic-zero identification is needed. On a toric special fiber we retain the constructed rank-four Lie bundle; no universal property or integral de Rham duality is asserted there.

Now pull back to a trait. The normal-base homomorphism extension theorem [FC90, Chapter I, Proposition 2.7], [Lan13, Proposition 3.3.1.5] applies directly to the semiabelian scheme $\mathcal{G}_R$. It extends both f and $f^\dagger = \lambda^{-1} f^\vee \lambda$ to $\mathcal{G}_R$; principal polarization is what makes $f^\dagger$ integral. The generic normalized biextension identity

$$(f, 1)^* \mathcal{B} \simeq (1, f^\dagger)^* \mathcal{B} \tag{5.5}$$

extends over R by the same divisor argument. Pulling an infinitesimal rigidification back along $f^\dagger|_N$ therefore sends the fiber of $\mathcal{E}$ over g to its fiber over $f(g)$. It defines an R -homomorphism $\mathcal{E} \rightarrow \mathcal{E}$. On the generic fiber it is the covariant map just described. Its differential preserves $\mathrm{Lie}(\mathcal{E}_R)$, proving the integrality assertion. After a finite trait extension the base-changed group scheme is again semiabelian and the new trait is normal. Apply the same homomorphism extension theorem and the same normalized biextension argument there. Compatibility of the constructed torsor with base change then proves the final assertion. $\square$

We apply Proposition 5.3 to each of the two regular integral toroidal models used in Lemma 4.3, after forgetting the other auxiliary level. For $i = 1, 2$ put $R_i = \mathcal{O}_K[1/\ell_i]$; the roots of unity required by the levels already belong to K . Over R_i , take a proper projective graph model $\mathcal{X}_i$ of $\bar{S}$ mapping to the level- ℓ_i toroidal compactification, and pull the rank-four bundle back to $\mathcal{X}_i$. The graph model need not be regular: local freeness survives pullback, and the proposition is applied on the regular toroidal base. A point t extends to $\mathcal{X}_i$ at every finite place whose residue characteristic is different from ℓ_i . Since $\ell_1 \neq \ell_2$, these two constructions supply a lattice at every finite place.

For the height calculation, extend each graph model separately as follows. Choose a projective embedding $\mathcal{X}_i \hookrightarrow \mathbf{P}_{R_i}^{M_i}$ and let $\overline{\mathcal{X}}_i$ be its reduced scheme-theoretic closure in $\mathbf{P}_{\mathcal{O}_K}^{M_i}$. This is an integral proper projective model with $\overline{\mathcal{X}}_i \times_{\mathcal{O}_K} R_i = \mathcal{X}_i$. The bundle is used only on this R_i -open. Fixed line-bundle denominators on $\overline{\mathcal{X}}_i$, constructed below, provide global height bounds even though the bundle is unavailable over ℓ_i .

5.3 Global coordinates and the sharper estimate near the center

Lemma 5.4. *For each finite place let $\mathcal{H}_i$ be the lattice obtained from an available model above. There are nonnegative local costs $\delta_v(t)$, depending only on the fixed frames and models, such that the coefficients in the chosen $\mathcal{P}_{\text{dR}}$ frame of any integral endomorphism $f \in \mathcal{P}_{\text{dR},t}$ satisfy*

$$\log^+ \|f\|_{v,\text{coord}} \leq \delta_v(t), \quad \sum_{v \nmid \infty} n_v \delta_v(t) \leq Ch. \quad (5.6)$$

At sufficiently close places about a potentially-good center, the costs have bounded weighted sum. Outside a fixed finite set of rational primes they can be taken to be zero on those disks. These assertions are uniform over finite extensions and over the finitely many conjugate centers.

Proof. Choose a rational frame $e_1, \dots, e_4$ for H_{dR} and its dual frame, and include the rational coefficient maps from $\text{End}(H_{\text{dR}})$ to the chosen coordinates on $\mathcal{P}_{\text{dR}}$. For each i , these form a finite list of rational sections of vector bundles on $\mathcal{X}_i$. Extend each of those bundles to a coherent sheaf $\mathcal{F}$ on $\overline{\mathcal{X}}_i$. Such an extension exists by clearing powers of ℓ_i in finite sets of local generators and relations on a finite affine cover. Removing its torsion preserves its restriction to $\mathcal{X}_i$. No local freeness of $\mathcal{F}$ over ℓ_i is needed.

Let r be one of the rational sections. The local functions b for which br belongs to $\mathcal{F}$ form a coherent nonzero ideal sheaf $\mathcal{I}_r$. Its stalk at the characteristic-zero center t_0 is the unit ideal, because the chosen frames and coefficient maps are regular there. For a sufficiently high power of the ample line bundle $\mathcal{L}_i = \mathcal{O}_{\mathbf{P}^{M_i}}(1)|_{\overline{\mathcal{X}}_i}$, $\mathcal{I}_r \otimes \mathcal{L}_i^{\otimes m}$ is generated by global sections. We may therefore choose a section nonzero at t_0 . Viewed as a section of $\mathcal{L}_i^{\otimes m}$, it gives a fixed $s \in H^0(\overline{\mathcal{X}}_i, \mathcal{L}_i^{\otimes m})$ such that sr is a regular section of $\mathcal{F} \otimes \mathcal{L}_i^{\otimes m}$. The zero divisor of s is an effective Cartier divisor, since $\overline{\mathcal{X}}_i$ is integral. Make these choices for the finite list of sections on both models and omit their finitely many generic-fiber zeros on the curve; the center remains in the resulting open. The same choices can be made for each of the fixed conjugate data.

At places of residue characteristic different from ℓ_i , equip the vector bundles with the lattice norms from $\mathcal{X}_i$. Use the model metric of $\mathcal{L}_i$ at *all* finite places, including those above ℓ_i , and continuous metrics at infinity. For a point off the zero divisor put $\lambda_{s,v}(t) = -\log \|s(t)\|_v$. Regularity of sr bounds the positive logarithmic norm of $r(t)$ by $\lambda_{s,v}(t)$ at the available places. Every finite-place $\lambda_{s,v}(t)$ is nonnegative, because s is an integral section on the proper model $\overline{\mathcal{X}}_i$. Moreover its archimedean terms are uniformly bounded below, since $\|s\|$ is bounded above on the compact complex curve. Consequently

$$\sum_{\substack{v \nmid \infty \\ v \nmid \ell_i}} n_v \lambda_{s,v}(t) \leq \sum_{v \nmid \infty} n_v \lambda_{s,v}(t) \leq h_{\mathcal{L}_i^{\otimes m}}(t) + C \leq Ch.$$

The height in this display uses the generic-fiber line bundle and the fixed metrics just specified. In particular no vector-bundle lattice over ℓ_i has been used to bound the partial sum. Adding the finitely many denominator bounds gives the required global frame cost.

If f preserves $\mathcal{H}_t$, then its operator norm for that lattice is at most one. Its matrix coefficients are evaluations $e_i^*(fe_j)$ and are bounded by the product of the norms of e_i^* and e_j . The fixed coordinate maps on $\mathcal{P}$ contribute their own denominator costs. Adding these finitely many terms proves (5.6). Applying the argument for both models and adding their bounds covers all finite places.

For the refinement, outside finitely many rational primes the center extends to the abelian interior and the rational frames, their duals, and the coordinate maps are integral and invertible throughout its formal neighborhood. Their costs are zero there. At one of the remaining potentially-good center places, first make a fixed finite local extension giving good reduction. The center then lies in the interior of a prime-to-characteristic integral model. On a sufficiently small disk in its actual branch, the family remains in that interior with the same specialization. Since the rational frames are regular and invertible at the characteristic-zero center, their matrices and inverse matrices are bounded on a smaller closed disk. This follows by evaluating their finitely many regular coordinate expressions; the disk radius and the bounds depend only on the center and the expressions. It remains valid for points over arbitrarily ramified extensions. Each such rational prime contributes a fixed constant, because its total place weight is at most one. There are only finitely many such primes and conjugate centers. $\square$

At an archimedean place use the Hodge metric on $H_1(\mathcal{A}_t, \mathbb{C})$ defined by its principal polarization. For an actual endomorphism, Rosati adjunction is adjunction for this metric, and therefore

$$\|f\|_{\text{op}, \text{Hdg}} \leq Q_t(f)^{1/2}. \quad (5.7)$$

The same trace integer computes $Q_t(f)$ after every field embedding, so this estimate and (5.2) are simultaneous at all archimedean places.

Here is the needed control of the frame norms. At a puncture with parameter u , make a fixed finite cover $u = z^b$ to make monodromy unipotent, using [Sch73, Monodromy Theorem (6.1)]. Regularity of the Gauss–Manin connection follows from [Del70, II, Proposition 6.14 and Theorem 7.9], applied to the smooth proper family and the trivial coefficient connection. Choose the *canonical unipotent extension*, with nilpotent residue, from [Del70, II, §5, Proposition 5.2]. If T is the unipotent monodromy, $N = \log T$, and $\mathbf{v}$ is a flat frame on the universal cover, this extension is locally framed by the single-valued sections obtained by cancelling monodromy:

$$\mathbf{w}(z) = \exp\left(-\frac{\log z}{2\pi i} N\right) \mathbf{v}(z).$$

Both this change-of-frame matrix and its inverse are polynomials in $\log z$, since N is nilpotent. A different holomorphic frame of this same extension differs from $\mathbf{w}$ by a holomorphic invertible matrix whose norm and inverse norm are bounded on a smaller disk. The algebraic rational frame differs from such a frame by meromorphic matrices with fixed finite pole orders; the agreement with the algebraic de Rham structure is also stated in [Sch73, Theorem (4.13)(a),(b)]. Schmid’s estimates for a polarized variation, applied also to the dual variation, bound the Hodge norms of flat frames and their duals by powers of $1 + |\log |z||$; see [Sch73, Theorem (6.6’), p. 252]. It follows that, for a fixed exponent M , all the frame norms and dual norms needed here are at most

$$C|u|^{-M}(1 + |\log |u||)^M \quad (5.8)$$

near each missing point. Finitely many sectors suffice, and the bound is independent of the target. The meromorphic coordinate maps from endomorphisms to $\mathcal{P}$ have the same

type of bound. After taking positive logarithms, (5.8) is at most a fixed multiple of a local height of the puncture plus a constant. Summing at the archimedean places is at most Ch , by the same effective divisor comparison as above. On a fixed small disk about an interior center these frame norms are simply bounded.

Proposition 5.5. *The labeled pair $\mathbf{P}$ of Proposition 5.2, expressed in the fixed rational $\mathcal{P}_{\text{dR}}$ frame, satisfies*

$$h_{\text{aff}}(\mathbf{P}) \leq C(h + \log d). \quad (5.9)$$

Let V be any subcollection of sufficiently close places in the actual center branches, with bad-center places excluded as in Proposition 4.5. Then

$$\sum_{v \in V} n_v \log \max\{1, \|P_1\|_{v, \text{coord}}, \|P_2\|_{v, \text{coord}}\} \leq C(1 + \log h + \log d). \quad (5.10)$$

Both estimates hold for simultaneously conjugated data. They also hold for a fixed number of endomorphisms F_j with $NF_j \in \mathcal{O}_t$ for a fixed positive integer N and $Q_t(F_j) \leq C(dh)^\kappa$. If $\phi : \mathcal{A}_t \rightarrow \mathcal{A}_{t'}$ is a polarized isogeny of one of the fixed multipliers in Proposition 3.4, write $I_t = \text{Ad}(\phi)$, so $I_t(D) = \phi D \phi^{-1}$. In particular the estimates hold for the transported pair $\hat{\mathbf{P}} = I_t^{-1} \mathbf{P}'$. The added choices in this last assertion have bounded relative field degree.

Proof. At finite places use Lemma 5.4. At infinity combine (5.7) with the frame and dual-frame bounds. The logarithm of the Rosati size is at most $C(1 + \log d + \log h)$. The global frame cost is at most Ch , and $\log h \leq h$; this proves (5.9). Near the good center branches, the finite-place cost is bounded and the archimedean frames have bounded norms. This gives (5.10). The construction is invariant under simultaneous conjugation, and all constants may be maximized over the fixed finite collection of conjugate models and frames.

For NF_j integral, the finite-place calculation acquires at most $\log^+ |N|_v^{-1}$. Its total weighted sum is $\log N$, and its sum on any subcollection is no larger. The Hodge calculation applies directly to F_j with its asserted Rosati bound. This proves the bounded-denominator assertion.

Finally let $\phi : (\mathcal{A}_t, \lambda_t) \rightarrow (\mathcal{A}_{t'}, \lambda_{t'})$ have fixed multiplier m , so $\phi^\vee \lambda_{t'} \phi = m \lambda_t$ and $\phi^{-1} = m^{-1} \phi^\dagger$. Its conjugation map on endomorphisms is a Rosati isometry. For P'_j integral, the transported endomorphism is

$$\hat{P}_j = \phi^{-1} P'_j \phi = m^{-1} \phi^\dagger P'_j \phi.$$

Thus $m \hat{P}_j$ is integral and $Q_t(\hat{P}_j) = Q_{t'}(P'_j)$. Apply the result just proved; this avoids estimating the coordinate height of a matrix of ϕ . There are uniformly boundedly many such ϕ for fixed m : its kernel is contained in $\mathcal{A}_t[m]$, and, for each kernel, the polarized identifications of the quotient with $\mathcal{A}_{t'}$ form a torsor under a finite polarized automorphism group. The latter has bounded order, since its faithful action on integral homology is a finite subgroup of $\text{GL}_4(\mathbb{Z})$. The kernel choices and hence the isogeny choices are bounded in number by a constant depending only on m . Their Galois orbits have that bound, so choosing ϕ requires only a bounded field extension of the compositum of the two fields of data. There are only finitely many allowed m . $\square$

All estimates in this section allow arbitrary d and h . In particular, (5.10) was obtained from integral actions and local frame bounds before establishing any upper bound for h in terms of d .

6 Local mass and equations

We retain the fixed data of [Propositions 3.1](#) and [4.2](#) and [Convention 5.1](#). The center is denoted by t_0 , its labeled pair by $\mathbf{B} = (B_1, B_2)$, and the chosen pair at a selected target t by $\mathbf{P} = (P_1, P_2)$. Throughout this section, $h = h(t) \geq 2$ and $d = \max\{2, [K(t) : K]\}$. We write C for positive constants depending only on the fixed data, allowing their value and exponent to increase. In particular, Cd^C always denotes a polynomial bound with fixed constants. The endomorphism pair is supplied by [Proposition 5.2](#), and its coordinate estimates, including the versions with fixed additional denominators, are those of [Proposition 5.5](#).

The argument has three steps. The divisor $[t_0]$ supplies a large weighted sum of local proximity to its own branch. At the retained finite places, good-reduction comparison converts horizontal transport into traces of actual endomorphisms of a common reduction. Grouping these equations then supplies all the numerical hypotheses of [Theorem 2.1](#). The nonidentity hypothesis of that theorem is proved separately in [Proposition 7.4](#).

6.1 The actual center and the amount of proximity

Choose a rational parameter x with a simple zero at t_0 , regular on the chosen affine neighborhood, and put $a = x(t)$. The finite omissions in [Proposition 3.1](#) and [Convention 5.1](#) ensure $a \neq 0, \infty$. Although x may have other zeros on the completed curve, only its inverse branch through t_0 will be used. The same convention applies to every conjugate of the fixed data.

All local sums below use the absolute-height weights of [Convention 5.1](#). For a field L of data, these can be written $w_v = [L_v : \mathbb{Q}_v]/[L : \mathbb{Q}]$ at places of L , or as the equivalent average over embeddings into an algebraically closed local field. Subsets of embeddings are allowed. Under a finite extension, the weights of the embeddings lying above a given one sum to its original weight. Thus local extensions used to realize reductions, or a global normal closure used to choose Frobenius, do not change any sum already defined.

Let $\mathcal{S}$ be a fixed finite set of rational primes containing the primes required for the models, frames, levels and fixed centered data. In particular, it contains the two auxiliary primes and the primes dividing the multipliers of [Proposition 3.4](#). It will also contain the finitely many primes specified in [Lemma 6.3](#) below. Outside $\mathcal{S}$, the center, its affine chart, parameter and regular frame spread out smoothly, with an étale parameter and an abelian scheme over that chart. The inverse branch of each curve coordinate at t_0 consequently has integral power-series coefficients. Write

$$\beta_p = \frac{\log p}{p-1}.$$

At such a prime we retain only embeddings for which t lies on that actual center branch and

$$s_v := -\log |a|_v > 2\beta_p. \tag{6.1}$$

At the archimedean places and at primes in $\mathcal{S}$, choose fixed sufficiently small disks on the actual branches. At a place where the center is potentially good, these disks are chosen to satisfy the good-reduction and convergence conditions below. At a place where it is not potentially good, [Proposition 4.5](#) excludes selected targets from a sufficiently small disk, so no such place is retained. Denote the resulting collection of nearby embeddings by V_0 .

Lemma 6.1 (Proximity mass). *For fixed constants $c > 0$ and C , every selected target satisfies*

$$\sum_{v \in V_0} w_v \log \frac{1}{|a|_v} \geq ch - C(1 + \log(dh))^2. \quad (6.2)$$

The constants are uniform over arbitrary ramification of the target fields. They may depend on the fixed disks and on $\mathcal{S}$.

Proof. Let $D_0 = [t_0]$ on the smooth projective completion of the selected curve. Its degree is positive. Choose local Weil functions $\lambda_{D_0, v}$ compatible with the fixed integral models away from $\mathcal{S}$. Their weighted sum is a height $h_{D_0}(t)$ up to a bounded constant. With the height convention using D_0 , this is $h + O(1)$; for any other fixed ample convention there is still a constant $c > 0$ with $h_{D_0}(t) \geq ch - O(1)$. For the latter assertion, choose a positive rational multiple of D_0 whose difference with the other ample divisor has positive degree; the height of that difference is bounded below. This justifies the comparison without introducing an error that grows linearly with h on the wrong side.

On the disk through t_0 , the ideal of D_0 is generated by x . Hence

$$\lambda_{D_0, v}(t) = \log \frac{1}{|x(t)|_v} + O_v(1).$$

At the spread-out primes the error is zero in the center tube, and the local Weil function is zero outside that tube. These assertions can also be read from projective distance to the fixed point: on the smooth chart its ideal is generated by the étale coordinate, whereas outside its residue tube one of the ideal generators is a unit. This uses the ideal of t_0 , and never assigns proximity to another zero of x to D_0 .

At each of the finitely many remaining local embeddings of the fixed data, $\lambda_{D_0, v}$ is bounded above outside any fixed small center disk. Within that disk it differs from $-\log|x|_v$ by a fixed bound. These bounds hold on algebraic points over every finite local extension: they follow from fixed analytic units and fixed strict inequalities defining the disks. At the bad center places the exclusion in [Proposition 4.5](#) therefore bounds the entire possible contribution by a constant. Shrinking the other fixed disks also loses only a constant in the weighted sum. Before imposing (6.1), the retained actual-branch contributions thus have sum at least $ch - O(1)$.

It remains to bound the cost of that cutoff. Take a field of degree $D \leq Cd^C$ containing a and the labeled target data, before taking any normal closure. We have $h(a) \leq Ch$ by [Convention 5.1](#). Let $\mathcal{T}(a)$ be the rational primes admitting an embedding with $|a|_v < 1$. If $v \mid p$ is one such place, then, with integral valuation ord_v and residue degree f_v ,

$$w_v \log \frac{1}{|a|_v} = \frac{f_v \text{ord}_v(a)}{D} \log p \geq \frac{\log p}{D}.$$

Summing positive valuations and using the product formula gives

$$\sum_{p \in \mathcal{T}(a)} \log p \leq Dh(a), \quad N := |\mathcal{T}(a)| \leq CDh. \quad (6.3)$$

This argument sums the positive valuations themselves; cancellations in the rational norm of a are irrelevant.

The total weight over each rational prime is at most one. The discarded mass there is at most $2\beta_p$. Order the N primes increasingly as $p_1, \dots, p_N$. The function $\log u/(u-1)$ decreases for $u > 1$, and $p_i \geq i+1$. Consequently

$$\sum_{p \in \mathcal{T}(a)} \beta_p \leq \sum_{i=1}^N \frac{\log(i+1)}{i} \leq C(1 + \log(N+1))^2 \leq C(1 + \log(dh))^2. \quad (6.4)$$

This proves (6.2). Both the support bound and the weight identity survive field extension, so no ramification degree or degree of a normal closure enters the estimate. $\square$

6.2 Good-reduction comparison on a common tube

In the rational frame of $\mathcal{P}_{\text{dR}}$ let $Y(a)$ be horizontal back-transport from t to t_0 , normalized by $Y(0) = 1$. Thus $Y(a)P$ is expressed in the fixed center frame. This convention is used for the primed data as well.

Proposition 6.2 (Rational comparison and transport). *At every retained finite-place embedding there is, after a finite local extension, a smooth proper abelian family over a smooth integral moduli chart such that A_{t_0} and A_t have the same polarized special fiber $\bar{A}$. Rational comparisons of their homologies with the cohomology of $\bar{A}$ identify their change of comparison on $\mathcal{P}$ with $Y(a)$. In particular,*

$$\langle B_i, Y(a)P_j \rangle = \text{Tr}(\bar{B}_i \bar{P}_j) \quad (1 \leq i, j \leq 2). \quad (6.5)$$

Here the bars are actual specializations of the endomorphisms of the two lifted fibers. The same comparison is functorial for polarized isogenies with specified reductions. The assertion allows arbitrary ramification of the local field of t .

Proof. Outside $\mathcal{S}$ the abelian family and the common specialization are part of the integral chart already chosen. At a retained prime in $\mathcal{S}$, the center is potentially good. Choose an auxiliary fine level invertible in that residue characteristic and a finite local extension over which the center has good reduction. The center then lies in the interior of the smooth integral moduli space. A sufficiently small fixed analytic neighborhood maps into the tube of its special point. Every algebraic point in that neighborhood, even over a further ramified extension, therefore gives a good-reduction lift with that same special fiber. The disks defining V_0 have been chosen inside these neighborhoods.

For clarity, the comparison used here is a comparison of rational cohomology spaces. For a complete mixed-characteristic DVR with perfect residue field k , fraction field L , and a smooth proper lift $\mathcal{X}$ of X_0 , it has the form

$$H_{\text{crys}}^1(X_0/W(k)) \otimes_{W(k)} L \simeq H_{\text{dR}}^1(\mathcal{X}_L/L). \quad (6.6)$$

The rational comparison of Berthelot–Ogus [BO83] is functorial and allows every ramification index; this form is stated in [Ber82, §1.2] and explicitly distinguished from integral comparison in [AY25, Introduction, equations (2)–(3)]. It does not assert preservation of integral lattices. We use its relative realization to compare the two lifts, as follows.

Embed the common proper special fiber $\bar{A}$ as the closed fiber over the base special point in the smooth formal total family. Its rigid cohomology is computed by the de Rham complex on its tube; properness eliminates a compactification boundary for this special fiber. Independence of the smooth embedding, functoriality and extension of coefficients identify restriction to each smooth proper lift with (6.6); these are the tube constructions of [Ber82, §§1.4–1.5 and 2.5(c)]. One may work on a smaller base disk, on which the relative Gauss–Manin connection is ordinary. An absolute de Rham hypercohomology class on the total tube maps to a horizontal relative class: the connecting morphism obtained by filtering the absolute de Rham complex by base differential forms is precisely the Gauss–Manin connection, and its composition with the absolute-to-relative map is zero. The evaluation isomorphism with the cohomology of a proper lifted fiber then shows that a basis of the common rigid cohomology gives a basis of horizontal relative classes on this disk. The change between two evaluations is therefore the Taylor isomorphism; the

convergent Taylor formalism is described in [Ber82, §4.1]. All these operations commute with extension of the coefficient field, so the same diagram holds for ramified evaluation fields. The unramified version of this diagram is displayed in [LV20, §3.3, diagram (3.9)].

After pullback to the parameter disk, the change of comparisons is therefore horizontal and has identity value at $a = 0$. Uniqueness for the ordinary differential equation identifies it with the Taylor solution throughout the retained convergence disk. Dualizing gives the statement for covariant homology, and taking endomorphisms gives the stated action on $\mathcal{P}$. Functoriality identifies an endomorphism of either lift with its specialization, and likewise identifies an isogeny with its specialized map. The polarization and trace pairing are preserved in these identifications. Formula (6.5) follows.

This argument uses a common good-reduction family. The disks at a center which is not potentially good have already been excluded by Proposition 4.5. Also, horizontal transport here identifies cohomology and its tensors; no assertion about horizontal constancy of the Hodge filtration is needed. $\square$

6.3 The three equations

For two labeled pairs in a quadratic space, use the matrix notation

$$\langle \mathbf{D}, T\mathbf{E} \rangle := (\langle D_i, TE_j \rangle)_{1 \leq i, j \leq 2}.$$

Thus each matrix equation below is a system of four scalar equations.

At a good-reduction place, Proposition 6.2 expresses this matrix as rational traces. Using those trace values as coefficients at every varying prime could give a number of distinct systems depending on the target height. We instead compare the matrix with its simultaneous Frobenius conjugate: conjugating both factors preserves each trace. For target pairs on the fixed Hecke list, we will rewrite that equality using only one horizontal transport. The next lemma makes this possible. A fixed-multiplier isogeny between the targets specializes to an isogeny between the center reductions; outside finitely many fixed primes, that specialized map lifts to one of finitely many center isogenies.

Lemma 6.3 (Bounded center isogenies under specialization). *Fix two of the conjugate centers and one multiplier m_* from Proposition 3.4. There are finitely many characteristic-zero polarized isogenies of multiplier m_* between these centers. After enlarging $\mathcal{S}$ by a fixed finite set, every polarized isogeny of that multiplier between their good special fibers is the specialization of one of these center isogenies. This assertion is simultaneous for the fixed finite list of centers and multipliers.*

Proof. A polarized isogeny ϕ of multiplier m_* satisfies $\phi^* \lambda' = m_* \lambda$. Since the fibers have dimension two, $\deg \phi = m_*^2$, and its kernel lies in $A[m_*]$. Away from primes dividing m_* , kernels are among a fixed finite set of subgroup schemes of this finite étale torsion scheme. For each such kernel, the quotient is fixed. Polarized identifications of that quotient with the second center form either an empty set or a torsor under its finite polarized automorphism group. This proves finiteness in characteristic zero.

Here is a finite-type construction that also includes all special-fiber possibilities. After a fixed extension and localization, choose relatively ample symmetric line bundles L_0, L'_0 inducing $N\lambda_0, N\lambda'_0$ for the same fixed integer $N > 0$. On the graph of any isogeny of multiplier m_* , the product bundle restricts to $L_0 \otimes \phi^* L'_0$, numerically equivalent to $L_0^{\otimes(1+m_*)}$. Riemann–Roch on the abelian surface gives the fixed Hilbert polynomial

$$\chi(A_{t_0}, (L_0 \otimes \phi^* L'_0)^{\otimes n}) = N^2(1 + m_*)^2 n^2.$$

All graphs therefore lie in one finite-type relative Hilbert scheme of the product of the two fixed abelian schemes. The locus on which the first projection of the graph is an isomorphism is open. On this graph locus, preservation of the origin is closed and makes the resulting morphism a homomorphism by rigidity for abelian schemes. The condition $\phi^\vee \lambda'_0 \phi = m_* \lambda_0$ is a closed equality of homomorphisms to the dual abelian scheme. This constructs a separated finite-type scheme of the desired polarized homomorphisms; the polarization identity forces each of them to be an isogeny. The construction uses the same Hilbert polynomial in every residue characteristic, so it includes any additional special-fiber polarized automorphisms as well.

After a fixed extension making its finitely many geometric generic points rational, finite presentation spreads each of them to a section over an open arithmetic base. A common localization works for all of them. Each section has closed image by separatedness. The complement of the finite union of these images is consequently open, hence constructible, and has empty generic fiber. Its image in the one-dimensional arithmetic base is constructible and misses the generic point, hence consists of finitely many closed points. Removing their residue characteristics proves the assertion. One can obtain the same conclusion by spreading out the finite kernel list and the polarized-identification schemes. Both constructions involve fixed centers and fixed multipliers, so the removed primes are independent of t and of any target isogeny subsequently chosen. $\square$

Proposition 6.4 (Local equations). *The nearby embeddings V_0 can be partitioned into groups using one of the following three systems, together with the base-coordinate equations in (6.9) below:*

$$\langle \mathbf{B}, Y(a)\mathbf{P} \rangle = \mathbf{r}, \quad \mathbf{r} \in \text{Mat}_{2 \times 2}(\mathbb{Q}), \quad (1)$$

$$\langle \mathbf{B}, Y(a)\mathbf{P} \rangle = \langle \mathbf{B}', Y'(aR)\mathbf{P}' \rangle, \quad R = a'/a, \quad (2)$$

$$\langle \mathbf{B}, Y(a)\mathbf{P} \rangle = \langle I_0^{-1}\mathbf{B}', Y(a)\widehat{\mathbf{P}} \rangle, \quad \widehat{\mathbf{P}} = I_t^{-1}\mathbf{P}'. \quad (3)$$

Here primes denote a simultaneous conjugate of the labeled data. Type (1) is used at infinity and at retained primes in $\mathcal{S}$. At the other retained primes, type (2) is used if the coarse target pair lies outside the closed finite Hecke list of Proposition 3.4, and type (3) is used if it lies on that list. In types (2) and (3), the following additional data and properties hold at every assigned embedding:

- (i) $|a'|_v = |a|_v$, hence $|R|_v = 1$, and $h(R) \leq Ch$.
- (ii) There are two actual common reductions $\overline{A}$ and $\overline{A}'$ of the unprimed and primed center-target pairs, respectively. Relative Frobenius $F : \overline{A} \rightarrow \overline{A}'$ has polarized multiplier p and conjugates both labeled pairs to their primed counterparts.
- (iii) On each common reduction, smooth proper specialization at $\ell_1 \neq p$ preserves the two separately saturated plane lattices. Their reductions span the independent degenerate four-space prescribed by Proposition 4.2.
- (iv) In type (3), $I_0 = \text{Ad}(\phi_0)$ and $I_t = \text{Ad}(\phi_t)$ are conjugation isometries associated to polarized isogenies of the same fixed multiplier m_* . Their reductions satisfy $\overline{\phi}_0 = \overline{\phi}_t$, and

$$I_0 Y(a) = Y'(a') I_t. \quad (6.7)$$

Here $p \nmid \ell_1 m_*$ and $\ell_1 \nmid m_*$. The maps induced by F and $\overline{\phi}_0$ on ℓ_1 -adic Tate modules are integral isomorphisms, up to their unit similitude multipliers.

The entries of $\mathbf{r}$ have bounded denominators and logarithmic heights at most $C(1 + \log h + \log d)$. The pairs $\widehat{\mathbf{P}}$ have fixed bounded denominators, polynomial Rosati norms, and all the coordinate bounds of Proposition 5.5.

Proof. At a retained finite place, (6.5) identifies the left-hand entries with traces of products of actual specialized endomorphisms. For an integral endomorphism f of an abelian variety, the degree polynomial $\deg(n - f)$ is integral and equals the characteristic polynomial on every prime-to-characteristic rational Tate module; see [Mil86, Proposition 12.9 and the preceding paragraph]. We explain the comparison with rational crystalline homology, since the product $\overline{B}_i \overline{P}_j$ need not lift to either characteristic-zero fiber. The cup-product map $\bigwedge^r H_{\text{crys}}^1(\overline{A}) \rightarrow H_{\text{crys}}^r(\overline{A})$ is an isomorphism. Indeed, the rational comparison with the center's good lift respects cup product: the restrictions and base-change maps of the de Rham complexes in the tube construction are multiplicative. It therefore identifies the displayed map, after a faithful scalar extension, with the de Rham cup-product map; that map is an isomorphism by comparison with the cohomology of a complex torus. Being defined by cup product, the crystalline isomorphism is natural for every endomorphism of $\overline{A}$, including those that do not lift.

For an actual integral endomorphism u of $\overline{A}$, the map $[n] - u$ is an isogeny for every sufficiently large integer n . Its pullback on first cohomology is $n - u^*$, by the Künneth description of first cohomology and the group law. On top cohomology its action is multiplication by $\deg([n] - u)$. One may verify this last assertion in rigid cohomology, using the trace fundamental class and compatibility of cycle classes with proper pushforward [Pet03, §2.1 and Proposition 6.4], and then use smooth-proper crystalline-rigid comparison [Ber82, §1.5, Proposition 2]. The degree statement includes inseparable isogenies: it is the degree of a finite flat map, not just the number of geometric kernel points; see also [Pet03, Lemma 6.5]. Taking the top exterior power therefore gives $\det(n - u^* | H_{\text{crys}}^1) = \deg([n] - u)$ for infinitely many n , hence as a polynomial identity. Dualization preserves the characteristic polynomial. Combining this with the Tate-module identity proves that all the homological traces used here agree. Thus these traces are rational and are integers when both factors are integral; fixed rational denominators introduce only fixed denominators in the trace.

All specialized operators here are symmetric for the common specialized polarization. Rosati positivity, valid in every characteristic [Mil86, Theorem 17.3], and Cauchy-Schwarz give

$$\left| \text{Tr}(\overline{B}_i \overline{P}_j) \right|^2 \leq \text{Tr}(\overline{B}_i^2) \text{Tr}(\overline{P}_j^2) = \text{Tr}(B_i^2) \text{Tr}(P_j^2) \leq C(dh)^C. \quad (6.8)$$

The absolute value in this inequality is the ordinary real absolute value of the rational trace. It is unrelated to the residue characteristic norm. At infinity, analytic Betti transport on the fixed simply connected center disk identifies integral homology lattices. The analogous cross traces are again rational with fixed denominators. The Hodge metrics on this fixed disk, in a flat trivialization, are uniformly comparable. Cauchy-Schwarz for the corresponding operator norms and the target Rosati bound give the same polynomial size bound. These observations establish (1) and its coefficient-height estimate at all of its assigned places.

Now let $p \notin \mathcal{S}$. Take a normal closure of a field containing the labeled data, embed it at the assigned place, and choose a lift of residue p -power Frobenius in its decomposition group. The lift preserves this valuation. Apply it simultaneously to the center, parameter, target, frames, polarization, levels and both endomorphism labels; denote the result by primes. This gives $|a'|_v = |a|_v$. Absolute heights are invariant under conjugation, so

$$h(R) = h(a'/a) \leq h(a') + h(a) = 2h(a) \leq Ch.$$

The primed special fiber is the p -power twist of the unprimed one. Functoriality of relative Frobenius gives, for the same isogeny F ,

$$\overline{B}'_i = F \overline{B}_i F^{-1}, \quad \overline{P}'_j = F \overline{P}_j F^{-1}.$$

These are conjugations in rational endomorphism algebras; no inverse of Frobenius on an integral crystalline lattice is asserted. Taking the trace of the product removes F . Applying [Proposition 6.2](#) on each of the two actual common reductions gives exactly (2).

At these primes all fixed levels are invertible. Integral smooth proper specialization identifies the ℓ_1 -adic homology lattices of each pair of lifts with the lattice of its common reduction, compatibly with the specified level marking. For either plane H this transports the intersection of $H \otimes \mathbb{Q}_{\ell_1}$ with the ambient integral operator lattice to the corresponding intersection after specialization. Thus the separately saturated lattices, and their marked reductions, retain the incidence in [Proposition 4.2](#). This statement concerns the rational planes and their intersections with lattices. It does not require the chosen small vectors B_i, P_j themselves to be bases after reduction modulo ℓ_1 .

If the coarse target pair belongs to the closed exceptional list, choose a polarized isogeny $\phi_t : A_t \rightarrow A_{t'}$ of one of its fixed multipliers m_* . The list is fixed before the auxiliary levels; only its pullback to those levels is being tested here. After a local extension, ϕ_t extends between the good-reduction abelian schemes. Its reduction is an isogeny of multiplier m_* between the same two special fibers as the centers. By [Lemma 6.3](#), it equals the reduction of one of the fixed center isogenies ϕ_0 . Conjugation by a polarized isogeny preserves trace and Rosati symmetry, hence induces the isometries I_t, I_0 . Applying the functorial comparisons to their identical reductions proves (6.7). Substitution into (2) gives (3).

The identity $\phi_t^{-1} = m_*^{-1} \phi_t^\dagger$ shows that $m_* \hat{P}_j = m_* \phi_t^{-1} P'_j \phi_t$ is integral. Since conjugation preserves the Rosati form, the norms of $\hat{P}_j$ are bounded by the same polynomial as those of P'_j . Apply [Proposition 5.5](#) to these actual rational endomorphisms of A_t with that fixed denominator. This supplies all their required coordinate bounds directly; no height estimate for the entries of a chosen isogeny matrix is needed. Finally p, m_* and ℓ_1 are pairwise coprime in the required positions, so the Tate-module integrality and invertibility in (iv) follow from the isogeny degrees p^2, m_*^2 . $\square$

The order of fixed choices is useful here. The list of coarse Hecke correspondences and its multipliers was chosen in [Proposition 3.4](#); the auxiliary levels were then chosen in [Proposition 4.2](#). The specialization lemma, or any later spreading out or shrinking of charts, may enlarge $\mathcal{S}$. A newly included prime is handled by (1) if the center is potentially good there, and by [Proposition 4.5](#) otherwise. Its contribution is therefore either retained with a single trace equation or bounded independently of the target. No new multiplier, level or coarse correspondence is chosen in this enlargement.

6.4 Uniform grouping and the numerical interpolation data

Fix affine coordinates $u_1, \dots, u_b$ on the chosen curve chart, with algebraic regular germs f_ν at the actual center so that $u_\nu(t) = f_\nu(a)$ on its disk. Write $\mathbf{u} = (u_1, \dots, u_b)$, and use primes for the corresponding coordinates on a conjugate chart. We now specify the coordinate tuple U in each system:

system	U	constant equation data
(1)	$(\mathbf{u}, \mathbf{P})$	$\mathbf{B}, \mathbf{r}$
(2)	$(\mathbf{u}, \mathbf{u}', \mathbf{P}, \mathbf{P}', R)$	$\mathbf{B}, \mathbf{B}'$
(3)	$(\mathbf{u}, \mathbf{P}, \hat{\mathbf{P}})$	$\mathbf{B}, I_0^{-1} \mathbf{B}'$

Every operator slot in U contributes its coordinates in the fixed rational five-dimensional frame. The listed constants remain fixed within a group, as do its centered germs; $\mathbf{r}$ may depend on the target. At the target the operator pairs have the endomorphism meanings

specified in [Proposition 6.4](#). On the ambient affine space their slots are unrestricted coordinates. In particular, an algebraic variety through the target used in [Theorem 2.1](#) is not assumed to parametrize Hodge endomorphisms. In the exceptional system the conjugation map I_t enters only through the target value of $\hat{\mathbf{P}}$.

Adjoin the scalar equations

$$u_\nu - f_\nu(a) = 0 \quad (1 \leq \nu \leq b), \quad u'_\nu - f'_\nu(aR) = 0 \quad \text{in type (2)}. \quad (6.9)$$

These equations explicitly retain the curve projections and branches; they will matter when testing identities on arbitrary algebraic subvarieties in [Proposition 7.4](#).

Proposition 6.5 (Groups and numerical bounds). *There are fixed constants C such that either $h \leq Cd^C$, or there is one system of type (1), (2) or (3), with (6.9), having the following properties. Its target is an algebraic point $z = (a, U)$ in an affine space of fixed dimension, and it holds at all embeddings in a weighted subcollection $V \subset V_0$. Writing each scalar expression as*

$$E(a, U) = \sum_{j \geq 0} a^j e_j(U),$$

the numerical and function hypotheses of [Theorem 2.1](#) hold:

$$h_{\text{aff}}(z) \leq Cd^C h, \quad \sum_{v \in V} w_v \log \frac{1}{|a|_v} \geq \frac{h}{Cd^C}, \quad (6.10)$$

$$\sum_{v \in V} w_v \log^+ \|U\|_v \leq Cd^C (1 + \log h)^C, \quad \deg e_j \leq C(1 + j), \quad (6.11)$$

$$h_{\text{aff}}(e_j) \leq C(1 + j + d)^C (1 + \log h)^C, \quad (6.12)$$

$$\left| \sum_{j \geq l} a^j e_j(U) \right|_v \leq |a|_v^l \exp((1 + l)b_v),$$

$$\sum_{v \in V} w_v b_v \leq Cd^C (1 + \log h)^C. \quad (6.13)$$

Here $l \geq 0$, $b_v \geq 0$, and one common choice of b_v works for the bounded number of expressions. Each expression is a polynomial combination of fixed bounded degree of the coordinates, fixed regular algebraic germs and fixed ordinary horizontal matrix germs, evaluated at a and, in type (2), at aR . Only a fixed finite list of such germs is used. Constants in the expressions may depend on the target, with the bounds displayed above.

There are at most Cd^C groups in the construction. This bound counts ordered conjugate tuples, not elements of the Galois group of a normal closure. The proposition asserts no nonidentity condition on an algebraic subvariety through z .

Proof. Counting the groups. Let τ comprise the fixed data, t and the labeled endomorphism coordinates, and choose a field of definition with absolute degree $D \leq Cd^C$, as supplied by [Proposition 5.2](#). There are at most D conjugate tuples. At infinity and at the finitely many primes in $\mathcal{S}$, group by the embedded tuple, the fixed branch and its resulting rational matrix $\mathbf{r}$. Each embedding gives one such matrix, hence this uses at most CD groups. This counts actual embeddings at fixed places; counting all rational matrices of the allowed size would give an unnecessary dependence on h .

For the Frobenius equations, group by the ordered pair (τ, τ') . There are at most D^2 values. The normal closure in which Frobenius was chosen can have much larger degree, but only these ordered tuple values enter an equation. The field generated by a pair has

degree at most D^2 . The primed fixed family belongs to the finite list of conjugates of the fixed family; its germs are independent of the varying target.

Split these groups according to whether the coarse pair is on the closed exceptional list. On it, refine by a choice of its fixed multiplier, an isogeny ϕ_t , and its specializing center isogeny ϕ_0 . The multiplier and center choices range over finite fixed sets. For a fixed multiplier, a target isogeny is specified by a subgroup of bounded torsion and a polarized identification of the quotient with the second target. The number of subgroups is bounded in dimension two and m_* . The order of a polarized automorphism group is uniformly bounded in fixed dimension: it is finite by positivity, and acts faithfully on an integral homology lattice of fixed rank, whose finite subgroups have bounded order. Thus only a fixed bounded relative extension is needed to choose ϕ_t , and the refinement still has at most CD^2 groups. The transported tuple also has polynomial degree. Enlarging fields and lifting the embedding weights in making this partition preserves their sum.

Selecting mass. By Lemma 6.1, the total mass is at least $ch - C(1 + \log(dh))^2$. Either $h \leq Cd^C$, or this is at least $ch/2$, after increasing the fixed constants. Indeed $(\log h)^2 \leq \varepsilon h + C_\varepsilon$ and $(1 + \log d)^2 \leq Cd^2$ absorb the error. In the latter case one of the at most Cd^C groups has mass at least $h/(Cd^C)$. Fix that group and call its embeddings V . Its algebraic tuple and constants give common equations at all of them.

Coordinate heights. Use the coordinate tuple U specified above for the chosen system. By Proposition 5.5, the endomorphism coordinates have global affine height $C(h + \log d)$ and weighted positive local logs at most $C(1 + \log h + \log d)$ at the chosen nearby embeddings. This also applies to $\hat{\mathbf{P}}$ by Proposition 6.4. Base-coordinate heights are $O(h)$, and their positive local logs are bounded on the fixed center disks, with zero cost away from the fixed primes. We have already proved $h(R) \leq Ch$, while $\log^+ |R|_v = 0$ on its group. These facts give (6.10) and the first bound in (6.11).

Fixed germs and their coefficients. The matrices Y, Y' solve fixed ordinary algebraic differential systems in the selected regular frames. Dual or inverse matrices, when needed to realize an operator system, satisfy fixed differential systems as well. Each f_ν is an algebraic regular germ on the fixed chart. There are only finitely many conjugate fixed charts and centers. Thus the number of systems, their ranks and the degrees of polynomial combinations are fixed. In particular, “ordinary” here refers to the absence of a pole of the differential equation at its center; it imposes no ordinarity condition on an abelian special fiber.

For completeness, consider a local system $Z' = A(x)Z$ at a spread-out prime, with $A(x) = \sum_{i \geq 0} A_i x^i$ integral and $Z(0)$ integral. Writing $Z(x) = \sum_{n \geq 0} Z_n x^n$ gives

$$(n+1)Z_{n+1} = \sum_{i=0}^n A_i Z_{n-i}.$$

Induction proves $n!Z_n$ integral, since $n!/(n-i)!$ is an integer. In the row convention for back-transport the same argument applies with multiplication on the other side. Therefore

$$\|Z_n\|_v \leq |n!|_v^{-1} \leq \exp(n\beta_p). \quad (6.14)$$

The last estimate follows from $v_p(n!) = \sum_{r \geq 1} \lfloor n/p^r \rfloor \leq n/(p-1)$. Regular algebraic coordinate germs are integral in the étale chart. A product of a fixed number of these series retains the factorial bound: multiplying its coefficient of degree n by $n!$ expresses it as a sum with the integral factors $n!/(n_1! \cdots n_k!)$.

At each of the finitely many other places, ordinary-point convergence gives fixed exponential coefficient bounds. At the archimedean embeddings this is Cauchy’s estimate

on a fixed smaller disk. At a fixed finite place it follows from the recurrence after rescaling the parameter to make the analytic connection bounded. Summing all local coefficient bounds yields, for each fixed germ,

$$h_{\text{aff}}(Z_n) \leq C(1 + n \log(n+1)).$$

Here the contribution from all factorial denominators is at most $\log(n!)$, and the exceptional-place contribution is $O(n+1)$. For the vector of coefficients up to index j the same reasoning gives a polynomial bound in $1+j$. Fixed products and sums preserve such a bound.

In $Y'(aR)$ the coefficient of a^j is $Y'_j R^j$. Thus the degree in the polynomial variables grows at most linearly in j , without altering its coefficient heights. The same holds for $f'_\nu(aR)$. The other coordinate factors occur with fixed bounded degree. The only varying constants of type (1) are $\mathbf{r}$, whose coefficient heights are $C(1 + \log h + \log d)$; the constants in type (3) are fixed centered isogeny data. This proves the remaining degree bound and (6.12).

Tails at the target. At $p \notin \mathcal{S}$, let $M_v \geq 1$ bound all fixed-degree coordinate and constant factors in the bounded collection of expressions. We may choose it so that

$$\log M_v \leq C \log^+ \|U\|_v + C_v,$$

where $C_v = 0$ outside fixed primes; a larger fixed bound includes constant terms when $l = 0$. The factorial argument gives $|e_j(U)|_v \leq M_v \exp(j\beta_p)$. The substitution aR introduces no further factor because $|R|_v = 1$. By (6.1), $|a|_v \exp(\beta_p) < 1$, and the ultrametric triangle inequality yields

$$\left| \sum_{j \geq l} a^j e_j(U) \right|_v \leq M_v \sup_{j \geq l} (|a|_v e^{\beta_p})^j = M_v |a|_v^l e^{l\beta_p}.$$

Thus $b_v = \beta_p + \log M_v$ works, including $l = 0$. At the fixed places choose the evaluation disk with radius at most one half of a fixed convergence radius in the archimedean case, and strictly smaller in the nonarchimedean case. The geometric-series estimate gives the same bound with $b_v = C_v + C \log^+ \|U\|_v$ and, where necessary, an additional $C \log^+ \|\mathbf{r}\|_v$. The sum of the latter contributions is bounded by the coefficient height of $\mathbf{r}$ and the fixed number of places. The positive coordinate logs have already been bounded. Finally (6.4) bounds the sum of β_p over every prime in this group. Enlarging b_v to the maximum for the bounded number of expressions proves (6.13).

The entire construction uses absolute weights. Selecting a subgroup of embeddings only decreases the nonnegative coordinate and tail costs; field extension preserves the previous sums. None of these estimates requires a bound in terms of d alone on the number of supersingular, or of any other, proximity primes. This completes the numerical verification and leaves precisely the nonidentity condition to [Proposition 7.4](#). $\square$

7 Nonidentities and the uniform height bound

We verify the last hypothesis of [Theorem 2.1](#): the equations selected in [Proposition 6.5](#) cannot all vanish identically on any algebraic branch that occurs in the interpolation argument. This assertion concerns arbitrary algebraic subvarieties of the coordinate space. Their endomorphism coordinates are not required to represent Hodge classes. Monodromy treats the single and paired equations. For the exceptional equation it leaves two signs. A Frobenius multiplier excludes the positive sign; the degenerate first auxiliary incidence excludes the negative sign through the integral Tate lattice.

7.1 Algebraic branches and continuation

Fix one group from [Proposition 6.5](#), its target $z = (a, U)$ with $a \neq 0$, and its equations, including the base-coordinate equations. Center data, conjugate families, and any matrix r or isometry I_0 are constants in these equations. The operator coordinates and, when used, R are coordinates of the ambient affine space. Write $\mathcal{E}$ for this finite collection of analytic expressions. Its algebraic function germs are taken on the specified branches through the centers.

Lemma 7.1 (Continuation from a normalized branch). *Let W be an irreducible closed algebraic subvariety of the ambient affine space, containing z , on which a is nonconstant. Let D be a prime divisor on the normalization of W lying over the generic point of a component of $W \cap \{a = 0\}$. If every expression in $\mathcal{E}$ vanishes as a germ on this branch, then the following hold.*

- (i) *The base coordinates give actual algebraic projections to the fixed curve charts; the first projection s is nonconstant and satisfies $x(s) = a$. In the paired case the second satisfies $x'(s') = aR$.*
- (ii) *On a connected smooth algebraic open W° all the identities continue along every path. At a fixed point of W° , continuation around a loop changes the horizontal transports by their pullback monodromy, while returning the algebraic coordinates to their original values.*
- (iii) *Every operator pair that is independent at z is independent at a general point of W° .*

The same conclusions hold if the pullbacks of all expressions vanish to infinite order along D .

Proof. At a general complex point of D the normalization and D are smooth. Take local coordinates $(u, v_2, \dots, v_k)$ there with $D = \{u = 0\}$. The regular function a is u^e times a unit, for some $e > 0$. All affine coordinates of W pull back to holomorphic functions; in particular R is finite there. Consequently both a and aR tend to zero, and the ordinary-point series defining our expressions converge on a sufficiently small neighborhood. An infinite-order zero along D makes every coefficient of their convergent expansion in u zero. Thus it gives an identity on the whole local branch. This is stronger than an identity obtained by setting $u = 0$.

Let $c_\alpha(x)$ be the algebraic coordinate germs of the first curve chart. Its added equations are $s_\alpha - c_\alpha(a) = 0$. Every polynomial relation defining the chart, and the relation $x(s) = a$, therefore holds on the analytic branch. Such a relation is an algebraic identity on W , since a nonempty analytic open of an irreducible complex algebraic variety is Zariski dense. This constructs the asserted projection to the actual chart; the same argument applies to $s'_\alpha - c'_\alpha(aR)$ when present. Relations involving regular functions on these affine charts can equivalently be checked after clearing denominators that are invertible on the chart. The target belongs to these charts by [Proposition 6.4](#), so the algebraic relations also hold at the specified target. Nonconstancy of a implies nonconstancy of s .

Remove singularities and the proper closed sets where the chosen frames or projections are undefined. Shrink further to avoid the branch loci of any parameter charts used to describe the germs. A nonempty part of our analytic branch lies over the resulting smooth open: normalization is an isomorphism over the smooth locus of W . This open is connected. Solutions of the pulled-back connections continue along its paths, and the identity theorem preserves the equations under this continuation. All their algebraic coordinate functions are single-valued there. This proves (ii), without requiring the transports themselves to be single-valued.

Finally, an independent pair at z has a two-by-two coordinate minor nonzero at z . That minor is a nonzero regular function on W . Deleting its zero set for each of the finitely many pairs proves (iii). No Hodge condition on these varying coordinates has entered the proof. $\square$

For use below, a nonconstant projection from W° to a fixed base curve has the full connected monodromy of that curve. Indeed, choose an algebraic curve in W° on which the projection is nonconstant. After normalization and omission of finitely many points it is a finite cover of an open of the base curve. Its monodromy has finite index in the monodromy on that open, and hence the same connected Zariski closure. The monodromy of W° contains this image. The joint assertion for two nonconstant projections is precisely [Proposition 3.3](#); its exceptional images are contained in the fixed closed correspondences of [Proposition 3.4](#).

Lemma 7.2 (Matrix coefficients and their span). *Let V be the complex five-dimensional quadratic representation of [Proposition 3.2](#), and let $G = \mathrm{SO}(V)$.*

- (i) *If $b, v \in V$ are nonzero, the function $g \mapsto \langle b, gv \rangle$ on G is not constant.*
- (ii) *The complex linear span of G is $\mathrm{End}_{\mathbb{C}}(V)$. Consequently, if $b, c \in V$, $p, q \in V_1$, and $M_0 : V_1 \rightarrow V$ is an isomorphism, the identities*

$$\langle b, gM_0p \rangle = \langle c, gM_0q \rangle \quad (g \in G)$$

imply $b \otimes p = c \otimes q$, where the first tensor factors are identified with their duals by the quadratic pairing.

The assertions remain valid if the identities are initially known only on a Zariski-dense monodromy subgroup.

Proof. For (i), let V' be the span of $(g - 1)v$ for $g \in G$. It is invariant, since

$$h(g - 1)v = (hg - 1)v - (h - 1)v.$$

It is nonzero, as the nontrivial irreducible representation V has no fixed nonzero vector. Irreducibility gives $V' = V$. Constancy of the displayed coefficient would make b perpendicular to V' , a contradiction.

Assertion (ii) is the matrix-span conclusion of Burnside's theorem; in this representation it also has the following direct verification. Choose an orthonormal basis. The diagonal matrices $D_\eta = \mathrm{diag}(\eta_1, \dots, \eta_5)$ with $\eta_i \in \{1, -1\}$ and $\prod_i \eta_i = 1$ lie in G . Their coordinate characters are distinct, so

$$E_{ii} = \frac{1}{16} \sum_{\prod_k \eta_k = 1} \eta_i D_\eta$$

lies in their span. The span of a matrix group is an algebra. Signed permutation matrices of determinant one, together with the E_{ii} , therefore give every matrix unit E_{ij} in this algebra. This proves (ii). The displayed identity is a linear functional of gM_0 ; nondegeneracy of the tensor-matrix pairing proves the tensor conclusion. Polynomial identities on a Zariski-dense subgroup extend to G . $\square$

7.2 The Frobenius obstruction

For the exceptional equation, a hypothetical analytic identity will give four tensor equalities by [Lemma 7.2](#). At the target, these force one common scalar λ : the center pair is multiplied

by λ , and the target pair by λ^{-1} . The center endomorphisms have nonzero rational trace norms, unchanged by conjugation and by the isometry I_0 , so $\lambda^2 = 1$. We isolate the arithmetic obstruction to these two signs first. The lemma concerns actual endomorphisms of the characteristic-zero fibers; we will apply it only after deriving their equalities at the target from the analytic identity.

Lemma 7.3 (Exclusion of both signs). *In an exceptional group of Proposition 6.4, the target data cannot satisfy, for either common sign $\epsilon \in \{1, -1\}$,*

$$I_0^{-1}B'_i = \epsilon B_i \quad (i = 1, 2), \quad I_t^{-1}P'_j = \epsilon P_j \quad (j = 1, 2) \quad (7.1)$$

as equalities of rational endomorphisms. Here I_0 and I_t are conjugation by the fixed-multiplier polarized isogenies specified in Proposition 6.4.

Proof. Choose any place in this group, of residue characteristic p . Proposition 6.4 gives a common unprimed polarized reduction A_0 and its primed reduction A'_0 , a relative Frobenius $F : A_0 \rightarrow A'_0$ of multiplier p , and a polarized isogeny $\phi : A_0 \rightarrow A'_0$ of multiplier m_* which is simultaneously the specialization of the center and target isogenies. In particular, $p \nmid m_*$ and $p \neq \ell_1$. Set

$$\psi = \phi^{-1}F \in \text{End}(A_0) \otimes \mathbb{Q}, \quad \nu(\psi) = \frac{p}{m_*}, \quad g = \text{Ad}(\psi)|_{\mathcal{P}_{\mathbb{Q}_{\ell_1}}}. \quad (7.2)$$

Functoriality of specialization preserves (7.1). Moreover F conjugates each unprimed labeled endomorphism to its primed specialization, for both the center and target labels using this same F . Composing with ϕ^{-1} therefore shows that g acts by ϵ on each of the two specialized rational endomorphism planes.

Let H_0, H_t be these planes in $\text{End}(A_0) \otimes \mathbb{Q}$ and put $H = H_0 + H_t$. On the common integral Tate module, write

$$\begin{aligned} \Lambda &= \{D \in \text{End}_{\mathbb{Z}_{\ell_1}}(T_{\ell_1}A_0) : D^\dagger = D, \text{Tr}(D) = 0\}, \\ L_0 &= (H_0 \otimes \mathbb{Q}_{\ell_1}) \cap \Lambda, \quad L_t = (H_t \otimes \mathbb{Q}_{\ell_1}) \cap \Lambda. \end{aligned}$$

The two plane lattices here are individually saturated in Λ . Their reductions inject into $\Lambda/\ell_1\Lambda$ and, by Propositions 4.2 and 6.4, are independent planes whose sum is a degenerate four-dimensional subspace. In particular the two $\mathbb{Q}_{\ell_1}$ -planes have four-dimensional sum, and $\dim_{\mathbb{Q}} H = 4$. These statements use the saturated planes, regardless of whether the chosen small pairs $\mathbf{B}, \mathbf{P}$ are integral bases at ℓ_1 .

Every element of H is an actual rational endomorphism of A_0 , self-adjoint for its Rosati involution. The endomorphism trace is rational and agrees with its Tate-module trace [Mil86, Proposition 12.9]. Rosati positivity in arbitrary characteristic [Mil86, Theorem 17.3] gives

$$\text{Tr}(D^2) = \text{Tr}(DD^\dagger) > 0 \quad (0 \neq D \in H).$$

Thus the Gram matrix of a rational basis of H is positive definite over $\mathbb{R}$, with nonzero rational determinant. It remains nonsingular over $\mathbb{Q}_{\ell_1}$. Write $H_{\ell_1} = H \otimes \mathbb{Q}_{\ell_1}$. This rational nondegeneracy is compatible with the prescribed degeneracy of the reduced four-space: its Gram determinant may be divisible by ℓ_1 .

Conjugation by a symplectic similitude acts through SO on $\mathcal{P}$, with scalar kernel on the four-dimensional first homology space, by Proposition 3.2. Since H_{ℓ_1} is nondegenerate, g preserves its one-dimensional orthogonal complement. On H_{ℓ_1} its determinant is $\epsilon^4 = 1$; hence $\det g = 1$ makes its action on the orthogonal complement $+1$.

If $\epsilon = 1$, this proves $g = 1$ on all of $\mathcal{P}_{\mathbb{Q}_{\ell_1}}$. The scalar-kernel assertion makes the action of ψ on $T_{\ell_1} A_0 \otimes \mathbb{Q}_{\ell_1}$ a scalar c . This scalar is rational: the actual rational quasi-endomorphism ψ has rational trace and $4c = \text{Tr}(\psi)$, by [Mil86, Proposition 12.9]. Its symplectic multiplier is consequently c^2 . Equation (7.2) would give $c^2 = p/m_*$, impossible because the p -adic valuation of the right side is one.

Suppose instead that $\epsilon = -1$. Then $g^2 = 1$ and its minus eigenspace is exactly H_{ℓ_1} . Both F and ϕ have degrees prime to ℓ_1 : their degrees are p^2 and m_*^2 , respectively. They induce integral isomorphisms on the ℓ_1 -adic Tate modules. Consequently ψ , its inverse, and their conjugation action preserve Λ . The trace pairing on Λ is unimodular, as in Proposition 3.2 and the choice of ℓ_1 in Proposition 4.2. Since ℓ_1 is odd, the integral idempotents

$$e_+ = \frac{1+g}{2}, \quad e_- = \frac{1-g}{2}$$

give an orthogonal direct sum

$$\Lambda = \Lambda_+ \perp \Lambda_-, \quad \Lambda_{\pm} = e_{\pm} \Lambda, \quad \text{rank } \Lambda_- = 4. \quad (7.3)$$

Orthogonality follows from $\langle v_+, v_- \rangle = \langle gv_+, gv_- \rangle = -\langle v_+, v_- \rangle$. In bases of these two direct summands the Gram determinant of Λ is the product of their integral Gram determinants. It is a unit, so both determinants are units. In particular $\Lambda_-/\ell_1 \Lambda_-$ is nondegenerate.

Because $g = -1$ on each rational plane, L_0 and L_t lie in Λ_- . Their independent reductions have total dimension four. The direct-summand property in (7.3) identifies $\Lambda_-/\ell_1 \Lambda_-$ with its image in $\Lambda/\ell_1 \Lambda$. Those two reductions therefore exhaust this image. Their sum must be nondegenerate, contrary to its imposed degenerate incidence. This excludes the second sign as well. $\square$

7.3 The three equation types

Proposition 7.4 (Nonidentity on every algebraic branch). *For every group selected in Proposition 6.5, every irreducible closed algebraic W containing its target and having nonconstant a , and every normalized branch over the generic point of a component of $W \cap \{a = 0\}$, at least one of its equations is a nonzero germ on that branch. This holds for each of the three types (1), (2), and (3), with their base-coordinate equations included.*

Proof. Suppose all equations were identities on one such branch. Apply Lemma 7.1, and fix a general point $w \in W^\circ$ at which the indicated operator pairs are independent. All vectors and coordinates at w are now fixed while we vary a continuation loop. The continued first transport $M : \mathcal{P}_{s(w)} \rightarrow \mathcal{P}_{t_0}$ ranges through the orbit of an invertible initial transport under monodromy whose connected Zariski closure is the full special orthogonal group.

The single equation. Equation (1) says, for all $i, j \in \{1, 2\}$,

$$\langle B_i, MP_j(w) \rangle = r_{ij}.$$

The right side is fixed under continuation. Each B_i and $P_j(w)$ is nonzero, and the first connected monodromy is full. This contradicts Lemma 7.2(i).

The paired equation. Here the equation is

$$\langle B_i, MP_j(w) \rangle = \langle B'_i, M'P'_j(w) \rangle.$$

If the second curve projection is constant, its pullback local system has trivial monodromy, so varying the first connected monodromy already contradicts [Lemma 7.2\(i\)](#). If both projections are nonconstant, [Propositions 3.3](#) and [3.4](#) give full product monodromy unless the joint image lies in one of the fixed closed Hecke correspondences. In that exception its algebraic closure also contains the target pair in that correspondence: the coordinate projections are defined at the target and the correspondence is closed. This is excluded by the definition of a paired group in [Proposition 6.4](#). We therefore have product monodromy. The equations are polynomial in the two transport matrices and hence hold on its Zariski closure. Hold the second factor fixed and vary the first; the same nonconstant matrix coefficient gives a contradiction.

The exceptional equation. Put $C_i = I_0^{-1}B'_i$, which is a constant center vector. The coordinate pair $\hat{\mathbf{P}}$ equals $I_t^{-1}\mathbf{P}'$ at the target; away from that target it is simply a pair of algebraic operator coordinates. Equation (3) and full first monodromy give

$$\langle B_i, MP_j(w) \rangle = \langle C_i, M\hat{P}_j(w) \rangle \quad (i, j \in \{1, 2\})$$

for the entire monodromy orbit of M . By [Lemma 7.2\(ii\)](#),

$$B_i \otimes P_j = C_i \otimes \hat{P}_j \quad (i, j \in \{1, 2\}) \tag{7.4}$$

at a general point of W . In the algebraic frames these are algebraic identities in the operator coordinates. Thus they hold everywhere on W , in particular at z .

At the target all vectors in the displayed pairs are nonzero. The equality for $(i, j) = (1, 1)$ gives a scalar $\lambda \neq 0$ with $C_1 = \lambda B_1$ and $\hat{P}_1 = \lambda^{-1}P_1$. The equality for $(2, 1)$ then gives $C_2 = \lambda B_2$, and that for $(1, 2)$ gives $\hat{P}_2 = \lambda^{-1}P_2$. This is one common scalar for all four cross entries.

Conjugation by I_0 preserves the trace pairing. Moreover each fixed center norm $\langle B_i, B_i \rangle$ is a positive rational number, since B_i is an actual nonzero Rosati-self-adjoint rational endomorphism. Conjugation of the number field preserves this rational trace, so

$$\langle C_i, C_i \rangle = \langle B'_i, B'_i \rangle = \langle B_i, B_i \rangle \neq 0.$$

It follows that $\lambda^2 = 1$. Write $\epsilon = \lambda$; then $\lambda^{-1} = \epsilon$ also, and the target equalities are exactly (7.1) in de Rham realization.

They are equalities of actual rational endomorphisms. Indeed all the target vectors used here represent such endomorphisms, and de Rham realization in characteristic zero is faithful: after embedding the field in $\mathbb{C}$, comparison gives their action on first Betti homology; a homomorphism of complex abelian varieties acting trivially on that lattice is zero. Clearing rational denominators gives the same assertion for quasi-endomorphisms. We may therefore apply [Lemma 7.3](#), which excludes both values of ϵ . This finishes all three cases. $\square$

7.4 Uniform height conclusion

Theorem 7.5 (Height of the selected lifts). *There are constants $C > 0$ and $\kappa > 0$, depending only on the fixed curve, covers, center, and auxiliary data, such that every selected non-CM elliptic-square lift t satisfies*

$$h(t) \leq C \max\{2, [K(t) : K]\}^\kappa. \quad (7.5)$$

The height is the shifted ample height of Convention 5.1.

Proof. Put $d = \max\{2, [K(t) : K]\}$ and $h = h(t) \geq 2$. All fixed omitted points may be absorbed by increasing C . For the remaining targets, Proposition 6.5 either already gives a polynomial bound for h in d , or supplies a target $z = (a, U)$, a weighted group V , and common equations to which we apply Theorem 2.1.

We match its hypotheses explicitly. The target has $a \neq 0$ and $|a|_v < 1$ on V . The global height, proximity mass, and positive coordinate bounds in (6.10)–(6.11) give condition (i). The coefficient bounds and common tail estimate in (6.12)–(6.13), together with the degree bound in (6.11), give (ii). The expressions are polynomial combinations of fixed bounded degree of ordinary horizontal systems and regular algebraic germs, evaluated at a and, in the paired case, at aR ; this is (iii). Here $|R|_v = 1$ on the selected group. The base-coordinate equations are included in this system.

These estimates use the same $d = \max\{2, [K(t) : K]\}$ for the fixed field K . The selected labeled data and isogenies have a field of definition of degree polynomial in d . The normalized weighted sums, including embedding multiplicities, are preserved by every further field extension. The three equation types involve only finitely many fixed choices of ambient dimension, germs, conjugate centers, and center isogenies. We may therefore choose the constants in Theorem 2.1 uniformly across them.

Finally, Proposition 7.4 gives condition (iv) for every irreducible algebraic variety through z and every normalized boundary branch, including those arising in the dimension descent. Varieties with empty boundary are already handled in the proof of Theorem 2.1. That theorem gives $h \leq Cd^\kappa$; taking the largest constants over the fixed alternatives proves (7.5). $\square$

8 From heights to finiteness

We first descend the height estimate to an arithmetic estimate on the original coarse curve. We then identify the entire unpolarized elliptic-square locus with the locus to which the counting theorem applies. The two companion branch theorems enter only in the final proof of the full Zilber–Pink theorem.

8.1 The discriminant and the Galois orbit

For an algebraic point s with $\text{End}_{\overline{\mathbb{Q}}}(A_s) \otimes \mathbb{Q} \simeq \text{Mat}_2(\mathbb{Q})$, put

$$\mathcal{O}_s = \text{End}_{\overline{\mathbb{Q}}}(A_s), \quad D_s = \mathcal{O}_s \otimes \mathbb{Q}.$$

Throughout this section, the discriminant of this order means

$$\Delta_s = \left| \det \left(\text{Tr}_{D_s/\mathbb{Q}}(e_i e_j) \right)_{1 \leq i, j \leq 4} \right|, \quad (8.1)$$

where (e_i) is any integral basis of $\mathcal{O}_s$ and $\text{Tr}_{D_s/\mathbb{Q}}(x)$ is the trace of left multiplication by x on D_s . In particular, the discriminant is that of the full order, and includes its conductor. It is a positive integer independent of the basis.

Proposition 8.1 (Orbit estimate). *Let $C \subset \mathcal{A}_2$ be a curve as in Theorem 1.1. There are a number field K defining C and constants $c, \delta > 0$ such that every $s \in \Sigma_{E^2}(C)$ satisfies*

$$\#(\text{Aut}(\mathbb{C}/K) \cdot s) = [K(s) : K] \geq c\Delta_s^\delta. \quad (8.2)$$

The constants are uniform over the whole indicated locus.

Proof. If this locus is finite, the assertion follows by taking the minimum of finitely many positive ratios; the empty case is vacuous. Otherwise, make the fixed choices in Propositions 3.1 and 4.2. All the resulting covers, auxiliary primes, and the center are fixed. Enlarge the fixed number field K to contain their fields of definition. After removing a finite set $F \subset C(\mathbb{Q})$, their composite gives a finite cover $q : S \rightarrow C \setminus F$ of fixed degree N . Further fixed deletions in the preceding sections are included in F by removing their images. We can restrict to the locus over which the cover has constant degree, and include its remaining exceptional points in F as well.

For $s \in \Sigma_{E^2}(C) \setminus F$, choose the lift $t \in S(\mathbb{Q})$ with the two markings used in Theorem 7.5. Write $r_s = [K(s) : K]$ and $d_t = \max\{2, [K(t) : K]\}$. A closed point in a fiber of a finite map of degree N has residue degree at most N . Thus, for every such chosen lift,

$$[K(t) : K] \leq Nr_s, \quad d_t \leq 2Nr_s. \quad (8.3)$$

This bound uses the degree of the fixed cover even though the chosen point in its fiber depends on s .

The height bound in Theorem 7.5 and the full endomorphism estimate in Proposition 5.2 give fixed positive constants C_1, C_2, u, v such that

$$h(t) \leq C_1 d_t^u, \quad \Delta_s = \Delta_t \leq C_2 (d_t h(t))^v. \quad (8.4)$$

Here forgetting a level structure leaves the geometric principally polarized abelian variety unchanged, so it leaves its endomorphism ring unchanged. The second inequality is the Masser–Wüstholz estimate established in Proposition 5.2; in particular its constants have polynomial dependence on the varying field degree. It estimates the full geometric order after an extension of bounded relative degree, which has already been accounted for there.

For clarity, the trace conventions agree exactly in this case. For $D_s \simeq \text{Mat}_2(\mathbb{Q})$, its action on $H_1(A_s, \mathbb{Q})$ is a sum of two standard two-dimensional modules. Left multiplication on D_s also acts on its two columns as two standard modules. Consequently

$$\text{Tr}_{H_1}(x) = 2 \text{tr}(x) = \text{Tr}_{D_s/\mathbb{Q}}(x). \quad (8.5)$$

Thus (8.1) is precisely the regular-trace discriminant used in Proposition 5.2 and in [DO22, §§3.1 and 5.4]. Bounding a smaller lattice generated by the two symmetric endomorphisms alone would not be this assertion.

Combining (8.3) and (8.4), and increasing constants, yields

$$\Delta_s \leq C_3 r_s^{v(u+1)}.$$

Each point of the finite set $F \cap \Sigma_{E^2}(C)$ has a finite positive discriminant. Increasing C_3 includes these points too. Taking $\delta = 1/(v(u+1))$ and $c = C_3^{-\delta}$ proves the lower bound in (8.2).

Finally, for an algebraic point of a variety defined over the number field K , its distinct K -embeddings are precisely its Galois conjugates. Every embedding of its residue field into $\mathbb{C}$ extends to an automorphism of $\mathbb{C}$ over K . Hence their number is $[K(s) : K]$, proving the asserted orbit equality. $\square$

8.2 All polarizations and all elliptic-square curves

An E^2 -special curve will mean a special curve whose generic rational endomorphism algebra is $\text{Mat}_2(\mathbb{Q})$, as in [DO22, §6.2]. We verify the geometric and integral identifications needed to apply its theorem.

Proposition 8.2 (Coverage of the square locus). *An algebraic point $s \in \mathcal{A}_2(\overline{\mathbb{Q}})$ is isogenous over $\overline{\mathbb{Q}}$ to E^2 with E non-CM if and only if it is Hodge generic on an E^2 -special curve. This statement allows the given arbitrary principal polarization on A_s and an arbitrary unpolarized isogeny. At such a point, the complexity $\Delta(Z_s)$ of its special curve in [DO22, §6.3] equals Δ_s .*

Proof. Fix an embedding into $\mathbb{C}$, put $V = H_1(E, \mathbb{Q})$, and let $W = \mathbb{Q}^2$ with its trivial Hodge structure. The given unpolarized isogeny provides a rational Hodge isomorphism

$$H_1(A_s, \mathbb{Q}) \simeq V \otimes W. \quad (8.6)$$

Choose an elliptic polarization ψ_E on V . Since E is non-CM, $\text{End}_{\text{Hdg}}(V) = \mathbb{Q}$. The polarization ψ_E identifies V with its appropriately twisted dual, so every Hodge pairing of two copies of V into $\mathbb{Q}(1)$ is a rational multiple of ψ_E . Apply this to the four matrix entries, on the two copies of V , of the alternating form ψ_s defined by the actual polarization λ_s . There is a rational bilinear form b on W with

$$\psi_s = \psi_E \otimes b. \quad (8.7)$$

Alternation of ψ_s and ψ_E makes b symmetric. Let J be the complex structure on $V_{\mathbb{R}}$, choosing the common sign convention for which $\psi_E(v, Jv) > 0$ for $v \neq 0$. The positivity of ψ_s gives

$$0 < \psi_s(v \otimes w, Jv \otimes w) = \psi_E(v, Jv)b(w, w) \quad (v \neq 0, w \neq 0),$$

so b is positive definite over $\mathbb{R}$.

Let G be the image of $\text{GL}(V) = \text{GSp}(V, \psi_E)$ acting on $V \otimes W$ through the first factor. In dimension two, g multiplies ψ_E by $\det(g)$; therefore (8.7) gives a rational inclusion

$$G \subset \text{GSp}(V \otimes W, \psi_s).$$

The elliptic Hodge homomorphism, acting trivially on W , is the Hodge homomorphism of A_s . Its $G(\mathbb{R})$ -conjugacy class is the elliptic upper and lower half-plane datum. Positivity of b ensures that this class lies in the Siegel datum for ψ_s . Thus it gives a one-dimensional rational Shimura subdatum through the period point of A_s .

Retain the actual lattice $\Lambda = H_1(A_s, \mathbb{Z})$ inside $V \otimes W$. The principal polarization makes (Λ, ψ_s) unimodular; a symplectic basis identifies it with the standard integral Siegel lattice. Intersecting its adelic stabilizer with $G(\mathbb{A}_f)$ gives a compact open subgroup of $G(\mathbb{A}_f)$. The associated morphism of Shimura varieties therefore has a special curve as the image of the component through our point. This construction uses Λ itself: neither a product lattice nor a product principal polarization has been substituted for it. Varying the rational identifications and the lattices also allows every Hecke translate.

The Mumford–Tate group of a non-CM elliptic curve is $\text{GL}(V)$. One can see the relevant dichotomy directly: its connected Hodge group is a reductive subgroup of $\text{SL}(V)$, and a proper such subgroup containing the elliptic Hodge circle is a torus. That torus case has an imaginary quadratic commutant and is precisely CM. Therefore the point in (8.6) has Mumford–Tate group G , and is Hodge generic on the special curve just constructed. The commutant of G on $V \otimes W$ is $\text{id}_V \otimes \text{End}(W) \simeq \text{Mat}_2(\mathbb{Q})$, so this is an E^2 -special curve.

Conversely, the defining rational representation of an E^2 -special curve has this same two-copy elliptic form. At a Hodge generic point its commutant is $\text{Mat}_2(\mathbb{Q})$ and its elliptic factor is non-CM. Equivalently, the rational matrix idempotents split its abelian surface, up to isogeny, into two isogenous elliptic factors; the off-diagonal matrix units identify their rational Hodge structures. The equivalence between abelian varieties up to isogeny and polarizable rational Hodge structures of type $\{(-1, 0), (0, -1)\}$ makes this a complex isogeny [Mil95, Theorem 4.1]. For an algebraic point, all these homomorphisms and abelian subvarieties descend to $\overline{\mathbb{Q}}$: each is defined over a finitely generated extension of $\overline{\mathbb{Q}}$, which is regular, and [Mil86, Corollary 20.4] applies. Thus the isogeny and the elliptic factors can be taken over $\overline{\mathbb{Q}}$.

It remains to check the order, rather than only its rational algebra. For the rational subgroup G defining the curve, its integral commutant is

$$R = \text{End}_G(H_1(A_s, \mathbb{Q})) \cap \text{End}_{\mathbb{Z}}(\Lambda). \quad (8.8)$$

At a Hodge generic point, commuting with G is equivalent to being a Hodge endomorphism. To check the integral refinement, let $u \in R$. The rational Hodge equivalence writes $u = f/n$ for an endomorphism f of A_s and a positive integer n . The inclusion $u\Lambda \subset \Lambda$ means $f\Lambda \subset n\Lambda$. Under $A_s[n](\mathbb{C}) \simeq n^{-1}\Lambda/\Lambda$, it follows that f annihilates $A_s[n]$. Factoring f through the quotient by $A_s[n]$, namely multiplication by n on A_s , makes u an actual endomorphism. Conversely every abelian-variety endomorphism preserves the integral Hodge structure. The same descent argument identifies its complex and geometric endomorphism rings. It follows that $R = \mathcal{O}_s$. Daw–Orr define $\Delta(Z_s)$ as the absolute discriminant of this integral commutant [DO22, §6.3]; (8.5) proves $\Delta(Z_s) = \Delta_s$. $\square$

8.3 The counting implication

We use the following precise E^2 case of the published conditional finiteness theorem [DO22, Theorem 1.3, Conjecture 6.2, and §§6.5–6.6]. Let $C \subset \mathcal{A}_2$ be an irreducible Hodge-generic algebraic curve. Let $\Sigma_{E^2}^{\text{DO}}$ be the points Hodge generic on E^2 -special curves. If, over a finitely generated field $L \subset \mathbb{C}$ defining C , there are $c_0, \delta_0 > 0$ such that

$$\#(\text{Aut}(\mathbb{C}/L) \cdot s) \geq c_0 \Delta(Z_s)^{\delta_0} \quad (s \in C \cap \Sigma_{E^2}^{\text{DO}}), \quad (8.9)$$

then $C \cap \Sigma_{E^2}^{\text{DO}}$ is finite. Here Z_s is the unique special curve containing the non-special point s . The constants may depend on C and L , and no boundary condition occurs in this implication.

Proof of Theorem 1.1. Every special curve is defined over $\overline{\mathbb{Q}}$. Since the closed Hodge-generic curve C is not contained in any such curve, its intersection with each one is a zero-dimensional algebraic set over $\overline{\mathbb{Q}}$. Consequently every point of $C(\mathbb{C}) \cap \Sigma_{E^2}^{\text{DO}}$ is algebraic. By Proposition 8.2, this intersection is exactly $\Sigma_{E^2}(C)$, and its complexity is Δ_s . Proposition 8.1 supplies (8.9) with $L = K$, a number field and therefore finitely generated. The stated counting implication proves that $\Sigma_{E^2}(C)$ is finite. In particular, in the infinite case used to choose the fixed center, it gives the required contradiction. No companion branch theorem is used. $\square$

8.4 The special-curve classification

The passage from the three branch statements to the full conjecture requires a classification of the special curves, including their Hecke translates. We record the precise consequence of the classification of Hodge groups of abelian surfaces by Moonen and Zarhin [MZ99,

Theorem (0.1)(iv), §2 and Corollary (3.9)]. The resulting list of special subvarieties is also given explicitly in [DO21, §2.F, Table 1].

Proposition 8.3 (The three special-curve types). *Let $Z \subset \mathcal{A}_2$ be a special curve and let $s \in Z(\overline{\mathbb{Q}})$ be Hodge generic on Z . Exactly one of the following holds:*

- (i) A_s is isogenous to $E \times F$, where E is non-CM and F is CM;
- (ii) A_s is isogenous to E^2 , where E is non-CM;
- (iii) the full algebra $\text{End}^0(A_s)$ is an indefinite quaternion division algebra over $\mathbb{Q}$.

Every point of a special curve that is not Hodge generic on it is a special point.

Proof. The Hodge group is the smallest connected $\mathbb{Q}$ -algebraic subgroup whose real points contain the Hodge circle; adjoining the weight homotheties gives the Mumford–Tate group. At a Hodge-generic point of a special subvariety, its associated Hermitian domain has the dimension of that subvariety. We apply this with dimension one, retaining the actual rational Hodge structure and polarization of A_s .

Suppose first that A_s is simple up to isogeny. The surface classification in [MZ99, §2(2.2)] gives four possibilities for its full rational endomorphism algebra: $\mathbb{Q}$, a real quadratic field, an indefinite quaternion division algebra, or a quartic CM field. The corresponding Hodge groups are respectively Sp_4 , a restriction of scalars of SL_2 from the real quadratic field, the norm-one quaternion group, and a torus. Their Hermitian domains have dimensions 3, 2, 1, 0: respectively the genus-two Siegel space, two copies of the upper half-plane, one upper half-plane, and a point. Thus only the quaternionic case can occur on a special curve at a Hodge-generic point. This gives (iii) as an equality of the full endomorphism algebra.

If A_s is not simple, Poincaré reducibility gives an isogeny $A_s \sim E_1 \times E_2$. When the two elliptic factors are isogenous, their product has the same Hodge group as either factor [MZ99, §1(1.2)]. For a non-CM factor this group is SL_2 , with one-dimensional domain, whereas a CM factor has a torus Hodge group and a zero-dimensional domain. This gives (ii). When the factors are nonisogenous, their Hodge group is the product of the two elliptic Hodge groups [MZ99, Corollary (3.9)]. Each non-CM factor contributes one domain dimension and each CM factor contributes zero. Dimension one therefore means precisely one CM factor, giving (i). The alternatives are disjoint, either by simplicity or by the full endomorphism algebras $\mathbb{Q} \times F$ and $M_2(\mathbb{Q})$.

These assertions are invariant under rational Hodge isomorphism, so they apply to every polarization and Hecke translate. The elliptic factors and isogenies at an algebraic point can be defined over $\overline{\mathbb{Q}}$, by the descent explained in Proposition 8.2. Finally the special closure of any point on Z is a special subvariety contained in Z . If it is proper, it has dimension zero and is a special point. This proves the last assertion. $\square$

8.5 The full Zilber–Pink theorem

Proof of Theorem 1.3. First the set of special points on C is finite. These are the CM points of $\mathcal{A}_2$. If there were infinitely many, their Zariski closure would be all of the irreducible curve C , since a proper closed subset of an irreducible algebraic curve is finite. The André–Oort theorem for the coarse moduli space of principally polarized abelian surfaces would then make C special [PT13, Theorem 1.1]. This contradicts its Hodge genericity.

By Proposition 8.3, after removing the already finite set of special points, every element of $\Sigma(C)$ belongs to one of the three indicated loci. The classification applies to all Hecke translates and to the original coarse moduli space.

Theorem 1.1 of the $E \times \text{CM}$ companion [Ope26b], restated as Theorem 1.2(A), gives finiteness of the entire first locus, with arbitrary CM fields, orders, and unpolarized isogenies. Theorem 1.1, proved above, gives finiteness of the second locus. Theorem 1.1 of the quaternionic companion [Ope26a], restated as Theorem 1.2(B), gives finiteness of the third locus: its full endomorphism algebra is an indefinite quaternion division algebra over $\mathbb{Q}$, exactly as the companion theorem requires. The union of these three finite sets and the finite set of special points contains $\Sigma(C)$, which is therefore finite.

All these conclusions concern the original coarse curve. The normalizations and fixed covers used for the second locus have already been descended in Proposition 8.1, and every fixed omission has been restored there. A singular original curve has only finitely many points removed in this process. The argument is also valid when C is complete, because none of these steps requires a point of its closure on a compactification boundary. $\square$

The special-closure formulation of Pink’s conjecture [Pin05, Conjecture 1.2] follows for every curve over $\overline{\mathbb{Q}}$.

Corollary 8.4 (The special-closure formulation). *Let $C \subset \mathcal{A}_2$ be a nonempty reduced irreducible closed algebraic curve over $\overline{\mathbb{Q}}$, and let S_C be the smallest Shimura-special subvariety containing C . Then*

$$C(\overline{\mathbb{Q}}) \cap \bigcup_{\substack{Z \subset \mathcal{A}_2 \text{ special} \\ \dim Z < \dim S_C - 1}} Z(\overline{\mathbb{Q}})$$

is finite.

Proof. Write $d = \dim S_C \in \{1, 2, 3\}$. If $d = 3$, then C is Hodge generic in $\mathcal{A}_2$, and the conclusion is Theorem 1.3. If $d = 2$, the union consists only of special points. Infinitely many such points on C would be Zariski dense, so Andr -Oort [PT13, Theorem 1.1] would make C special, contradicting $\dim S_C = 2$. Finally, if $d = 1$, the union is empty. $\square$

A Proof of the interpolation theorem

We prove Theorem 2.1 by the graph construction explained in Section 2. The algebraic bounds below control every new variety and every ideal-membership certificate. An ordinary differential estimate supplies the pole along each normalized boundary divisor; layered interpolation on its graph then gives the polynomial used in the product-formula descent.

A.1 Effective algebra with absolute heights

We use polynomial bounds in a fixed number of variables. More precisely, in the lemmas below each assertion of such a bound means an inequality $c(1+D)^c$ or $c(1+D+T)^c(1+H)$; the constant and exponent depend only on the fixed number of variables. After a fixed number of operations these are still polynomial bounds. Heights are logarithmic throughout.

Lemma A.1 (Algebraic degree and height bounds). *Fix M . Suppose at most D polynomials in M variables have degree at most $D \geq 2$ and affine coefficient height at most H .*

- (a) *Their ideal has a degree-compatible Gr bner basis of polynomially bounded degree and coefficient height at most $\text{poly}_M(D)(1+H)$. Each basis element has a representation in the original generators whose degree and coefficient height satisfy the same types of bounds. Normal-form reduction can be chosen linear. For an input of degree T , its representation by the remainder and the original generators has degree at most $T + \Delta$, where $\Delta = \text{poly}_M(D)$ is independent of T .*

- (b) *If a polynomial of degree T belongs to the ideal, it has a certificate of membership of degree polynomial in $D + T$. The coefficient height of a certificate can be bounded polynomially in $D + T$ times one plus the heights of the given polynomial and generators.*
- (c) *Every irreducible component of the common zero set has degree polynomial in D , and admits set-theoretic defining equations whose number and degrees are polynomial in D and whose affine coefficient heights are at most $\text{poly}_M(D)(1 + H)$.*

All bounds are absolute and are unchanged by enlarging the number field. The ideals in (a) and (b) need not be radical.

Proof. The fixed-variable degree inputs are the affine Gröbner basis degree bound of [Dub90, Corollary 8.3] and the polynomial ideal-membership degree bound of [Asc04, Theorem 3.4], which gives a proof of the Hermann–Seidenberg method. Their degree bounds become polynomials in D when M is fixed. For example, using one additional homogenizing variable, a larger permissible bound for the first input is

$$\Gamma_M(D) = \left\lceil 2(D^2/2 + D)^{2^M} \right\rceil.$$

Only this polynomial character is needed here.

We spell out the height consequence, since it must not introduce the degree or discriminant of a field of definition. A polynomial identity of a prescribed degree becomes a linear system by equating coefficients. There are at most $\binom{T+M}{M}$ monomials of degree at most T . After the degree bounds just quoted, the number of entries of every such system is polynomial in $D + T$. For a matrix with u entries, each of height at most H_1 , the affine height of the full matrix is at most uH_1 . A determinant of size r then has height at most $ruH_1 + \log(r!)$: this follows place by place from the determinant expansion, using the ultrametric inequality at finite places. Choosing a nonzero pivot minor and applying Cramer’s rule gives solution and kernel bases of height polynomial in the matrix size times $1 + H_1$. The product formula for the pivot denominator proves these assertions with normalized absolute heights. There is no factor $[F : \mathbb{Q}]$.

The reduced Gröbner basis can be recovered by these linear systems: each required polynomial has bounded degree, and membership in the generator ideal can be imposed with the bounded-degree certificates. Leading coefficients and the coefficients of forbidden leading monomials are fixed by linear equations. This gives the stated height bounds for the basis and its representations. Fix those representations. With a degree-compatible order, division of a monomial of degree at most T uses multiples of basis elements of degree at most T . Replacing a basis element by its fixed representation increases this degree by at most a fixed additive amount Δ . Reducing each monomial and extending linearly gives the required linear reduction map. This proves (a); the same coefficient argument and the membership degree bound prove (b).

For (c), pass to projective closures. Follow a component of interest by proper hyper-surface cuts, starting with projective space. Whenever the current component is larger than the desired one, some given defining polynomial is nonzero on the current component; cut by it and choose a component still containing the desired one. There are at most M proper cuts. Geometric Bézout bounds the degrees at each step. For heights, apply the global intersection bound [KPS01, Corollary 2.11] to the affine parts of these cuts. It bounds the height of each affine intersection by a polynomial in the degrees times one plus the preceding height and the coefficient heights of the cutting polynomials. The height of an affine variety in that result is defined through its projective closure [KPS01, §1.2.4]. This absolute Chow height is nonnegative and additive on components, so it bounds each

retained component separately. Its comparison with the projective coefficient height of a Chow form costs a polynomial in the degree; this follows also by comparing the coefficient norm with the logarithmic integral norm in the definition of Chow height. Components supported at infinity do not enter these affine intersections.

Here is an explicit way to return to equations. Let X be the projective closure of a component, of dimension r and degree δ , and let Φ_X be its Chow form. In each of its $r + 1$ hyperplane groups substitute the coefficients of a hyperplane through $[1 : x]$:

$$(u_{i0}, u_{i1}, \dots, u_{iM}) = \left(-\sum_{j=1}^M u_{ij}x_j, u_{i1}, \dots, u_{iM} \right).$$

Set every coefficient in the remaining u_{ij} equal to zero. These are polynomials in x , of degree at most $(r + 1)\delta$, with polynomially many coefficients. They all vanish if $[1 : x] \in X$. If $[1 : x] \notin X$, projection from this point has image of dimension at most r ; $r + 1$ general hyperplanes through the point have no common intersection with X . The substituted Chow form is then not the zero polynomial. Thus the coefficient equations define exactly the affine component. Normalize a nonzero coefficient of Φ_X to one. Its affine coefficient height is its projective coefficient height by the product formula, and the indicated substitutions increase height by a polynomial in δ . This proves the equation-height and equation-count assertions, including after a field extension needed to define the component. $\square$

A.2 A uniform boundary multiplicity bound

The next lemma supplies a truncation $F = E_{<m}$ for which F/a^m has a pole along every normalized boundary divisor. This is the input to [Lemma A.3](#); an identity or nonidentity merely on the divisor would not determine that pole. Only degrees enter the estimate. In particular, the coefficients of the transverse slices used in its proof need no height bound. The differential method uses the span of successive derivatives; compare [\[BB85, Theorem 1\]](#). We prove the version needed here because the algebraic curve carrying the pulled-back system varies during the dimension descent.

Lemma A.2 (Ordinary differential multiplicity). *Use the fixed function data of [Theorem 2.1](#). There is a polynomial $M_0(D)$, depending only on those data and N , with the following property. Let $W \subseteq \mathbb{A}^N$ be irreducible of degree at most D , with a nonconstant, and suppose its normalized boundary branches satisfy condition (iv). There are integers $0 \leq c_\alpha \leq M_0(D)$ and an integer $1 \leq m \leq M_0(D)$ such that, for*

$$E = \sum_{\alpha=1}^s c_\alpha E_\alpha = \sum_{j \geq 0} a^j e_j(U), \quad E_{<m} = \sum_{j=0}^{m-1} a^j e_j(U),$$

every prime divisor D_0 of the normalization over $W \cap \{a = 0\}$ satisfies

$$\text{ord}_{D_0} E_{<m} < m \text{ord}_{D_0} a. \tag{A.1}$$

The constants do not depend on the coefficient sizes of W or of the polynomial combinations defining E_α .

Proof. Put $k = \dim W$ and let $\nu : \widetilde{W} \rightarrow W$ be its finite normalization. Since a is a nonzero nonunit whenever the boundary is nonempty, its zero set is pure of codimension one. Finiteness of ν implies that every divisor of a upstairs maps onto a component of this zero set. At a general point of such a divisor D_0 , both $\widetilde{W}$ and D_0 are smooth, and the map on

D_0 has rank $k - 1$. Indeed the induced extension of function fields is finite and separable in characteristic zero.

Choose $k - 1$ general affine hyperplanes in the original ambient space. Their pullbacks cut smooth curve germs transverse to D_0 at its general intersection points. They can also be chosen to avoid the zero set of the first nonzero transverse coefficient of any nonzero E_α . To see the latter assertion explicitly, use local coordinates (t, s) with $D_0 = \{t = 0\}$. A nonzero germ has the expansion $t^r u(s) + O(t^{r+1})$ with u nonzero on an open subset of D_0 . A transverse curve through a point of that open subset preserves its order r . Infinite order would instead mean that all transverse coefficients vanish on a local open set; analyticity would make the germ identically zero. Thus the hypothesis supplies a finite first order on each branch to be tested. The excluded set may be taken Zariski closed: once this finite order is known, choose a truncation longer than its ratio to $\text{ord}_{D_0} a$. That algebraic truncation has the same leading transverse coefficient. Its nonvanishing is the nonvanishing of an algebraic section of a power of the conormal line bundle on a smooth open subset of D_0 . This justifies using general algebraic hyperplanes to make all these choices simultaneously.

The sliced curve has degree at most D , after removing components that do not meet the chosen open sets. The bound on the number of branches can be seen directly on the finite normalization $\bar{\nu}$ of the projective closure of W . Put $L = \bar{\nu}^* \mathcal{O}(1)$. The homogeneous coordinate defining $a = 0$ is a nonzero section of L ; if its divisor is $\sum_T m_T T$, then

$$\deg W = L^k = \sum_T m_T [\bar{\mathbb{Q}}(T) : \bar{\mathbb{Q}}(\bar{\nu}(T))] \deg \bar{\nu}(T).$$

Each term is a positive integer. Every affine normalized divisor under consideration occurs in this sum, so there are at most D of them. Equivalently the transverse curve branches are counted with their positive hyperplane-intersection multiplicities. This also explains why no bound for equations of $\widetilde{W}$ is needed.

On a tested curve, adjoin the algebraic curve coordinates implicit in the fixed germs. A parameter on each fixed curve gives a fixed finite map of its smooth projective completion to $\mathbb{P}^1$. Pulling these maps back by the functions a or aR and taking the component of the specified germ increases degree by a fixed factor. Repeating this for the fixed finite list still gives an algebraic curve of degree polynomial in D in a fixed product of projective spaces. A fixed Segre embedding and normalization produce a smooth projective curve X with genus polynomial in D . For example a general plane projection bounds its genus by the arithmetic genus of a plane curve of polynomial degree. The degrees of a , R , the original coordinate functions, and each adjoined coordinate on X are polynomial in D . The chosen points of X lie over the specified ordinary centers. Ramification in these maps does not destroy regularity of the pulled back connections. If an argument is constant, its germ contributes a constant solution vector instead.

Direct sums, tensor products, and duals encode every degree- b polynomial combination of the fixed germs as

$$f = r y, \tag{A.2}$$

where r is a rational row on X and y is a horizontal analytic vector for one algebraic connection of rank at most a fixed integer r_* . The coordinates and the target-dependent scalar coefficients occur in r . The integer r_* is determined before W, d, h are chosen: it is the sum of the ranks of the finitely many tensor and dual systems needed for all monomials of degree at most b . Choose the tensor rational frame obtained from the specified fixed frames. It is regular and invertible at the tested point, and the connection is ordinary there. The polar degrees of its rational connection matrix and of r are polynomial in D , since all constituent rational maps and functions have those degree bounds.

We give the zero estimate for (A.2). Let $\delta = d/da$ on the function field of X . If g is rational, then

$$(\delta g)_\infty \leq (g)_\infty + ((g)_\infty)_{\text{red}} + (da)_0, \quad \deg(da)_0 \leq 2g(X) + 2\deg(a).$$

The first inequality follows by writing $\delta g = dg/da$ in a local parameter; the second uses the canonical divisor degree and the bound $\deg(da)_\infty \leq 2\deg(a)$. Write the horizontal equation as $\delta y = By$ and define derivative rows

$$r_0 = r, \quad r_{j+1} = \delta r_j + r_j B.$$

Then $r_j y = \delta^j f$. The entries of B have a common polar divisor of polynomial degree: divide the rational connection one-form by da , whose zero divisor has just been bounded. If P_j bounds the poles of the entries of r_j and Q bounds the poles of B , then

$$P_{j+1} = 2P_j + (da)_0 + Q$$

is a valid common bound. For $j \leq r_*$ all these divisors therefore have polynomial degree, independently of coefficient sizes.

Let s' be the first index for which $r_{s'}$ lies in the function-field span of $r_0, \dots, r_{s'-1}$. Unless $r = 0$, we have $1 \leq s' \leq r_*$. Differentiating this dependence shows that the span is stable under covariant differentiation; all subsequent rows lie in it. Let K_η be its common kernel. At the tested point take the local ring of convergent germs $\mathcal{O} = \mathbb{C}\{t\}$, the ordinary frame $\mathcal{O}^{r'_*}$ of the actual rank $r'_* \leq r_*$, and the saturation

$$K = (K_\eta \otimes_{\overline{\mathbb{Q}}(X)} \text{Frac } \mathcal{O}) \cap \mathcal{O}^{r'_*}.$$

This is a direct summand, since the quotient is torsion free over a discrete valuation ring. It is preserved by the regular connection with respect to d/dt : generic preservation follows from that for δ , and the derivative of a regular section is regular. Thus the quotient has an ordinary connection. If f is nonzero, the projection of y to the quotient is nonzero and cannot vanish at $t = 0$, by uniqueness for an ordinary linear differential equation. It is therefore primitive in the quotient lattice.

This primitivity can be used in the original frame, without a height or pole bound on any quotient frame. Let A be the matrix with rows $r_0, \dots, r_{s'-1}$, and let P be the degree of a common polar divisor for its entries. Its entries have order at least $-P$ at $t = 0$. Some $s' \times s'$ minor Δ is nonzero, with polar degree at most $s'P$, and hence with $\text{ord}_0 \Delta \leq s'P$. Smith normal form over $\mathcal{O}$, after clearing the entry poles, has nonzero exponents $\sigma_1 \leq \dots \leq \sigma_{s'}$ satisfying

$$\sigma_i \geq -P, \quad \sum_{i=1}^{s'} \sigma_i = \min_{\Delta' \neq 0} \text{ord}_0 \Delta' \leq s'P.$$

Consequently $\sigma_{s'} \leq (2s' - 1)P$. The invertible row and column matrices in Smith normal form preserve the local lattices and their maximal ideals. The saturated kernel corresponds to the zero columns, so primitivity modulo K means that some coordinate in a nonzero column is a unit. It follows that

$$\min_{0 \leq j < s'} \text{ord}_0(\delta^j f) \leq (2s' - 1)P. \quad (\text{A.3})$$

If $e = \text{ord}_0 a \geq 1$ and $n_0 = \text{ord}_0 f$, local differentiation gives

$$\text{ord}_0(\delta^j f) \geq n_0 - je.$$

Cancellation of leading terms only increases the order, so the inequality also covers that case. Combining it with (A.3) bounds n_0/e by a polynomial in D . Finite pullback multiplies both orders by its ramification index; the same bound holds on the original normalized divisor. The transverse slice was chosen to preserve its first nonzero coefficient, so this proves the desired order bound for each nonzero original expression on each divisor.

For each divisor take the least order among the nonzero E_α . Cancellation at that order in $\sum c_\alpha E_\alpha$ is a proper linear condition on $(c_1, \dots, c_s)$, with coefficients in a residue function field. There are at most D such conditions. A proper hyperplane meets the grid $\{0, \dots, D\}^s$ in at most $(D+1)^{s-1}$ points: fix all coordinates except one whose coefficient is nonzero. The union of the forbidden hyperplanes cannot fill the grid. Choose a vector outside it. The resulting E has the stated polynomial order bound on every divisor. Choose an integer m larger than all the bounded order ratios. All coordinate functions are regular at these divisors, and $E - E_{<m} = a^m \sum_{j \geq m} a^{j-m} e_j(U)$ is an analytic multiple of a^m . Its order is at least $m \operatorname{ord} a$. The smaller order of E therefore equals that of $E_{<m}$, proving (A.1) and the lemma. $\square$

A.3 The graph and its projected boundary

Lemma A.3 (Graph boundary). *Let W be an irreducible affine variety of dimension $k \geq 1$, and let a be a nonconstant regular coordinate. Suppose a polynomial F and an integer $m \geq 1$ satisfy*

$$\operatorname{ord}_D F < m \operatorname{ord}_D a$$

on every normalized divisor above $W \cap \{a = 0\}$. Let W' be the affine closure of the graph

$$q = g := a^{-m} F \quad \text{over } W \setminus \{a = 0\},$$

and let $\pi : W' \rightarrow W$ be the projection. Then

$$\dim \pi(\overline{W' \cap \{a = 0\}}) \leq k - 2. \quad (\text{A.4})$$

In particular, for any polynomial $f(U)$,

$$\dim(W' \cap \{a = 0, q + f(U) = 0\}) \leq k - 2. \quad (\text{A.5})$$

When $k = 1$ these sets are empty. Degrees and heights of W' admit the bounds of Lemma A.1 in terms of defining equations of W , F , and m .

Proof. The graph is irreducible of dimension k . Over $a \neq 0$ it is the whole zero set of $a^m q - F$ in $W \times \mathbb{A}^1$, and its closure is an irreducible component of that zero set. This proves the complexity assertion by Lemma A.1.

Let $\nu : \widetilde{W} \rightarrow W$ be the finite normalization and let $\widetilde{\Gamma}$ be the closure of the same rational graph in $\widetilde{W} \times \mathbb{A}^1$. Its map to $W \times \mathbb{A}^1$ is finite, so its image is closed. That image equals W' : it contains the dense original graph and is contained in its closure. At the generic point of each normalized boundary divisor D , the reciprocal g^{-1} is regular and vanishes, because g has a pole. On a neighborhood of that point, the upstairs graph closure satisfies

$$qg^{-1} = 1.$$

It therefore has no point above a dense open subset of D .

Fix a boundary component $B \subset W$. All points of the finite normalization above its generic point lie on the finitely many divisors dominating B . For each of these divisors remove the proper closed subset not covered by the neighborhood just used. The images of

the removed subsets are closed under the finite map and have dimension at most $k - 2$. Outside their union, and outside intersections with the other boundary components, no upstairs graph point lies over B . The finite surjection $\tilde{\Gamma} \rightarrow W'$ then gives the same exclusion downstairs. Doing this for every B proves (A.4).

On $q + f(U) = 0$ the projection has inverse $(a, U) \mapsto (a, U, -f(U))$. Thus the set in (A.5) is isomorphic to a closed subvariety of the projected boundary and has its asserted dimension. This step fixes the graph coordinate; no bound on the dimension of $W' \cap \{a = 0\}$ itself was used. $\square$

A.4 The auxiliary polynomial in the actual ideal

The following construction keeps a high power of a in the membership certificate. Working with the actual defining ideal is essential for retaining that power when the certificate is evaluated.

Lemma A.4 (Layered interpolation). *Fix M . In $\overline{\mathbb{Q}}[a, U, q]$, with M variables in total, let $I = (I_1, \dots, I_r)$ be generated by polynomials of controlled number, degree, and height. Suppose its zero set W' is irreducible of dimension $k \geq 1$. Let*

$$f_0 = q + e_m(U), \quad J = (I, a, f_0), \quad F_n = f_0 + \sum_{j=1}^{n-1} a^j e_{m+j}(U),$$

where $\deg e_{m+j} \leq C_0(1 + m + j)$ and $\dim V(J) \leq k - 2$, interpreted as emptiness for $k = 1$. Let $D \geq 2$ bound r, m , the generator degrees, and C_0 . There are constants B, B' , polynomial in D for fixed M , such that the following holds. Whenever $n, L \geq 1$ and

$$B'n(1 + L + Bn + B)^{k-2} < \binom{L+k}{k}, \quad (\text{A.6})$$

there is a polynomial P with $\deg P \leq L$, nonzero on W' , and

$$P \in (I, a^n, F_n). \quad (\text{A.7})$$

For $k = 1$ the left side of (A.6) is replaced by zero. Both P and a certificate for (A.7) can have degree polynomial in $D + L + n$ and coefficient height polynomial in $D + L + n$ times one plus the coefficient heights of $I_1, \dots, I_r, e_m, \dots, e_{m+n-1}$.

In particular, for any prescribed positive integer c , one can take $n = cL$ with L polynomial in $D + c$.

Proof. Fix a degree-compatible Gröbner basis of J and bounded-degree representations of its elements in $I_1, \dots, I_r, a, f_0$ using Lemma A.1. Reducing each monomial and extending linearly gives fixed linear maps, with

$$D_1 = \text{NF}_J(D_1) + \sum_{\rho=1}^r I_\rho A_\rho(D_1) + aA(D_1) + f_0H(D_1). \quad (\text{A.8})$$

If $\deg D_1 \leq T$, every coefficient polynomial on the right has degree at most $T + \Delta$, where Δ is polynomial in D and independent of T . The normal form has degree at most T .

Start with a general polynomial P of degree at most L . Set the residual in layer zero to $D_0 = P$ and the other residuals to zero. For $i = 0, \dots, n - 1$, impose the linear condition

$$\text{NF}_J(D_i) = 0.$$

Use (A.8) for D_i . Record $a^i \sum I_\rho A_\rho(D_i)$ in the certificate. The term $a^{i+1}A(D_i)$ is added to the residual in layer $i + 1$, or recorded as a multiple of a^n if $i + 1 = n$. In the other term use the exact identity

$$a^i f_0 H(D_i) = a^i F_n H(D_i) - \sum_{j=1}^{n-1} a^{i+j} e_{m+j}(U) H(D_i).$$

Record the first summand as a multiple of F_n . For $i + j < n$ add $-e_{m+j}H(D_i)$ to layer $i + j$; for $i + j \geq n$ record the term as a multiple of a^n . After the last layer no residual remains, so these conditions imply the actual membership (A.7). This procedure never takes a radical or raises P to a power.

All the residuals and conditions depend linearly on the coefficients of P . Choose B so large that

$$\Delta + C_0(1 + m + j) \leq Bj \quad (j \geq 1), \quad \Delta \leq B.$$

Such B is polynomial in D . A contribution advancing by j layers then increases residual degree by at most Bj ; advancement using the generator a has the same bound with $j = 1$. Induction gives

$$\deg D_i \leq L + Bi \quad (0 \leq i < n). \quad (\text{A.9})$$

Every recorded term in the certificate also has degree polynomial in $D + L + n$, including the terms advanced past the last layer.

To count the conditions, let G be a polynomial bound for the degrees of generators of the initial monomial ideal of J , and put $r_0 = \dim V(J)$. In a surviving monomial, let S be the set of variables whose exponents are at least G . No generator of the initial ideal can be supported in S , since it would then divide the monomial. The coordinate space in the variables S is therefore contained in the zero set of that monomial ideal, so $|S| \leq r_0$. There are at most 2^M choices of S , at most G^M choices for the remaining bounded exponents, and at most $(1 + T)^{r_0}$ choices of total degree at most T in the unbounded exponents. Hence the space of normal forms of degree at most T has dimension at most

$$2^M G^M (1 + T)^{r_0} \leq B'(1 + T)^{k-2}.$$

The equality of the dimensions of an ideal and its initial ideal follows from equality of their degree filtrations under a degree-compatible order. This count applies to nonradical ideals as well. If J is the unit ideal its normal-form space is zero. Combining the count with (A.9), the entire construction imposes at most the left side of (A.6) independent linear conditions.

On the other hand, the restrictions of degree- $\leq L$ polynomials to W' span a space of dimension at least $\binom{L+k}{k}$. Indeed, k generic linear coordinates are algebraically independent on W' ; their monomials of total degree at most L are linearly independent. If every polynomial satisfying the layer conditions vanished on W' , the codimension of the solution space in the degree- $\leq L$ polynomial space would be at least this number. That contradicts (A.6). A polynomial nonzero on W' satisfying (A.7) therefore exists.

For the asserted height control, let T be the polynomial certificate degree bound just proved. Introduce unknown coefficients for P of degree at most L and for the certificate polynomials of degree at most T , and equate coefficients in

$$P = \sum_{\rho} B_{\rho} I_{\rho} + B_a a^n + B_f F_n.$$

This is a homogeneous linear system of polynomial size in $D + L + n$. The height of its coefficient matrix is polynomial in these quantities times the stated input heights. The

minors argument in [Lemma A.1](#) supplies a basis of solutions with the claimed height bound. At least one basis vector has its P component nonzero on W' ; otherwise every solution would vanish there, contradicting the existence already established.

Finally set $n = cL$. For $k \geq 2$ and $L \geq 1$, the left side of (A.6) is at most

$$B'c(2 + Bc + B)^{k-2}L^{k-1},$$

whereas the right side is at least $L^k/k!$. Thus it suffices to choose the integer

$$L > k!B'c(2 + Bc + B)^{k-2}.$$

This is polynomial in $D + c$, because $k \leq M$ is fixed. For $k = 1$ there are no conditions and any $L \geq 1$ suffices. This proves all assertions. $\square$

A.5 The product formula and dimension descent

Proof of Theorem 2.1. All implicit constants in this proof depend on the fixed data in the theorem. We keep an irreducible closed $W \ni z$ whose defining equation count and degrees are bounded by a polynomial in d and whose defining coefficient heights are bounded by

$$D_i(d)(1 + \log h)^{B_i}, \quad (\text{A.10})$$

where D_i is a fixed polynomial and B_i is a fixed nonnegative integer for the current descent step i . Its degree has a polynomial bound as well. Initially $W = \mathbb{A}^N$ and these assertions hold. At most N strict dimension drops will be made. Consequently enlarging the polynomials and exponents at each step still yields constants depending only on the initial fixed data.

First suppose $W \cap \{a = 0\}$ is empty. If $F_1, \dots, F_r$ define W set theoretically, the Nullstellensatz says $(F_1, \dots, F_r, a) = (1)$. This assertion is true for the displayed ideal itself: an ideal with empty zero set is the unit ideal. By [Lemma A.1](#) there is an identity

$$1 = aA + \sum_{\rho=1}^r F_\rho A_\rho$$

whose degrees are polynomial in d and whose coefficient heights have the form (A.10). At z we get $1 = aA(z)$. If τ_v is one at an archimedean place and zero otherwise, a polynomial Q of degree T in M variables satisfies

$$\log |Q(x)|_v \leq \log \max(1, \|Q\|_v) + T \log \max(1, \|x\|_v) + \tau_v \log \binom{T+M}{M}, \quad (\text{A.11})$$

where $\|Q\|_v$ denotes its maximum coefficient absolute value. On V , $|a|_v < 1$, so only the positive logarithms of U enter this bound. Evaluation of A and (2.3) give

$$\frac{h}{C_0 d^{C_0}} \leq \sum_{v \in V} w_v \log |a|_v^{-1} = \sum_{v \in V} w_v \log |A(z)|_v \leq D_*(d)(1 + \log h)^{B_*}.$$

An inequality $h \leq D(d)(1 + \log h)^B$ implies a polynomial bound for h : for fixed B there is a constant c_B such that $(1 + \log h)^B \leq c_B \sqrt{h}$ for all $h \geq 2$, and hence $h \leq c_B^2 D(d)^2$. This proves the theorem in the present case. It includes every case where a is constant on W , since its value is the nonzero target value, and every zero-dimensional W .

We may therefore suppose a varies on W and its boundary is nonempty. Condition (iv) and [Lemma A.2](#) give an integer combination E and an integer m , with bounds polynomial

in d , satisfying (A.1). Its coefficient and tail bounds retain the forms required in the theorem. More explicitly, if $c_{\max} = \max_{\alpha} c_{\alpha}$, the ultrametric inequality and integrality of the c_{α} preserve the original tail bound at finite places. At infinity replace b_v by

$$b_v + \tau_v \log \max(1, sc_{\max}).$$

The total extra weighted cost is at most $\log \max(1, sc_{\max})$, since the total archimedean weight is one. Combining the coefficient tuples likewise preserves their bounds. We use b_v for these enlarged errors.

Put $F = E_{<m}$ and form the graph variety W' of Lemma A.3. Its dimension is $k = \dim W$ and its equation count, degrees, and heights have the bounds (A.10), by Lemma A.1. Lift the target to $z' = (a, U, q)$, with $q = a^{-m}F(z)$. The rational expression for q gives

$$h_{\text{aff}}(z') \leq C_1(d)h + C_2(d)(1 + \log h)^{B_2} \leq C_3(d)h,$$

where C_3 is polynomial in d . For the second inequality use $(1 + \log h)^{B_2} \leq c_{B_2}h$; its constant is fixed at this descent step. The polynomial C_3 is chosen before the interpolation degrees n, L .

The local bound for q is much smaller. Since $E(z) = 0$ on V ,

$$q = -a^{-m} \sum_{j \geq m} a^j e_j(U), \quad \log^+ |q|_v \leq (m+1)b_v.$$

Consequently

$$\sum_{v \in V} w_v \log \max(1, \|z'\|_v) \leq C_4(d)(1 + \log h)^{B_4}. \quad (\text{A.12})$$

The exact equation at the lifted target is

$$q + \sum_{j \geq 0} a^j e_{m+j}(U) = 0.$$

Take set-theoretic defining equations for W' and let I be the ideal they generate. The ideal is kept as given, without assuming it radical. By Lemma A.3,

$$J = (I, a, q + e_m(U)) \quad \text{satisfies} \quad \dim V(J) \leq k - 2.$$

Set $A_0(d) = C_0 d^{C_0}$, and first choose an integer

$$c \geq 2A_0(d)(C_3(d) + 1). \quad (\text{A.13})$$

Choose L by Lemma A.4 and set $n = cL$. Both choices are polynomial in d , independent of h . That lemma gives P of degree at most L , nonzero on W' , with a certificate

$$P = \sum_{\rho} B_{\rho} I_{\rho} + B_a a^n + B_f \left(q + \sum_{j=0}^{n-1} a^j e_{m+j}(U) \right). \quad (\text{A.14})$$

Its degrees are polynomial in d . The coefficient heights of P and every B_{ρ}, B_a, B_f are at most $C_5(d)(1 + \log h)^{B_5}$. To apply the lemma with many coefficients, bound the height of their joint tuple by the sum of the individual heights in (2.4); there are polynomially many indices $m, \dots, m+n-1$, so this still has the asserted form.

We now test whether $P(z')$ can be nonzero. At a selected place, the parenthesized expression in (A.14) is

$$-a^{-m} \sum_{j \geq m+n} a^j e_j(U).$$

Its absolute value is at most $|a|_v^n \exp((m+n+1)b_v)$. Evaluate the two remaining certificate terms by (A.11), use (A.12), and include $\log 2$ at infinity for their sum. If $P(z') \neq 0$, the result is

$$\sum_{v \in V} w_v \log |P(z')|_v \leq -n \sum_{v \in V} w_v \log |a|_v^{-1} + C_6(d)(1 + \log h)^{B_6}. \quad (\text{A.15})$$

Here polynomial certificate degrees multiply only the small positive local logarithms, and $(m+n+1) \sum_V w_v b_v$ is of the same permitted size. This explains precisely where (2.3) is needed in addition to the global height bound.

At all other places evaluate P itself, of degree at most L , using (A.11). Its certificate is unnecessary for this estimate. Summing gives

$$\sum_{v \notin V} w_v \log |P(z')|_v \leq LC_3(d)h + C_7(d)(1 + \log h)^{B_7}. \quad (\text{A.16})$$

Enlarge the number field to contain W , W' , their chosen components, and all coefficients in (A.14); the normalized sums and all previously established estimates are unchanged. The product formula applied to the nonzero algebraic number $P(z')$ combines (A.15) and (A.16) into

$$0 \leq \left(-\frac{n}{A_0(d)} + LC_3(d) \right) h + C_8(d)(1 + \log h)^{B_8}.$$

By (A.13) the coefficient of h is at most $-L$. It follows that $h \leq C_8(d)(1 + \log h)^{B_8}$ and hence that h is polynomially bounded in d as above.

If this polynomial height bound does not hold, we have proved $P(z') = 0$. Substitute $q = F/a^m$ and clear denominators:

$$H(a, U) = a^{mL} P(a, U, F(a, U)/a^m).$$

This is a polynomial. It vanishes at z because $a(z) \neq 0$, and it is not identically zero on W , because P is nonzero on the irreducible graph closure and its graph over $a \neq 0$ is dense. Its degree is at most $mL + L \max(1, \deg F)$; expanding the substitution bounds its coefficient height by $h_{\text{aff}}(P) + Lh_{\text{aff}}(F)$ plus a polynomial degree error. Thus it has exactly the controls (A.10). Choose an irreducible component containing z of the proper intersection $W \cap \{H = 0\}$. By Lemma A.1 its defining equations have the same types of degree and height bounds, and its dimension is strictly smaller.

Repeat with this component as W . Condition (iv) applies to it because it is an irreducible algebraic variety containing the same target. After at most N drops one either has obtained the polynomial height bound or reaches the empty-boundary case, which gives that bound. There are finitely many fixed-step polynomial bounds in this procedure, so their maximum is bounded by Cd^C for a constant depending only on the fixed data. This completes the proof. $\square$

References

- [And89] Yves André. *G-Functions and Geometry*, volume E13 of *Aspects of Mathematics*. Friedr. Vieweg & Sohn, Braunschweig, 1989.
- [And92] Yves André. Mumford–Tate groups of mixed Hodge structures and the theorem of the fixed part. *Compositio Mathematica*, 82(1):1–24, 1992.
- [And95] Yves André. Théorie des motifs et interprétation géométrique des valeurs p -adiques de G -functions (une introduction). In Sinnou David, editor, *Number Theory: Paris 1992–3*, volume 215 of *London Mathematical Society Lecture Note Series*, pages 37–60. Cambridge University Press, Cambridge, 1995.
- [And25] Yves André. G -functions, motives, and unlikely intersections—old and new. *Rendiconti Lincei. Matematica e Applicazioni*, 36:381–392, 2025.
- [Asc04] Matthias Aschenbrenner. Ideal membership in polynomial rings over the integers. *Journal of the American Mathematical Society*, 17(2):407–441, 2004.
- [AY25] Abhinandan and Alex Youcis. An integral comparison of crystalline and de Rham cohomology. Preprint, arXiv:2507.17631, 2025. The introductory restatement of the classical rational comparison is used.
- [BB85] Daniel Bertrand and Frits Beukers. Équations différentielles linéaires et majorations de multiplicités. *Annales scientifiques de l’École Normale Supérieure*, 18(1):181–192, 1985.
- [Ber82] Pierre Berthelot. Géométrie rigide et cohomologie des variétés algébriques de caractéristique p . *Groupe d’étude d’analyse ultramétrique*, 9(3):J1–J18, 1981–1982. Exposé J2.
- [BLR90] Siegfried Bosch, Werner Lütkebohmert, and Michel Raynaud. *Néron Models*, volume 21 of *Ergebnisse der Mathematik und ihrer Grenzgebiete (3)*. Springer-Verlag, Berlin, 1990.
- [BO83] Pierre Berthelot and Arthur Ogus. F -isocrystals and de Rham cohomology. I. *Inventiones Mathematicae*, 72(2):159–199, 1983.
- [Bom81] Enrico Bombieri. On G -functions. In Heini Halberstam and Christopher Hooley, editors, *Recent Progress in Analytic Number Theory, Vol. 2 (Durham, 1979)*, pages 1–67. Academic Press, London–New York, 1981.
- [Cai10] Bryden Cais. Canonical extensions of Néron models of Jacobians. *Algebra & Number Theory*, 4(2):111–150, 2010.
- [Del70] Pierre Deligne. *Équations différentielles à points singuliers réguliers*, volume 163 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin–Heidelberg–New York, 1970.
- [DO21] Christopher Daw and Martin Orr. Unlikely intersections with $E \times \text{CM}$ curves in $\mathcal{A}_2$. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*, 22(4):1705–1745, 2021.

- [DO22] Christopher Daw and Martin Orr. Quantitative reduction theory and unlikely intersections. *International Mathematics Research Notices*, 2022(20):16138–16195, 2022.
- [DO25a] Christopher Daw and Martin Orr. The large Galois orbits conjecture under multiplicative degeneration. Preprint, arXiv:2306.13463v2, 2025.
- [DO25b] Christopher Daw and Martin Orr. Zilber–Pink in a product of modular curves assuming multiplicative degeneration. *Duke Mathematical Journal*, 174(13):2877–2926, 2025.
- [DOP26] Christopher Daw, Martin Orr, and Georgios Papas. Some new cases of Zilber–Pink in $Y(1)^3$. Preprint, arXiv:2510.09603v2, 2026.
- [Dub90] Thomas W. Dubé. The structure of polynomial ideals and Gröbner bases. *SIAM Journal on Computing*, 19(4):750–773, 1990.
- [EL23] Bas Edixhoven and Guido Lido. Geometric quadratic Chabauty. *Journal of the Institute of Mathematics of Jussieu*, 22(1):279–333, 2023.
- [FC90] Gerd Faltings and Ching-Li Chai. *Degeneration of Abelian Varieties*, volume 22 of *Ergebnisse der Mathematik und ihrer Grenzgebiete (3)*. Springer-Verlag, Berlin, 1990. With an appendix by David Mumford.
- [Her26] Grete Hermann. Die frage der endlich vielen schritte in der theorie der polynomideale. *Mathematische Annalen*, 95:736–788, 1926.
- [KPS01] Teresa Krick, Luis Miguel Pardo, and Martín Sombra. Sharp estimates for the arithmetic Nullstellensatz. *Duke Mathematical Journal*, 109(3):521–598, 2001.
- [Lan13] Kai-Wen Lan. *Arithmetic Compactifications of PEL-Type Shimura Varieties*, volume 36 of *London Mathematical Society Monographs*. Princeton University Press, Princeton, 2013. Numbered statements checked in the author’s revised thesis, March 14, 2021; <https://www.kwlan.org/articles/cpt-PEL-type-thesis-revision.pdf>.
- [LV20] Brian Lawrence and Akshay Venkatesh. Diophantine problems and p -adic period mappings. *Inventiones Mathematicae*, 221(3):893–999, 2020.
- [Mil86] James S. Milne. Abelian varieties. In Gary Cornell and Joseph H. Silverman, editors, *Arithmetic Geometry*, pages 103–150. Springer-Verlag, New York, 1986. Author’s revised version, January 2, 2022; theorem numbering unchanged; revised text: <https://www.jmilne.org/math/xnotes/AVs.pdf>.
- [Mil95] James S. Milne. Shimura varieties: The geometric side of the zeta function. Notes for four talks at the Institute for Advanced Study, March 6, 1995. <https://www.jmilne.org/math/articles/1995a.pdf>.
- [Mil13] James S. Milne. Shimura varieties and moduli. In Gavril Farkas and Ian Morrison, editors, *Handbook of Moduli. Vol. II*, volume 25 of *Advanced Lectures in Mathematics*, pages 467–548. International Press, Somerville, MA, 2013. Citations follow the published numbering.

- [MM74] Barry Mazur and William Messing. *Universal Extensions and One Dimensional Crystalline Cohomology*, volume 370 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin–Heidelberg, 1974.
- [MVW84] Colin R. Matthews, Leonid N. Vaserstein, and Boris Weisfeiler. Congruence properties of Zariski-dense subgroups I. *Proceedings of the London Mathematical Society. Third Series*, 48(3):514–532, 1984.
- [MW94] David W. Masser and Gisbert Wüstholz. Endomorphism estimates for abelian varieties. *Mathematische Zeitschrift*, 215:641–653, 1994. Author preprint: MPI 1991/51.
- [MZ99] Ben Moonen and Yuri G. Zarhin. Hodge classes on abelian varieties of low dimension. *Mathematische Annalen*, 315(4):711–733, 1999.
- [Ope26a] OpenAI. Quaternionic division points on curves in the Siegel threefold. OpenAI Math Release preprint [OAI:Quaternionic-Division-Points-on-Curves-in-the-Siegel-Threefold-September-24-2026](#), 2026.
- [Ope26b] OpenAI. The $E \times \text{CM}$ component of Zilber–Pink for curves in $\mathcal{A}_2$. OpenAI Math Release preprint [OAI:The-E-times-CM-Component-of-Zilber-Pink-for-Curves-in-A2-September-24-2026](#), 2026.
- [Pap25] Georgios Papas. On the v -adic values of G-functions I: Splittings in $\mathcal{A}_2$. Preprint, arXiv:2510.11806v1, 2025.
- [Paz12] Fabien Pazuki. Theta height and Faltings height. *Bulletin de la Société Mathématique de France*, 140(1):19–49, 2012.
- [Pet03] Denis Petrequin. Classes de Chern et classes de cycles en cohomologie rigide. *Bulletin de la Société Mathématique de France*, 131(1):59–121, 2003.
- [Pin05] Richard Pink. A common generalization of the conjectures of André–Oort, Manin–Mumford, and Mordell–Lang. Preprint, April 17, 2005. <https://people.math.ethz.ch/~pink/ftp/AOMMML.pdf>.
- [PT13] Jonathan Pila and Jacob Tsimerman. The André–Oort conjecture for the moduli space of abelian surfaces. *Compositio Mathematica*, 149(2):204–216, 2013.
- [PW06] Jonathan Pila and Alex J. Wilkie. The rational points of a definable set. *Duke Mathematical Journal*, 133(3):591–616, 2006.
- [PZ08] Jonathan Pila and Umberto Zannier. Rational points in periodic analytic sets and the Manin–Mumford conjecture. *Rendiconti Lincei. Matematica e Applicazioni*, 19(2):149–162, 2008.
- [Sch73] Wilfried Schmid. Variation of Hodge structure: The singularities of the period mapping. *Inventiones Mathematicae*, 22:211–319, 1973.
- [Zil02] Boris Zilber. Exponential sums equations and the Schanuel conjecture. *Journal of the London Mathematical Society*, 65(1):27–44, 2002.